

Glossary

ABC transporters A large class of membrane transport proteins that use the energy of ATP hydrolysis to transfer peptides or small molecules across membranes. (Chapter 11) (Figure 11–16)

acetyl CoA Small water-soluble activated carrier molecule. Consists of an acetyl group linked to coenzyme A (CoA) by an easily hydrolyzable, energy-rich thioester bond. (Chapter 2) (Figure 2–38)

acetylcholine receptor (AChR) Membrane protein that responds to binding of acetylcholine (ACh). The nicotinic AChR is a transmitter-gated ion channel that opens in response to ACh. The muscarinic AChR is not an ion channel, but a G-protein-coupled cell-surface receptor. (Chapter 11)

acid A proton donor. Substance that releases protons (H^+) when dissolved in water, forming hydronium ions (H_3O^+) and lowering the pH. (Chapter 2) (Panel 2–2, pp. 96–97)

acid hydrolases Hydrolytic enzymes—including proteases, nucleases, glycosidases, lipases, phospholipases, phosphatases, and sulfatases—that work best at acidic pH; these enzymes are found within the lysosome. (Chapter 13)

action potential Rapid, transient, self-propagating electrical excitation in the plasma membrane of a cell such as a neuron or muscle cell. Action potentials, or nerve impulses, make possible long-distance signaling in the nervous system. (Chapter 11) (Figure 11–33)

activated carrier Small diffusible molecule that stores easily exchangeable energy in the form of one or more energy-rich covalent bonds. Examples are ATP, acetyl CoA, $FADH_2$, NADH, and NADPH. (Chapter 2) (Figure 2–31)

activation energy The extra energy that must be acquired by atoms or molecules in addition to their ground-state energy in order to reach the transition state required for them to undergo a particular chemical reaction. (Figure 2–21)

activation-induced deaminase (AID) The enzyme catalyzing the processes of somatic hypermutation and immunoglobulin class switching in activated B cells. (Chapter 24)

active site Region of an enzyme surface to which a substrate molecule binds in order to undergo a catalyzed reaction. (Chapter 3)

active transport Movement of a molecule across a membrane or other barrier driven by energy other than that stored in the electrochemical or concentration gradient of the transported molecule itself. (Chapter 11)

adaptation (1) Adaptation (desensitization): adjustment of sensitivity following repeated stimulation. The mechanism that allows a cell to react to small changes in stimuli even against a high background level of stimulation. (2) Evolutionary adaptation: an evolved trait. (Chapters 11, 15)

adaptive immune system System of lymphocytes providing highly specific and long-lasting defense against pathogens in vertebrates. It consists of two major classes of lymphocytes: B lymphocytes (B cells), which secrete antibodies that bind specifically to the pathogen or its products, and T lymphocytes (T cells), which can either directly kill cells infected with the pathogen or produce secreted or cell-surface signal proteins that stimulate other host cells to help eliminate the pathogen. (Chapter 24) (Figure 24–2)

adaptor protein, adaptor General term for a protein that functions solely to link two or more different proteins together in an intracellular signaling pathway or protein complex. (Chapters 13, 15) (Figure 15–11)

adenylyl cyclase (adenylate cyclase) Membrane-bound enzyme that catalyzes the formation of cyclic AMP from ATP. An important component of some intracellular signaling pathways. (Chapter 15)

adherens junction Cell junction in which the cytoplasmic face of the plasma membrane is attached to actin filaments. Examples include adhesion belts linking adjacent epithelial cells and focal contacts on the lower surface of cultured fibroblasts. (Chapter 19)

adhesins Specific proteins or protein complexes of pathogenic bacteria that recognize and bind cell-surface molecules on the host cells to enable tight adhesion and colonization of tissues. (Chapter 23)

adhesion belt Adherens junctions in epithelia that form a continuous belt (*zonula adherens*) just beneath the apical face of the epithelium, encircling each of the interacting cells in the sheet. (Chapter 19)

ADP (adenosine 5'-diphosphate) Nucleotide derivative (a nucleoside diphosphate) produced by hydrolysis of the terminal phosphate of ATP. Regenerates ATP when phosphorylated by an energy-generating process such as oxidative phosphorylation. (Chapter 2) (Figure 2–33)

adult stem cells Undifferentiated cells found throughout the body that undergo cell divisions to maintain homeostasis of adult tissues and replenish damaged tissues; also known as somatic stem cells. (Chapter 22)

aerobic respiration Process by which a cell obtains energy from sugars or other organic molecules by allowing their carbon and hydrogen atoms to combine with the oxygen in air to produce CO_2 and H_2O , respectively. (Chapters 2 and 14)

affinity constant (K_a) The equilibrium constant for a simple binding interaction, when expressed as $[AB]/[A][B]$. Also known as the *association constant*. (Chapter 3)

affinity maturation Progressive increase in the affinity of antibodies for the immunizing antigen with the passage of time after immunization. (Chapter 24)

AIRE (autoimmune regulator) A protein expressed by a subpopulation of epithelial cells in the thymus that stimulates the production of small amounts of self proteins characteristic of other organs, exposing developing thymocytes to these proteins for the purpose of self-tolerance. (Chapter 24)

Akt Serine/threonine protein kinase that acts in the PI-3-kinase–Akt intracellular signaling pathway; involved especially in signaling cells to grow and survive. Also called protein kinase B (PKB). (Chapter 15)

allele One of several alternative forms of a gene. In a diploid cell, each gene will typically have two alleles, occupying the same corresponding position (locus) on homologous chromosomes. (Chapters 5, 8)

allosteric protein A protein that can adopt at least two distinct conformations, and for which the binding of a ligand at one site causes a conformational change that alters the activity of the protein at a second site; this allows one type of molecule in a cell to alter the fate of a molecule of another type, a feature widely exploited in enzyme regulation. (Chapter 3)

allostery (adjective **allosteric**) Change in a protein's conformation brought about by the binding of a regulatory ligand (at a site other than the protein's catalytic site) or by covalent modification. The change in conformation alters the activity of the protein; it can also form the basis of directed movement. (Chapter 3) (Figures 3–71 and 16–53)

alpha helix (α helix) Common folding pattern in proteins, in which a linear sequence of amino acids folds into a right-handed helix stabilized by internal hydrogen-bonding between backbone atoms. (Chapter 3) (Figure 3–6)

alternative RNA splicing Production of different RNAs from the same gene by splicing the transcript in different ways. (Chapter 7) (Figure 7–59)

amino acid Organic molecule containing both an amino group and a carboxyl group. Those that serve as building blocks of proteins are alpha amino acids, having both the amino and carboxyl groups linked to the same carbon atom. (Chapter 1) (Figure 3–1 and Panel 3–1, pp. 118–119)

aminoacyl-tRNA synthetase Enzyme that attaches the correct amino acid to a tRNA molecule to form an aminoacyl-tRNA. (Chapter 6) (Figure 6–58)

amoeboid cell migration A rapid mode of cell locomotion typical of amoebae and white blood cells that is characterized by protrusion of actin-rich pseudopodia at the leading edge, formation of weak attachments to the substratum, and actomyosin-based contraction at the rear of the cell. (Chapter 16)

AMPA receptor Glutamate-gated ion channel in the mammalian central nervous system that carries most of the depolarizing current responsible for excitatory postsynaptic potentials. (Chapter 11)

amphiphilic Having both hydrophobic and hydrophilic regions, as in a phospholipid or a detergent molecule. (Chapter 10)

amyloid fibrils Self-propagating, stable β -sheet aggregates built from hundreds of identical polypeptide chains that become layered one over the other to create a continuous stack of β sheets. The unbranched fibrous structure can contribute to human diseases when not controlled. (Chapter 3)

anaphase (1) Stage of mitosis during which sister chromatids separate and move away from each other. (2) In meiosis, anaphase I and II are the stages during which chromosome homolog pairs separate (I), and then sister chromatids separate (II). (Panel 17–1, pp. 1048–1049)

anaphase A The stage of mitosis during which chromosome segregation occurs as chromosomes move toward the two spindle poles. (Chapter 17)

anaphase B The stage of mitosis during which chromosome segregation occurs as spindle poles separate and move apart. (Chapter 17)

anaphase-promoting complex, or cyclosome (APC/C) Ubiquitin ligase that catalyzes the ubiquitylation and destruction of securin and M- and S-cyclins, initiating the separation of sister chromatids in the metaphase-to-anaphase transition during mitosis. (Chapter 17)

anchorage dependence Dependence of cell growth, proliferation, and survival on attachment to a substratum. (Chapter 19)

anchoring junction Cell junction that attaches cells to neighboring cells or to the extracellular matrix. (Chapter 19) (Table 19–1, p. 1107)

angiogenesis Growth of new blood vessels by sprouting from existing ones. (Chapter 20)

antenna complex Part of a photosystem that captures light energy and channels it into the photochemical reaction center. It consists of protein complexes that bind large numbers of chlorophyll molecules and other pigments. (Chapter 14)

Antennapedia complex One of two gene clusters in *Drosophila* that contain *Hox* genes; genes in the *Antennapedia* complex control the differences among the thoracic and head segments of the body. (Chapter 21)

anti-apoptotic Bcl2 family proteins Proteins (for example, Bcl2, BclxL) on the cytosolic surface of the outer mitochondrial membrane that bind and inhibit pro-apoptotic Bcl2 family proteins and thereby help prevent inappropriate activation of the intrinsic pathway of apoptosis. (Chapter 18)

anti-IAP proteins Produced in response to various apoptotic stimuli and, by binding to IAPs (inhibitory antiapoptotic proteins), prevent their binding to a caspase—thereby blocking the inhibition of apoptosis provided by IAPs. (Chapter 18)

antibiotic Substance such as penicillin or streptomycin that is toxic to microorganisms. Often a natural product of a particular microorganism or plant. (Chapter 23)

antibody (or immunoglobulin) Protein secreted by activated B cells in response to a pathogen or foreign molecule. Binds tightly to the pathogen or foreign molecule, inactivating it or marking it for destruction by phagocytosis or complement-induced lysis. (Chapters 3, 24) (Figure 24–23)

antibody response Adaptive immune response in which B cells are activated to secrete antibodies that circulate in the bloodstream or enter other body fluids, where they can bind specifically to the foreign antigen that stimulated their production. (Chapter 24)

anticodon Sequence of three nucleotides in a transfer RNA (tRNA) molecule that is complementary to a three-nucleotide codon in a messenger RNA (mRNA) molecule. (Chapter 6)

antigen Any molecule that can bind specifically to an antibody or B cell receptor, or any protein fragment bound to an MHC protein that can bind specifically to a T cell receptor. (Chapters 3, 24)

antigen-presenting cell (APC) Cell that displays foreign antigen complexed with an MHC protein on its surface for presentation to T lymphocytes. (Chapter 24)

antigenic determinant Specific region of an antigen that binds to an antibody or a complementary receptor on the surface of a B cell (BCR) or T cell (TCR). (Chapter 24)

antigenic variation Ability to change the antigens displayed on the cell surface; a property of some pathogenic microorganisms that enables them to evade attack by the adaptive immune system. (Chapter 23)

antiparallel Describes the relative orientation of the two strands in a DNA double helix or two paired regions of a polypeptide chain; the chemical polarity of one strand is opposite to that of the other. (Chapter 4)

antiporter Carrier protein that transports two different ions or small molecules across a membrane in opposite directions, either simultaneously or in sequence. (Chapter 11) (Figure 11–8)

Apaf1 Adaptor protein of the intrinsic apoptotic pathway; on binding cytochrome c, oligomerizes to form an apoptosome. (Chapter 18)

apical Referring to the tip of a cell, a structure, or an organ. The apical surface of an epithelial cell is the exposed free surface, opposite to the basal surface. The basal surface rests on the basal lamina that separates the epithelium from other tissue. (Chapter 19)

apoptosis Form of programmed cell death, in which a “suicide” program is activated within an animal cell, leading to rapid cell death mediated by intracellular proteolytic enzymes called caspases. (Chapter 18)

apoptosome Heptamer of Apaf1 proteins that forms on activation of the intrinsic apoptotic pathway; it recruits and activates initiator caspases that subsequently activate downstream executioner caspases to induce apoptosis. (Chapter 18)

aquaporin (water channel) Channel protein embedded in the plasma membrane that greatly increases the cell's permeability to water, allowing transport of water, but not ions, at a high rate across the membrane. (Chapter 11)

archaea (archaeobacteria) Single-celled organisms without a nucleus, superficially similar to bacteria. At a molecular level, more closely related to eukaryotes in genetic machinery than are bacteria. Archaea and bacteria together make up the prokaryotes. (Figure 1–9)

ARF proteins Monomeric GTPase in the Ras superfamily responsible for regulating both COPI coat assembly and clathrin coat assembly. (Chapter 13) (Table 15–5, p. 915)

Arp2/3 complex Complex containing two actin-related proteins that binds to an actin filament and then nucleates actin filament growth from the minus end. (Chapter 16)

arrestin Member of a family of proteins that contributes to GPCR desensitization by preventing the activated receptor from interacting with G proteins; also serves as an adaptor to couple the receptor to clathrin-dependent endocytosis. (Chapter 15) (Figure 15–43)

association constant (K_a) The equilibrium constant for a simple binding interaction, when expressed as $[AB]/[A][B]$. Also known as the *affinity constant*. (Chapter 3)

astral microtubule In the mitotic spindle, any of the microtubules radiating from the aster that are not attached to a kinetochore of a chromosome. (Chapter 17)

asymmetric cell division Cell division in which some important molecule or molecules are distributed unequally

between the two daughter cells, causing these cells to become different from each other. (Figures 21–35 and 22–16)

ATM (ataxia telangiectasia mutated protein) Protein kinase activated by double-strand DNA breaks. If breaks are not repaired, ATM initiates a signal cascade that culminates in cell-cycle arrest. Related to ATR. (Chapter 17)

ATP (adenosine 5'-triphosphate) Nucleoside triphosphate composed of adenine, ribose, and three phosphate groups. The principal carrier of chemical energy in cells. The terminal phosphate groups are highly reactive in the sense that their hydrolysis, or transfer to another molecule, takes place with the release of a large amount of free energy. (Chapter 2) (Figure 2–33)

ATP synthase (F_1F_0 ATP synthase) Abundant transmembrane enzyme complex in the inner membrane of mitochondria and the thylakoid membrane of chloroplasts. Driven by an electrochemical proton gradient, it catalyzes the formation of ATP from ADP and phosphate during oxidative phosphorylation and photosynthesis. Also present in the plasma membrane of bacteria. (Chapter 14)

ATR (ataxia telangiectasia and Rad3 related protein) Protein kinase activated by DNA damage. If damage remains unrepaired, ATR helps initiate a signal cascade that culminates in cell-cycle arrest. Related to ATM. (Chapter 17)

autoimmune disease Pathological state in which the body mounts a disabling adaptive immune response against one or more of its own molecules. (Chapter 24)

autophagosome Organelle surrounded by a double membrane that contains engulfed cytoplasmic cargo in the initial stages of autophagy. (Chapter 13)

autophagy Digestion of cytoplasm and worn-out organelles by the cell's own lysosomes. (Chapter 13)

auxin Plant hormone, indole-3-acetic acid, with numerous roles in plant growth and development. (Chapter 15)

axon Long nerve-cell projection that can rapidly conduct nerve impulses over long distances so as to deliver signals to other cells. (Chapter 11)

axoneme Bundle of microtubules and associated proteins that forms the core of a cilium or a flagellum in eukaryotic cells and is responsible for their movements. (Chapter 16)

B cell receptor (BCR) The transmembrane immunoglobulin protein on the surface of a B cell that serves as its receptor for antigen. (Chapter 24)

bacterial artificial chromosome (BAC) Cloning vector that can accommodate large pieces of DNA, typically up to 1 million base pairs. (Chapter 8)

bacteriorhodopsin Pigmented protein found in the plasma membrane of a salt-loving archaeon, *Halobacterium salinarum* (*Halobacterium halobium*). Pumps protons out of the cell in response to light. (Chapter 10)

bacterium (plural bacteria) (eubacterium) Member of the domain bacteria, one of the three main branches of the tree of life (archaea, bacteria, and eukaryotes). Bacteria and archaea both lack a distinct nuclear compartment and together comprise the prokaryotes. (Chapter 23) (Figure 1–9)

Bak A main effector Bcl2 family protein of the intrinsic pathway of apoptosis in mammalian cells that is bound to the mitochondrial outer membrane even in the absence of an apoptotic signal; activation is usually by activated pro-apoptotic BH3-only proteins. (Chapter 18)

basal Situated near the base. Opposite the apical surface. (Chapter 19)

basal lamina (plural **basal laminae**) Thin mat of extracellular matrix that separates epithelial sheets, and many other types of cells such as muscle or fat cells, from connective tissue. Sometimes called basement membrane. (Chapter 19) (Figure 19–51)

base (1) A substance that can reduce the number of protons in solution, either by accepting H^+ ions directly or by releasing OH^- ions, which then combine with H^+ to form H_2O . (Chapter 2) (2) The purines and pyrimidines in DNA and RNA are organic nitrogenous bases and are commonly referred to simply as bases. (Panel 2–2, pp. 96–97)

base excision repair DNA repair pathway in which single faulty bases are removed from the DNA helix and replaced. Compare **nucleotide excision repair**. (Chapter 5) (Figure 5–41)

base pair Two nucleotides in an RNA or DNA molecule that are held together by hydrogen bonds; for example, G paired with C, and A paired with T or U. (Chapter 4)

basement membrane Thin mat of extracellular matrix that separates epithelial sheets, and many other types of cells such as muscle or fat cells, from connective tissue. Also called basal lamina. (Chapter 19) (Figure 19–51)

Bax A main effector Bcl2 family protein of the intrinsic pathway of apoptosis in mammalian cells; located mainly in the cytosol, it translocates to the mitochondria only after activation, usually by activated pro-apoptotic BH3-only proteins. (Chapter 18)

Bcl2 Anti-apoptotic Bcl2 family protein of the outer mitochondrial membrane that binds and inhibits pro-apoptotic Bcl2 family proteins and prevents inappropriate activation of the intrinsic pathway of apoptosis. (Chapter 18)

Bcl2 family Family of intracellular proteins that either promote or inhibit apoptosis by regulating the release of cytochrome c and other mitochondrial proteins from the intermembrane space into the cytosol. (Chapter 18)

BclxL Anti-apoptotic Bcl2 family protein of the outer mitochondrial membrane that binds and inhibits pro-apoptotic Bcl2 family proteins and prevents inappropriate activation of the intrinsic pathway of apoptosis. (Chapter 18)

benign Of tumors: self-limiting in growth, and noninvasive. (Chapter 20)

beta sheet (β sheet) Common structural motif in proteins in which different sections of the polypeptide chain run alongside each other, joined together by hydrogen-bonding between atoms of the polypeptide backbone. Also known as a β pleated sheet. (Chapter 3) (Figure 3–6)

beta-catenin (β -catenin) Multifunctional cytoplasmic protein involved in cadherin-mediated cell–cell adhesion, linking cadherins to the actin cytoskeleton. Can also act independently as a transcription regulatory protein. Has an important role in animal development as part of a Wnt signaling pathway. (Chapter 15)

BH3-only proteins The largest subclass of Bcl2 family proteins. Produced or activated in response to an apoptotic stimulus, these proteins promote apoptosis mainly by inhibiting anti-apoptotic proteins in the Bcl2 family. (Chapter 18)

bi-orientation In mitosis, the attachment of sister chromatids to opposite poles of the mitotic spindle, so that they move to opposite ends of the cell when they separate in anaphase. (Chapter 17)

binding site Region on the surface of one molecule (usually a protein or nucleic acid) that can interact with another molecule through noncovalent bonding. (Chapter 3)

biomolecular condensate An aggregate inside cells, formed by a process analogous to liquid–liquid phase separation and based on fluctuating weak interactions between scaffold proteins; concentrates selected protein and RNA molecules in a membraneless compartment. (Chapters 1, 3, 12) (Figures 3–77 and 12–5)

BiP Endoplasmic reticulum (ER)-resident chaperone protein. Member in the family of hsp70-type chaperones. (Chapter 12)

Bithorax complex One of two gene clusters in *Drosophila* that contain *Hox* genes; genes in the *Bithorax* complex control the differences among the abdominal and thoracic segments of the body. (Chapter 21)

bivalent A four-chromatid structure formed during meiosis, consisting of a duplicated chromosome tightly paired with its homologous duplicated chromosome. (Chapter 17)

blastomere One of the many similar cells formed by the early cleavages of a fertilized egg. (Chapter 21)

blastula Early stage of an animal embryo, usually consisting of a hollow ball of epithelial cells surrounding a fluid-filled cavity, before gastrulation begins. (Chapter 21)

blebbing Membrane protrusion formed when the plasma membrane detaches locally from the underlying actin cortex, allowing cytoplasmic flow and hydrostatic pressure within the cell to push the membrane outward. (Chapter 16)

bone Dense and rigid connective tissue comprising a mixture of tough fibers (type I collagen fibrils), which resist pulling forces, plus solid particles (calcium phosphate as hydroxylapatite crystals), which resist compression. (Chapter 21)

brassinosteroids Class of steroid signal molecules in plants that regulate the growth and differentiation of plants throughout their life cycle via binding to a cell-surface receptor kinase to initiate a signaling cascade. (Chapter 15)

bright-field microscope Normal light microscope in which the image is obtained by simple transmission of light through the object being viewed. (Chapter 9)

Brownian motion The random movement of particles or molecules suspended in a liquid or gas, caused by molecular collisions. (Chapter 1)

buffer Solution of weak acid or weak base that resists the pH change that would otherwise occur when small quantities of acid or base are added. (Chapter 2)

C3 The pivotal complement protein that is activated by the early components of all three complement pathways (the classical pathway, the lectin pathway, and the alternative pathway). (Figure 24–7)

Ca^{2+} pump (calcium pump, Ca^{2+} ATPase) Transport protein in the membrane of sarcoplasmic reticulum of muscle cells (and elsewhere). Pumps Ca^{2+} out of the cytoplasm into the sarcoplasmic reticulum using the energy of ATP hydrolysis. (Chapter 11)

Ca^{2+} -activated K^+ channel Opens in response to the raised concentration of Ca^{2+} in nerve cells that occurs in response to an action potential. Increased K^+ permeability makes the membrane harder to depolarize, increasing the delay between action potentials and decreasing the response of the cell to constant, prolonged stimulation (adaptation). (Chapter 11)

Ca²⁺/calmodulin-dependent kinase (CaM-kinase) Serine/threonine protein kinase that is activated by Ca²⁺/calmodulin. Indirectly mediates the effects of an increase in cytosolic Ca²⁺ by phosphorylating specific target proteins. (Chapter 15) (Figure 15–34)

cadherin Member of the large cadherin superfamily of transmembrane adhesion proteins. Mediates homophilic Ca²⁺-dependent cell–cell adhesion in animal tissues. (Chapter 19) (Figure 19–3 and Table 19–1, p. 1107)

cadherin superfamily Family of classical and nonclassical cadherin proteins with more than 180 members in humans. (Chapter 19)

calmodulin Ubiquitous intracellular Ca²⁺-binding protein that undergoes a large conformation change when it binds Ca²⁺, allowing it to regulate the activity of many target proteins. In its activated (Ca²⁺-bound) form, it is called Ca²⁺/calmodulin. (Chapter 15) (Figure 15–34)

calnexin Carbohydrate-binding chaperone protein in the endoplasmic reticulum (ER) membrane; binds to oligosaccharides on incompletely folded proteins and retains them in the ER. (Chapter 12)

calreticulin Carbohydrate-binding chaperone protein in the endoplasmic reticulum (ER) lumen; binds to oligosaccharides on incompletely folded proteins and retains them in the ER. (Chapter 12)

CaM-kinase II Multifunctional Ca²⁺/calmodulin-dependent protein kinase that phosphorylates itself and various target proteins when activated. Found in most animal cells but is especially abundant at synapses in the brain; it is involved in some forms of synaptic plasticity in vertebrates. (Chapter 15) (Figure 15–35)

cancer stem cells Rare cancer cells capable of dividing indefinitely. (Chapter 20)

cancer-critical genes Genes whose alteration contributes to the causation or evolution of cancer by driving tumorigenesis. (Chapter 20)

capsid Protein coat of a virus, formed by the self-assembly of one or more types of protein subunit into a geometrically regular structure. (Chapter 23) (Figure 3–27)

carbohydrate layer The carbohydrate-rich zone on the eukaryotic cell surface attributable to glycoproteins, glycolipids, and proteoglycans of the plasma membrane. (Chapter 10)

carbon-fixation reaction Process by which inorganic carbon (as atmospheric CO₂) is incorporated into organic molecules. The second stage of photosynthesis. (Chapter 14) (Figure 14–40)

carcinogenesis The generation of cancer. (Chapter 20)

carcinoma Cancer of epithelial cells. The most common form of human cancer. (Chapter 20)

cargo The membrane components and soluble molecules carried by transport vesicles. (Chapter 13)

cartilage Form of connective tissue composed of cells (chondrocytes) embedded in a matrix rich in type II collagen and chondroitin sulfate proteoglycan. (Chapters 19, 21)

caspase Intracellular protease that is involved in mediating the intracellular events of apoptosis. (Figures 18–3 and 18–4)

catalyst Substance that can lower the activation energy of a reaction (thus increasing its rate), without itself being consumed by the reaction. (Chapters 1, 2, 3)

caveola (plural **caveolae**) Invaginations at the cell surface that bud off internally to form pinocytic vesicles. Thought to form from lipid rafts, regions of membrane rich in certain lipids. (Chapter 13)

caveolins Family of unusual integral membrane proteins that are the major structural proteins in caveolae. (Chapter 13)

CD4 Co-receptor protein on helper T cells and regulatory T cells that binds to a nonvariable part of class II MHC proteins (on antigen-presenting cells) outside the peptide-binding groove. (Chapter 24) (Figure 24–40)

CD8 Co-receptor protein on cytotoxic T cells that binds to a nonvariable part of class I MHC proteins (on antigen-presenting cells and infected target cells) outside the peptide-binding groove. (Chapter 24) (Figure 24–40)

Cdc20 Activating subunit of the anaphase-promoting complex/cyclosome (APC/C). (Chapter 17)

Cdc25 Protein phosphatase that dephosphorylates Cdk and increases their activity. (Chapter 17)

Cdc42 Member of the Rho family of monomeric GTPases that regulate the actin and microtubule cytoskeletons, cell-cycle progression, gene transcription, and membrane transport. (Chapter 15)

Cdc6 Protein essential in the preparation of DNA for replication. With Cdt1 it binds to an origin recognition complex on chromosomal DNA and helps load the Mcm proteins onto the DNA. (Figure 5–31)

Cdh1 Activating subunit of the anaphase-promoting complex/cyclosome (APC/C). (Chapter 17)

Cdk inhibitor protein (CKI) Protein that binds to and inhibits cyclin–Cdk complexes, primarily involved in the control of G₁ and S phases. (Chapter 17)

Cdt1 Protein essential in the preparation of DNA for replication. With Cdc6 it binds to origin recognition complexes on chromosomes and helps load the Mcm proteins on to the DNA. (Chapter 17) (Figure 5–31)

cDNA clone Clone containing double-stranded cDNA molecules derived from the protein-coding mRNA molecules present in a cell. (Chapter 8)

cDNA library Collection of cloned DNA molecules representing complementary DNA copies of the mRNA produced by a cell. (Chapter 8)

cell cortex Specialized layer of cytoplasm on the inner face of the plasma membrane. In animal cells it is an actin-rich layer responsible for movements of the cell surface. (Chapter 16)

cell cycle (cell-division cycle) Reproductive cycle of a cell: the orderly sequence of events by which a cell duplicates its chromosomes and other cell contents, and then divides into two. (Chapters 4, 17) (Figure 17–4)

cell-cycle control system Network of regulatory proteins that governs progression of a eukaryotic cell through the cell cycle. (Chapter 17)

cell determination Process whereby a cell progressively loses the potential to form other cell types, as development proceeds. (Chapter 21)

cell doctrine The nineteenth-century proposal that all living organisms are composed of one or more cells and that all cells arise from the division of other living cells. (Chapter 9)

cell memory Retention by cells and their descendants of persistently altered patterns of gene expression, without any

change in DNA sequence. *See also* **epigenetic inheritance**. (Chapters 7, 21)

cell plate Flattened membrane-bounded structure that forms by fusing vesicles in the cytoplasm of a dividing plant cell and is the precursor of the new cell wall. (Chapter 17)

cellulose Long, unbranched chains of glucose; major constituent of plant cell walls. (Chapter 19)

cellulose microfibril Highly ordered crystalline aggregate formed from bundles of about 18 cellulose chains, arranged with the same polarity and stuck together in overlapping parallel arrays by hydrogen bonds between adjacent cellulose molecules. (Chapter 19)

central (primary) lymphoid organ Organ in which T or B lymphocytes are produced from precursor cells. In adult mammals, these are the thymus and bone marrow, respectively. (Chapter 24) (Figure 24–12)

centriole Short cylindrical array of microtubules, closely similar in structure to a basal body. A pair of centrioles is usually found at the center of a centrosome in animal cells. (Chapter 16) (Figure 16–42)

centromere Constricted region of a mitotic chromosome that holds sister chromatids together. This is also the site on the DNA where the kinetochore forms so as to capture microtubules from the mitotic spindle. (Chapter 4) (Figure 4–43)

centrosome Centrally located organelle of animal cells that is the primary microtubule-organizing center (MTOC) and acts as the spindle pole during mitosis. In most animal cells it contains a pair of centrioles. (Chapter 17) (Figures 16–41 and 17–27)

cerebral cortex Outermost layer of the hemispheres of the brain; the most complex structure in the human body. (Figure 21–41)

CG island Region of DNA in vertebrate genomes with a greater than average density of CG sequences; the C nucleotides in these regions generally remain unmethylated. (Chapter 7)

channel (membrane channel) Transmembrane protein complex that allows inorganic ions or other small molecules to diffuse passively across the lipid bilayer. (Chapter 11) (Figure 11–3)

channelrhodopsin Photosensitive protein forming a cation channel across the membrane that opens in response to light. (Chapter 11)

charge separation In photosynthesis, the light-induced transfer of a high-energy electron from chlorophyll to an acceptor molecule resulting in the formation of a positive charge on the chlorophyll and a negative charge on a mobile electron carrier. (Figure 14–45)

chemical carcinogens Disparate chemicals that are carcinogenic—due to the ability to cause mutations—when fed to experimental animals or painted repeatedly on their skin. (Chapter 20)

chemical group Certain common combinations of atoms—such as methyl (–CH₃), hydroxyl (–OH), carboxyl (–COOH), carbonyl (–C=O), phosphate (–PO₃^{2–}), sulfhydryl (–SH), and amino (–NH₂) groups—that have distinct chemical and physical properties and influence the behavior of the molecule in which the group occurs. (Chapter 2)

chemiosmotic coupling (chemiosmosis) Mechanism in which an electrochemical proton gradient across a membrane (composed of a pH gradient plus a membrane potential) is used

to drive an energy-requiring process, such as ATP production or the rotation of bacterial flagella. (Chapter 14)

chemotaxis Movement of a cell toward or away from some diffusible chemical. (Chapter 16)

chiasma (plural **chiasmata**) X-shaped connection visible between paired homologous chromosomes during meiosis. Represents a site of chromosomal crossing-over, a form of genetic recombination. (Chapter 17)

chlorophyll Light-absorbing green pigment that plays a central part in photosynthesis in bacteria, plants, and algae. (Chapter 14)

chloroplast Membrane-bounded organelle in green algae and plants that contains chlorophyll and carries out photosynthesis. (Chapters 12, 14)

cholera toxin Secreted toxic protein of *Vibrio cholerae* responsible for causing the watery diarrhea associated with cholera. Comprises an A subunit with enzymatic activity and a B subunit that binds to host-cell receptors to direct subunit A to the host-cell cytosol. (Chapter 23)

cholesterol An abundant lipid molecule with a characteristic four-ring steroid structure. An important component of the plasma membranes of animal cells. (Chapter 10) (Figure 10–4)

chromatin Complex of DNA, histones, and non-histone proteins found in the nucleus of a eukaryotic cell. The material of which chromosomes are made. (Chapter 4)

chromatin immunoprecipitation Technique by which chromosomal DNA bound by a particular protein can be isolated and identified by precipitating it by means of an antibody against the protein. (Chapter 8) (Figures 8–67 and 8–68)

chromosome Structure composed of a very long DNA molecule and associated proteins that carries part (or all) of the hereditary information of an organism. Especially evident in plant and animal cells undergoing mitosis or meiosis, during which each chromosome becomes condensed into a compact rodlike structure visible in the light microscope. (Chapter 4)

cilium (plural **cilia**) Hairlike extension of a eukaryotic cell containing a core bundle of microtubules. Many cells contain a single nonmotile cilium, while others contain large numbers that perform repeated beating movements. *Compare* **flagellum**. (Chapter 16)

circadian clock Internal cyclical process that produces a particular change in a cell or organism with a period of around 24 hours; for example, the sleep–wakefulness cycle in humans. (Chapter 15)

cis face Face on the same or near side. (Chapter 13)

cis Golgi network (CGN) Network of fused vesicular tubular clusters that is closely associated with the *cis* face of the Golgi apparatus and is the compartment at which proteins and lipids enter the Golgi from the ER. (Chapter 13)

cis-regulatory sequences DNA sequences to which transcription regulators bind to control the rate of gene transcription. In nearly all cases, these sequences must be on the same chromosome (that is, *in cis*) to the genes they control. (Chapter 7) (Figure 7–18)

cisternal maturation mechanism One hypothesis for how the Golgi apparatus achieves and maintains its polarized structure and how molecules move from one cisterna to another. This model views the cisternae as dynamic structures that mature from early to late by acquiring and then losing

specific Golgi-resident proteins as they move through the Golgi stack with cargo. (Chapter 13)

citric acid cycle [tricarboxylic acid (TCA) cycle; Krebs cycle] Central metabolic pathway found in aerobic organisms. Oxidizes acetyl groups derived from food molecules, generating the activated carriers NADH and FADH₂, some GTP, and waste CO₂. In eukaryotic cells, it occurs in the mitochondria. (Chapter 2) (Panel 2–9, pp. 110–111)

clamp loader Protein complex that utilizes ATP hydrolysis to load the sliding clamp onto a primer–template junction in the process of DNA replication. (Chapter 5)

class I MHC protein One of two classes of major histocompatibility complex (MHC) protein. Found on the surface of almost all vertebrate cell types, where it can present foreign peptides derived from a pathogen such as a virus to cytotoxic T cells. (Chapter 24) (Figures 24–35 and 24–36A)

class II MHC protein One of two classes of major histocompatibility complex (MHC) protein. Found on the surface of various antigen-presenting cells, where it presents peptides to helper and regulatory T cells. (Chapter 24) (Figures 24–35 and 24–36B)

class switch recombination An irreversible change at the DNA level when a B cell switches from making IgM and IgD to making one of the secondary classes of immunoglobulin. (Chapter 24)

class switching Change from making one class of immunoglobulin (for example, IgM) to making another class (for example, IgG) that many B cells undergo during the course of an adaptive immune response. Involves DNA rearrangements called class switch recombination. (Chapter 24) (Figure 24–30)

classical cadherins Family of cadherin proteins, including E-cadherin, N-cadherin, and P-cadherin, that are closely related in sequence throughout their extracellular and intracellular domains. (Chapter 19)

clathrin Protein that assembles into a polyhedral cage on the cytosolic side of a membrane so as to form a clathrin-coated pit, which buds off by endocytosis to form an intracellular clathrin-coated vesicle. (Chapter 13) (Figure 13–6)

clathrin-coated pits Specialized regions typically occupying about 2% of the total plasma membrane area at which the endocytic pathway often begins. (Chapter 13)

clathrin-coated vesicles Coated vesicles inside the cell that transport material from the plasma membrane and between endosomal and Golgi compartments. (Chapter 13)

cleave, cleavage (1) Physical splitting of a cell into two. (2) Specialized type of cell division seen in many early embryos whereby a large cell becomes subdivided into many smaller cells without growth. (Chapter 21)

clonal selection The process whereby, from a population of T and B lymphocytes with a vast repertoire of randomly generated antigen-specific receptors, a given foreign antigen activates (selects) only those lymphocyte clones that display a receptor that fits the antigen. Explains how the adaptive immune system can respond to millions of different antigens in a highly specific way. (Chapter 24) (Figure 24–15)

co-receptor In immunology: an accessory receptor on B cells or T cells that does not bind antigen but binds to a co-stimulatory signal and helps activate the lymphocyte, by helping to activate an intracellular signaling pathway. (Chapter 24)

co-stimulatory signal In immunology: a secreted or membrane-bound signal protein that helps activate an antigen-responding B cell or T cell. (Chapter 24)

co-translational Occurring as translation proceeds. Examples include the import of a protein into the endoplasmic reticulum before the polypeptide chain is completely synthesized (co-translational translocation; Figure 12–24), and the folding of a nascent protein into its secondary and tertiary structure as it emerges from a ribosome (Figure 6–83C). (Chapter 12)

coat-recruitment GTPases Members of a family of monomeric GTPases that have important roles in vesicle transport, being responsible for coat assembly at the membrane. (Chapter 13)

coated vesicle Small membrane-enclosed organelle with a cage of proteins (the coat) on its cytosolic surface. Formed by the pinching off of a coated region of membrane (coated pit). Some coats are made of clathrin, others are made from other proteins. (Chapter 13)

codon Sequence of three nucleotides in a DNA or mRNA molecule that represents the instruction for incorporation of a specific amino acid into a growing polypeptide chain. (Chapter 6)

coenzyme Small molecule tightly associated with an enzyme that participates in the reaction that the enzyme catalyzes, often by forming a covalent bond to the substrate. Examples include biotin, NAD⁺, and coenzyme A. (Chapter 3)

cohesin, cohesin complex Complex of proteins that uses ATP hydrolysis energy to organize an interphase chromosome into a series of looped domains; during mitosis, cohesins also hold sister chromatids together along their length before their separation. (Chapters 4, 17) (Figures 4–57 and 17–23)

coiled-coil Especially stable rodlike protein structure formed by two or more α helices coiled around each other. (Chapter 3) (Figure 3–8)

collagen Fibrous protein rich in glycine and proline that is a major component of the extracellular matrix in animals, conferring tensile strength. Exists in many forms: type I, the most common, is found in skin, tendon, and bone; type II is found in cartilage; type IV is present in basal laminae. (Chapter 19) (Figures 3–24 and 19–38)

collagen fibril A higher-order collagen polymer of fibrillar collagens that assemble into thin structures (10–300 nm in diameter) many hundreds of micrometers long in mature tissues. (Chapter 19)

colony-stimulating factor (CSF) General name for numerous signal molecules that control differentiation of blood cells. (Chapter 15)

colorectal cancer Cancer arising from the epithelium lining the colon (the large intestine) and rectum (the terminal segment of the gut). (Chapter 20)

column chromatography Technique for separation of a mixture of substances in solution by passage through a column containing a porous solid matrix. Substances are retarded to different extents by their interaction with the matrix and can be collected separately from the column. Depending on the matrix, separation can be according to charge, hydrophobicity, size, or the ability to bind to other molecules. (Chapter 8)

commensalism Ecologic relationship between microbes and their host in which the microbe benefits but offers no benefit and causes no harm. (Chapter 23)

committed progenitor Cell derived from a stem cell that divides for a limited number of times before terminally differentiating; also known as a transit amplifying cell.

complement system System of blood proteins that can be activated by antibody–antigen complexes or pathogens to help eliminate the pathogens by directly causing their lysis, by promoting their phagocytosis, or by activating an inflammatory response. (Chapter 24) (Figure 24–7)

complementary (1) Of nucleic acid sequences: capable of forming a perfect base-paired duplex with each other (Figure 4–5). (2) Of other interacting molecules, such as an enzyme and its substrate: having biochemical or structural features that marry up, so that noncovalent bonding is facilitated. (Chapter 4) (Figure 2–4)

complementation test Test to determine whether two mutations that produce similar phenotypes are in the same or different genes. (Chapter 8) (Panel 8–1, pp. 520–521)

complex oligosaccharides Broad class of *N*-linked oligosaccharides, attached to mammalian glycoproteins in the endoplasmic reticulum and modified in the Golgi apparatus, containing *N*-acetylglucosamine, galactose, sialic acid, and fucose residues. (Chapter 13)

condensin, condensin complex Complex of proteins involved in chromosome condensation prior to mitosis. Target for M-Cdk. (Chapters 4, 17) (Figure 17–25)

conditional mutation Mutation that changes a protein or RNA molecule so that its function is altered only under some conditions, such as at an unusually high or unusually low temperature. (Chapter 8)

cone photoreceptor (cone) Photoreceptor cell in the vertebrate retina that is responsible for color vision in bright light. (Chapter 15)

confocal microscope Type of light microscope that produces a clear image of a given plane within a solid object. It uses a laser beam as a pinpoint source of illumination and scans across the plane to produce a two-dimensional “optical section.” (Chapter 9) (Figure 9–24)

conformation The folded, three-dimensional structure of a polypeptide chain. (Chapter 3)

connective tissue Any supporting tissue that lies between other tissues and consists of cells embedded in a relatively large amount of extracellular matrix. Includes bone, cartilage, and loose connective tissue. (Chapter 19)

connexin Protein component of gap junctions, a four-pass transmembrane protein. Six connexins assemble in the plasma membrane to form a connexon, or “hemichannel.” (Chapter 19) (Figure 19–25)

connexon Water-filled pore in the plasma membrane formed by a ring of six connexin protein subunits. Half of a gap junction: connexons from two adjoining cells join to form a continuous channel through which ions and small molecules can pass. (Chapter 19) (Figure 19–25)

consensus nucleotide sequence A summary or “average” of a large number of individual nucleotide sequences derived by comparing many sequences with the same basic function and tallying up the most common nucleotides found at each position. (Chapter 6) (Figure 6–12)

conservative site-specific recombination A type of DNA recombination that takes place between short, specific sequences of DNA and occurs without the gain or loss of

nucleotides. Unlike homologous recombination, it does not require an extensive region of homology between the two recombining DNA molecules. (Chapter 5)

constant region In immunology: region of an immunoglobulin or T cell receptor chain that has a constant amino acid sequence. (Chapter 24)

constitutive secretory pathway Pathway present in all cells by which molecules such as plasma membrane proteins are continually delivered to the plasma membrane from the Golgi apparatus in vesicles that fuse with the plasma membrane. The default route to the plasma membrane if no other sorting signals are present. (Chapter 13) (Figure 13–38)

contact-dependent secretion system Specialized bacterial systems that secrete effector proteins directly into host cell targets. (Chapter 23) (Figure 23–7)

contact-dependent signaling Form of intercellular signaling in which signal molecules remain bound to the surface of the signaling cell and influence only cells that contact it. (Chapter 15)

contractile ring Ring containing actin and myosin that forms under the surface of animal cells undergoing cell division. It contracts to pinch the two daughter cells apart. (Chapter 17) (Figure 17–43)

convergent extension Rearrangement of cells within a tissue that causes it to extend in one dimension and shrink in another. (Chapter 21) (Figure 21–48)

COPI-coated vesicles Coated vesicles that transport material in the secretory pathway, budding from Golgi compartments. (Chapter 13)

COPII-coated vesicles Coated vesicles that transport material early in the secretory pathway, budding from the endoplasmic reticulum. (Chapter 13)

copy number variations (CNVs) A difference between two individuals in the same population in the number of copies of a particular block of DNA sequence. This variation arises from occasional duplications and deletions of these sequences.

cortex The cytoskeletal network in the cortical region of the cytosol just beneath the plasma membrane. (Chapter 10)

coupled reaction Linked pair of chemical reactions in which the free energy released by one reaction serves to drive the other. (Chapter 2) (Figure 2–29)

covalent bond Stable chemical link between two atoms produced by sharing one or more pairs of electrons. (Chapter 2) (Panel 2–1, pp. 94–95)

CRE-binding (CREB) protein Transcription regulator that recognizes the cyclic AMP response element (CRE) in the regulatory region of genes activated by cAMP. On activation by PKA, phosphorylated CREB recruits a transcriptional coactivator (CREB-binding protein; CBP) to stimulate transcription of target genes. (Chapter 15) (Figure 15–28)

CRISPR A defense mechanism in bacteria using small noncoding RNA molecules (crRNAs) to seek out and destroy invading viral genomes through complementary base-pairing and targeted nuclease digestion. (Chapter 7)

crista (plural **cristae**) A specialized invagination of the inner mitochondrial membrane. (Chapter 14)

cross-linking glycan One of a heterogeneous group of branched polysaccharides that help to cross-link cellulose

microfibrils into a complex network. Has a long linear backbone of one sugar type (glucose, xylose, or mannose) with short side chains of other sugars. (Chapter 19)

cross-presentation A process in which extracellular proteins taken up by specialized dendritic cells can give rise to peptides that can be presented by their class I MHC proteins to cytotoxic T cells. (Chapter 24)

crRNAs Small noncoding RNAs (~30 nucleotides) that are the effectors of CRISPR-mediated immunity in bacteria. (Chapter 7)

cryo-electron microscopy Technique for examining a thin film of an aqueous suspension of biological material that has been frozen rapidly enough to create vitreous ice. The specimen is then kept frozen and transferred to the electron microscope. Image contrast is low, but this type of microscopy permits the determination of atomic-level structures because the image is generated solely by the macromolecular structures present. (Chapters 9, 12)

cryptochrome Plant flavoprotein sensitive to blue light. Structurally related to blue-light-sensitive enzymes called photolyases (involved in the repair of ultraviolet-induced DNA damage). Also found in animals, where they have an important role in circadian clocks. (Chapter 15)

cyclic AMP (cAMP) Nucleotide that is generated from ATP by adenylyl cyclase in response to various extracellular signals. It acts as a small intracellular signaling molecule, mainly by activating cAMP-dependent protein kinase (PKA). It is hydrolyzed to AMP by a phosphodiesterase. (Chapter 15) (Figure 15–25)

cyclic-AMP-dependent protein kinase (protein kinase A; PKA) Enzyme that phosphorylates target proteins in response to a rise in intracellular cyclic AMP. (Chapter 15) (Figure 15–27)

cyclic AMP phosphodiesterase Specific enzyme that rapidly and continually destroys cyclic AMP, forming 5'-AMP. (Chapter 15) (Figure 15–26)

cyclic GMP (cGMP) Nucleotide that is generated from GTP by guanylyl cyclase in response to various extracellular signals. (Chapter 15)

cyclic GMP phosphodiesterase Specific enzyme that rapidly hydrolyzes and degrades cyclic GMP. (Chapter 15)

cyclin Protein that periodically rises and falls in concentration in step with the eukaryotic cell cycle. Cyclins activate crucial protein kinases (called cyclin-dependent protein kinases, or Cdks) and thereby help control progression from one stage of the cell cycle to the next. (Chapter 17)

cyclin-Cdk complex Protein complex formed periodically during the eukaryotic cell cycle as the level of a particular cyclin increases. A cyclin-dependent kinase (Cdk) thereby becomes activated. (Chapter 17) (Figures 17–9 and 17–10, and Table 17–1, p. 1034)

cyclin-dependent kinase (Cdk) Protein kinase that has to be complexed with a cyclin protein in order to act. Different cyclin-Cdk complexes trigger different steps in the cell-division cycle by phosphorylating specific target proteins. (Chapter 17) (Figure 17–9)

cyclosome See **anaphase-promoting complex**. (Chapter 17)

cytochrome Family of colored heme-containing proteins that transfer electrons during respiration and photosynthesis. (Chapter 14)

cytochrome c Soluble component of the mitochondrial electron-transport chain. As a second type of function, its release into the cytosol from the mitochondrial intermembrane space initiates apoptosis. (Chapters 14, 18)

cytochrome c oxidase complex Third of the three electron-driven proton pumps in the respiratory chain. It accepts electrons from cytochrome c and generates water using molecular oxygen as an electron acceptor. (Chapter 14) (Figure 14–18)

cytochrome c reductase Second of the three electron-driven proton pumps in the respiratory chain. Accepts electrons from ubiquinone and passes them to cytochrome c. (Chapter 14) (Figure 14–18)

cytokine Extracellular signal protein or peptide that acts as a local mediator in cell–cell communication. (Chapter 24)

cytokine receptor Cell-surface receptor that binds a specific cytokine or hormone and acts through the JAK–STAT signaling pathway. (Chapter 15) (Figure 15–57)

cytokinesis Division of the cytoplasm of a plant or animal cell into two, as distinct from the associated division of its nucleus (which is mitosis). Part of M phase. (Chapter 17) (Panel 17–1, pp. 1048–1049)

cytoplasm Contents of a cell that are contained within its plasma membrane but, in the case of eukaryotic cells, outside the nucleus. (Chapter 12)

cytoplasmic tyrosine kinase Enzyme activated by certain cell-surface receptors (tyrosine-kinase-associated receptors) that transmits the receptor signal onward by phosphorylating target cytoplasmic proteins on tyrosine side chains. (Chapter 15)

cytoskeleton System of protein filaments in the cytoplasm of a eukaryotic cell that gives the cell shape and the capacity for directed movement. Its most abundant components are actin filaments, microtubules, and intermediate filaments. (Chapter 16)

cytosol Contents of the main compartment of the cytoplasm, defined as excluding membrane-bounded organelles such as endoplasmic reticulum and mitochondria. (Chapter 12)

cytotoxic T cell (T_C cell) Type of T cell responsible for killing host cells infected with a virus or another type of intracellular pathogen. (Chapter 24) (Figure 24–42)

D gene segment A short DNA sequence that encodes a part of the variable region of an immunoglobulin heavy chain or the β chain of a T cell receptor (TCR). (Chapter 24)

dark-field microscopy Type of light microscopy in which oblique rays of light focused on the specimen do not enter the objective lens, but light that is scattered by components in the living cell can be collected to produce a bright image on a dark background. (Chapter 9) (Figure 9–6)

death-inducing signaling complex (DISC) Complex in which initiator caspases interact and are activated after extracellular ligands bind to cell-surface death receptors in the extrinsic pathway of apoptosis. (Chapter 18)

death receptor Transmembrane receptor protein that can signal the cell to undergo apoptosis when it binds its extracellular ligand. (Chapter 18) (Figure 18–6)

default pathway The transport pathway of proteins directly to the cell surface via the nonselective constitutive secretory pathway, entry into which does not require a particular signal. (Chapter 13)

defensin Positively charged, amphipathic, antimicrobial peptide—secreted by epithelial cells—that binds to and disrupts the membranes of many pathogens. (Chapter 24)

delayed K⁺ channel Neuronal voltage-gated K⁺ channel that opens after membrane depolarization during the falling phase of an action potential, being delayed because of its slower activation kinetics than that of Na⁺ channels. Its opening causes a K⁺ efflux that drives the membrane potential back toward its original negative value, enabling it to transmit a second impulse. (Chapter 11)

Delta Single-pass transmembrane signal protein displayed on the surface of cells that binds to the Notch receptor protein on a neighboring cell, activating a contact-dependent signaling mechanism. (Chapter 15)

dendrite Extension of a nerve cell, often elaborately branched, that receives stimuli from other nerve cells. (Chapter 11)

dendritic cell The most potent type of antigen-presenting cell, which takes up antigen and processes it for presentation to T cells. It is required for activating naïve T cells. (Chapter 24) (Figure 24–11)

deoxyribonucleic acid (DNA) Polynucleotide formed from covalently linked deoxyribonucleotide units. The store of hereditary information within a cell and the carrier of this information from generation to generation. (Figure 4–3 and Panel 2–6, pp. 104–105)

depolarization Deviation in the electrical potential across the plasma membrane toward a positive value. A depolarized cell has a potential that is positive outside and negative inside. (Chapter 11)

desensitization See **adaptation**. (Chapter 15)

desmosome Anchoring cell–cell junction, usually formed between two epithelial cells. Characterized by dense plaques of protein into which intermediate filaments in the two adjoining cells insert. (Chapter 19) (Figure 19–2)

detergent Small amphiphilic molecule, more soluble in water than in lipids, that disrupts hydrophobic associations and destroys the lipid bilayer thereby solubilizing membrane proteins. (Chapter 10)

diacylglycerol (DAG) Lipid produced by the cleavage of inositol phospholipids in response to extracellular signals. Composed of two fatty acid chains linked to glycerol, it serves as a small signaling molecule to help activate protein kinase C (PKC). (Chapter 15) (Figure 15–29)

dideoxy sequencing The standard enzymatic method of DNA sequencing. (Chapter 8)

differential-interference-contrast microscope Type of light microscope that exploits the interference effects that occur when light passes through parts of a cell of different refractive indices. Used to view unstained living cells. (Chapter 9)

differentiation Process by which a cell undergoes a change to an overtly specialized cell type. (Chapter 21)

diffusion The net drift of molecules through space due to random thermal movements. (Chapter 2)

Dishevelled Scaffold protein recruited to the Frizzled family of cell-surface receptors upon their activation by Wnt binding that helps relay the signal to other signaling molecules. (Chapter 15)

dissociation constant The reciprocal of the association constant, with units of moles/liter. (Chapter 3)

DNA cloning (1) The act of making many identical copies (typically billions) of a DNA molecule—the amplification of a particular DNA sequence. (2) Also, the isolation of a particular stretch of DNA (often a particular gene) from the rest of the cell's genome. (Chapter 8)

DNA helicase Enzyme that harnesses ATP hydrolysis energy to open a region of the DNA helix into its single strands as an aid to DNA replication or DNA repair. (Chapter 5)

DNA library Collection of cloned DNA molecules, representing either an entire genome (genomic library) or complementary DNA copies of the mRNA produced by a cell (cDNA library). (Chapter 8)

DNA ligase Enzyme that joins the ends of two strands of DNA together with a covalent bond to produce one continuous DNA strand. (Chapter 5)

DNA methylation Addition of methyl groups to DNA. Extensive methylation of the cytosine base in CG sequences is used in plants and animals to help keep genes in an inactive state. (Chapter 7)

DNA-only transposon Transposable element that exists as DNA throughout its life cycle. Many of these elements move by cut-and-paste transposition. See also **transposon**. (Chapter 5)

DNA polymerase Enzyme that synthesizes DNA by joining nucleotides together using a DNA template as a guide; its substrates are the four nucleoside triphosphates: A, G, C, and T. (Chapter 5)

DNA primase Enzyme that synthesizes a short strand of RNA on a DNA template, producing an RNA primer for DNA synthesis. (Chapter 5) (Figure 5–10)

DNA repair A set of different enzymatic processes for repairing the many accidental lesions that occur continually in DNA. (Chapter 5)

DNA replication Process by which a copy of a DNA molecule is made. (Chapter 1)

DNA supercoiling A conformation with loops or coils that DNA adopts in response to superhelical tension; conversely, creating various loops or coils in the helix can create such tension. (Chapter 6)

DNA topoisomerase (topoisomerase) Enzyme that binds to DNA and reversibly breaks a phosphodiester bond in one or both strands. Topoisomerase I creates transient single-strand breaks, allowing the double helix to swivel and relieving superhelical tension. Topoisomerase II creates transient double-strand breaks, allowing one double helix to pass through another and thus resolving knots and tangles. (Chapter 5) (Figures 5–22 and 5–23)

DNA transcription See **transcription**. (Chapter 6)

DNA tumor virus General term for a variety of different DNA viruses that can cause tumors. (Chapter 20)

dolichol Isoprenoid lipid molecule that anchors the precursor oligosaccharide in the endoplasmic reticulum membrane during protein glycosylation. (Chapter 12)

domain (protein domain) Portion of a protein that has a tertiary structure of its own. Larger proteins are generally composed of several domains, each connected to the next by short flexible regions of polypeptide chain. Homologous domains are recognized in many different proteins. (Chapter 3)

Dorsal Transcription regulator of the NF- κ B family regulating gene expression and involved in establishing the dorsoventral axis in an embryo. (Chapter 21)

double helix The three-dimensional structure of DNA, in which two antiparallel DNA chains, held together by hydrogen-bonding between the bases, are wound into a helix. (Chapter 4) (Figure 4–5)

drivers Mutations that are causal factors in the development of cancer. (Chapter 20)

dynamic instability Sudden conversion from growth to shrinkage, and vice versa, in a protein filament such as a microtubule or actin filament. (Chapter 16) (Panel 16–2, pp. 960–961, and Figure 16–40)

dynamain Cytosolic GTPase that binds to the neck of a clathrin-coated vesicle in the process of budding from the membrane and which is involved in completing vesicle formation. (Chapter 13)

dynein Large motor protein that undergoes ATP-dependent movement along microtubules. (Chapter 16)

E2F protein Transcription regulatory protein that switches on many genes that encode proteins required for entry into the S phase of the cell cycle. (Chapter 17)

early endosome Common receiving compartment with which most endocytic vesicles fuse and where internalized cargo is sorted either for return to the plasma membrane or for degradation by inclusion in a late endosome. (Chapter 13)

ectoderm Embryonic epithelial tissue that is the precursor of the epidermis and nervous system. (Chapter 21)

edema factor One of the two A subunits of anthrax toxin; an adenyl cyclase that catalyzes production of cAMP, leading to ion imbalance and consequent edema in the skin or lung. (Chapter 23)

effector cell Cell that carries out the final response or function in a particular process. The main effector cells of the immune system, for example, are activated lymphocytes and phagocytes that help eliminate pathogens. (Chapter 24)

egg-polarity genes Genes in the *Drosophila* egg that define the anteroposterior and dorsoventral axes of the future embryo through the creation of landmarks (mRNA or protein) in the egg that provide signals organizing the developmental process. (Chapter 21)

elastic fiber Extensible fiber formed by the protein elastin in many animal connective tissues, such as in skin, blood vessels, and lungs, which gives them their stretchability and resilience. (Chapter 19)

elastin Extracellular protein that forms extensible fibers (elastic fibers) in connective tissues. (Chapters 3, 19)

electrochemical gradient Combined influence of a difference in the concentration of an ion on two sides of a membrane and the electrical charge difference across the membrane (membrane potential). Ions or charged molecules can move passively only down their own electrochemical gradient. (Chapter 11)

electrochemical proton gradient The driving force for the production of ATP by ATP synthase. See also **electrochemical gradient**. (Chapter 14)

electron microscope Microscope that uses a beam of electrons to create the image. (Chapter 9)

electron microscope (EM) tomography Technique for viewing three-dimensional specimens in the electron microscope in which multiple views are taken from different directions by tilting the specimen holder. The views are

combined computationally to give a three-dimensional image. (Chapter 9)

electron-transport chain Series of reactions in which electron carrier molecules pass electrons “down the chain” from higher to successively lower energy levels. The energy released during such electron movement can be used to power various processes. Electron-transport chains present in the inner mitochondrial membrane (called the respiratory chain) and in the thylakoid membrane of chloroplasts generate an electrochemical proton gradient across the membrane that is used to drive ATP synthesis. See especially Figures 14–18 and 14–52. (Chapters 2, 14)

electrostatic attraction A noncovalent, ionic attraction between two molecules carrying groups of opposite charge. (Chapter 2) (Panel 2–3, pp. 98–99)

embryonic stem cells (ES cells) Cells derived from the inner cell mass of the early mammalian embryo. Capable of giving rise to all the cells in the body. Can be grown in culture, genetically modified, and inserted into a blastocyst to develop a transgenic animal. (Chapter 22)

endocrine cell Specialized animal cell that secretes a hormone into the blood. Usually part of a gland, such as the thyroid or pituitary gland. (Chapter 15)

endocycle Variation of the cell cycle in which multiple rounds of DNA replication occur without intervening M phases. (Chapter 17)

endocytic vesicle Vesicle formed as material ingested by the cell during endocytosis is enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form the vesicle. (Chapter 13)

endocytosis Uptake of material into a cell by an invagination of the plasma membrane and its internalization in a membrane-enclosed vesicle. See also **pinocytosis** and **phagocytosis**. (Chapter 13)

endoderm Embryonic tissue that is the precursor of the gut and associated organs. (Chapter 21)

endoplasmic reticulum (ER) Extensive, net-like membrane-bounded compartment in the cytoplasm of eukaryotic cells, where lipids are synthesized and membrane-bound proteins and secretory proteins are made. (Chapter 12) (Figure 12–15)

endosome maturation Process by which early endosomes mature to late endosomes and endolysosomes; in the conversion process, the endosome membrane protein composition changes, the endosome moves away from the cell periphery and close to the nucleus, and the endosome ceases to recycle material to the plasma membrane and irreversibly commits its remaining contents to degradation. (Chapter 13)

endothelial cell Flattened cell type that forms a sheet (the endothelium) lining all blood and lymphatic vessels. (Chapter 21)

engulfment Process by which a portion of the cytoplasm becomes enclosed by a double membrane, such as during autophagy. (Chapter 12)

entropy (S) Thermodynamic quantity that measures the degree of disorder or randomness in a system; the higher the entropy, the greater the disorder. (Chapter 2) (Panel 2–7, pp. 106–107)

enveloped virus Virus with a capsid surrounded by a lipid bilayer membrane (the envelope), which is often derived from the host-cell plasma membrane when the virus buds from the cell. (Chapter 23) (Figure 23–12)

enzyme Protein that catalyzes a specific chemical reaction. (Chapters 2, 3)

enzyme-coupled receptor A major type of cell-surface receptor that has a cytoplasmic domain that either has enzymatic activity or is associated with an intracellular enzyme. In either case, the enzymatic activity is stimulated by an extracellular ligand binding to the receptor. (Chapter 15) (Figure 15–6)

ephrin One of a family of membrane-bound protein ligands for the Eph receptor tyrosine kinases (RTKs) that, among many other functions, stimulate repulsion or attraction responses that guide the migration of cells and nerve cell axons during animal development. (Chapter 15)

epidermis Epithelial layer covering the outer surface of the body. Has different structures in different animal groups. The outer layer of plant tissue is also called the epidermis. (Chapter 22)

epigenetic inheritance Inheritance of phenotypic changes in a cell or organism that do not result from changes in the nucleotide sequence of DNA. Can be due to positive feedback loops of transcription regulators or to heritable modifications in chromatin such as DNA methylation or histone modifications. (Chapters 4, 7) (Figure 7–56)

epistasis analysis Analysis to discover the order in which genes act, by investigating if a mutation in one gene can mask the effect of a mutation in another gene when both mutations are present in the same organism or cell. (Chapter 8)

epithelial tissue A tissue, such as the lining of the gut or the epidermal covering of the skin, in which cells are closely bound together into sheets called epithelia. (Chapter 19)

epithelium (plural **epithelia**) Sheet of cells covering the outer surface of a structure or lining a cavity. (Chapter 19)

equilibrium State in a chemical reaction where there is no net change in free energy to drive the reaction in either direction. The ratio of product to substrate reaches a constant value at chemical equilibrium. (Chapters 2, 3) (Figure 2–30)

equilibrium constant (*K*) The ratio of forward and reverse rate constants for a reaction. Equal to the association or affinity constant (K_a) for a simple binding reaction ($A + B \rightarrow AB$). See *also* **affinity constant**, **association constant**, **dissociation constant**. (Chapters 2, 3) (See p. 146.)

ER lumen The space enclosed by the membrane of the endoplasmic reticulum (ER). (Chapter 12)

ER resident protein Protein that remains in the lumen of the endoplasmic reticulum (ER) or its membranes and carries out its function there, as opposed to the many proteins that are present in the ER only in transit. (Chapter 12)

ER retention signal Short amino acid sequence on a protein that prevents it from moving out of the endoplasmic reticulum (ER). Found on those proteins that are resident in the ER and function there. (Chapter 12)

ER signal sequence N-terminal signal sequence that directs proteins to enter the endoplasmic reticulum (ER). Cleaved off by signal peptidase after entry. (Chapter 12)

erythrocyte Small hemoglobin-containing blood cell of vertebrates that transports oxygen to, and carbon dioxide from, tissues. Also called a red blood cell. (Chapter 22)

erythropoietin A hormone produced by the kidney that stimulates the production of red blood cells in bone marrow. (Chapter 15)

ESCRT protein complexes Four protein complexes (ESCRT-0, ESCRT-I, ESCRT-II, and ESCRT-III) that act sequentially to shepherd mono-ubiquitylated membrane proteins on endosomal membranes into intraluminal vesicles. The ESCRT-III complex catalyzes the pinching-off reaction. (Chapter 13)

ethylene Small gas molecule that is a plant growth regulator influencing plant development in various ways including promoting fruit ripening, leaf abscission, and plant senescence and functioning as a stress signal in response to wounding, infection, and flooding. (Chapter 15)

euchromatin Region of an interphase chromosome that stains diffusely; “normal” chromatin, as opposed to the more condensed heterochromatin. (Chapter 4)

eukaryote Organism composed of one or more cells that have a distinct nucleus. Member of one of the three main divisions of the living world, the other two being bacteria and archaea. (Chapter 1) (Figures 1–9 and 1–21)

eukaryotic initiation factor (eIF) Protein that helps load initiator tRNA on to the ribosome, thus initiating translation. (Chapter 6)

excitatory neurotransmitter Neurotransmitter that opens cation channels in the postsynaptic membrane, causing an influx of Na^+ , and in many cases Ca^{2+} , that depolarizes the postsynaptic membrane toward the threshold potential for firing an action potential. (Chapter 11)

executioner caspases Apoptotic caspases that catalyze the widespread cleavage events during apoptosis that kill the cell. (Chapter 18)

exocytosis Excretion of material from the cell by vesicle fusion with the plasma membrane; can occur constitutively or be regulated. (Chapter 13)

exon Segment of a eukaryotic gene that consists of a sequence of nucleotides that will be represented in mRNA or in a final transfer, ribosomal, or other mature RNA molecule. In protein-coding genes, exons encode the amino acids in the protein. An exon is usually adjacent to a noncoding DNA segment called an intron. (Chapters 4, 6) (Figure 4–15)

expansion microscopy (ExM) A microscopy technique in which superresolution is achieved by physically enlarging the specimen. Biological material is labeled with fluorescent probes before being embedded in a polymer gel. The sample is then gently swollen before examination in a light microscope. (Chapter 9)

extracellular pathogens Pathogens that disturb host cells and can cause serious disease without replicating in host cells. (Chapter 23)

extracellular signal molecule Any secreted or cell-surface chemical signal that binds to receptors and regulates the activity of the cell expressing the receptor. (Chapter 15)

extrinsic pathway Pathway of apoptosis triggered by extracellular signal proteins binding to cell-surface death receptors. (Chapter 18)

facultative pathogens Bacteria that replicate in an environmental reservoir such as water or soil and only cause disease if they happen to encounter a susceptible host. (Chapter 23)

FAD/FADH₂ (flavin adenine dinucleotide/reduced flavin adenine dinucleotide) Electron carrier pair that

functions in the citric acid cycle and fatty acid oxidation. One molecule of FAD gains two electrons plus two protons in becoming the activated carrier FADH₂. (Chapter 2) (Figure 2–39)

Fas (Fas protein, Fas death receptor) Transmembrane death receptor that initiates apoptosis when it binds its extracellular ligand (Fas ligand). (Chapter 18) (Figure 18–6)

Fas ligand Ligand that activates the cell-surface death receptor, Fas, triggering the extrinsic pathway of apoptosis. (Chapter 18)

fat Energy-storage lipid in cells. Composed of triglycerides—fatty acids esterified with glycerol. (Chapter 2)

fate map Representation showing which cell types will later derive from which regions of a developing tissue; for example, from the blastula. (Chapter 21) (Figure 21–27)

Fc receptor One of a family of cell-surface receptors that bind the tail region (Fc region) of an antibody molecule. Different Fc receptors are specific for different classes of antibodies, such as IgG, IgA, or IgE. (Chapter 24)

feedback inhibition The process in which a product of a reaction feeds back to inhibit a previous reaction in the same pathway. (Chapter 3) (Figures 3–52 and 3–53)

fermentation Anaerobic energy-yielding metabolic pathway involving the oxidation of organic molecules. Anaerobic glycolysis refers to the process whereby pyruvate is converted into lactate or ethanol, with the conversion of NADH to NAD⁺. (Chapter 2) (Figure 2–50)

fibril-associated collagen A collagen (including types IX and XII) that has a flexible triple-stranded helical structure and binds to the surface of collagen fibrils. Mediates the interactions of these collagen fibrils with one another and with other matrix macromolecules to help determine their organization in the matrix. (Chapter 19)

fibrillar collagen Class of fibril-forming collagens (including type I collagen, the most common type and the principal collagen of skin and bone) that have long rope-like structures with few or no interruptions and which assemble into collagen fibrils. (Chapter 19)

fibroblast Common cell type found in connective tissue. Secretes an extracellular matrix rich in collagen and other extracellular matrix macromolecules. Migrates and proliferates readily in wounded tissue and in tissue culture. (Chapter 19)

fibronectin Extracellular matrix protein involved in adhesion of cells to the matrix and guidance of migrating cells during embryogenesis. Integrins on the cell surface are receptors for fibronectin. (Chapter 19)

filopodium (plural **filopodia**) (**microspike**) Thin, spike-like protrusion with an actin filament core, generated on the leading edge of a crawling animal cell. (Chapter 16) (Figure 16–17)

flagellum (plural **flagella**) Long, whiplike protrusion whose undulations drive a cell through a fluid medium. Eukaryotic flagella are longer versions of cilia. Bacterial flagella are smaller and completely different in construction and mechanism of action. *Compare cilium.* (Chapter 16)

fluorescence microscope Microscope designed to view material stained with fluorescent dyes or proteins. Similar to a light microscope but the illuminating light is passed through one set of filters before the specimen, to select those wavelengths that excite the dye, and through another set of filters before the

light reaches the eye, to select only those wavelengths emitted when the dye fluoresces. (Chapter 9) (Figure 9–10)

fluorescence recovery after photobleaching (FRAP)

Technique for monitoring the kinetic parameters of a protein by analyzing how fluorescent protein molecules move into an area of the cell bleached by a beam of laser light. (Chapter 9) (Figure 9–20)

fluorescence resonance energy transfer (FRET)

Technique for monitoring the closeness of two fluorescently labeled molecules (and thus their interaction) in cells. Also known as Förster resonance energy transfer. (Chapter 9) (Figure 9–19)

focal adhesion kinase (FAK) Cytoplasmic tyrosine kinase present at cell–matrix junctions (focal adhesions) in association with the cytoplasmic tails of integrins. (Chapter 15)

follicular helper T cell (T_{FH} cell) Type of T cell located in lymphoid follicles that secretes various cytokines to stimulate B cells to undergo antibody class switching and somatic hypermutation. (Chapter 24)

formin Dimeric protein that nucleates the growth of straight, unbranched actin filaments that can be cross-linked by other proteins to form parallel bundles. (Chapter 16)

Förster resonance energy transfer See **fluorescence resonance energy transfer (FRET)**. (Chapter 9)

FRAP See **fluorescence recovery after photobleaching**. (Chapter 9)

free energy (G) (Gibbs free energy) The energy that can be extracted from a system to drive reactions. Takes into account changes in both energy and entropy. (Chapter 2) (Panel 2–7, pp. 106–107)

free-energy change (ΔG) Change in the free energy during a reaction: the free energy of the product molecules minus the free energy of the starting molecules. A large negative value of ΔG indicates that the reaction has a strong tendency to occur. (Chapter 2) (Panel 2–7, pp. 106–107)

free ribosome Ribosome that is free in the cytosol, unattached to any membrane. (Chapter 12)

FRET See **fluorescence resonance energy transfer**. (Chapter 9)

Frizzled Family of cell-surface receptors that are seven-pass transmembrane proteins that resemble GPCRs in structure but do not generally work through the activation of G proteins. Activated by Wnt binding to recruit the scaffold protein Dishevelled, which helps relay the signal to other signaling molecules. (Chapter 15)

fungus (plural **fungi**) Kingdom of eukaryotic organisms that includes the yeasts, molds, and mushrooms. Many plant diseases and a relatively small number of animal diseases are caused by fungi. (Chapter 23)

fusion protein Engineered protein that combines two or more normally separate polypeptides. Produced from a recombinant gene. (Chapter 8)

ΔG Change in the free energy during a reaction: the free energy of the product molecules minus the free energy of the starting molecules. A large negative value of ΔG indicates that the reaction has a strong tendency to occur. (Chapter 2) (Panel 2–7, pp. 106–107)

G₀ State of withdrawal from the eukaryotic cell-division cycle by entry into a quiescent digression from the G₁ phase. A

common, sometimes permanent, state for differentiated cells. (Chapter 17)

G₁ phase Gap 1 phase of the eukaryotic cell-division cycle, between the end of mitosis and the start of DNA synthesis. (Chapter 17) (Figure 17–4)

G₁-Cdk Cyclin–Cdk complex formed in vertebrate cells by a G₁-cyclin and the corresponding cyclin-dependent kinase (Cdk). (Chapter 17) (Table 17–1, p. 1034)

G₁-cyclin Cyclin present in the G₁ phase of the eukaryotic cell cycle. Forms complexes with Cdks that help govern the activity of the G₁/S-cyclins, which control progression to S phase. (Chapter 17)

G₁/S-Cdk Cyclin–Cdk complex formed in vertebrate cells by a G₁/S-cyclin and the corresponding cyclin-dependent kinase (Cdk). (Chapter 17) (Figure 17–10 and Table 17–1, p. 1034)

G₁/S-cyclin Cyclin that activates Cdks in late G₁ of the eukaryotic cell cycle and thereby helps trigger progression through Start, resulting in a commitment to cell-cycle entry. Its level falls at the start of S phase. (Chapter 17) (Figure 17–10)

G₂ phase Gap 2 phase of the eukaryotic cell-division cycle, between the end of DNA synthesis and the beginning of mitosis. (Chapter 17) (Figure 17–4)

G₂/M transition Point in the eukaryotic cell cycle at which the cell checks for completion of DNA replication before triggering the early mitotic events that lead to chromosome alignment on the spindle. (Chapter 17) (Figure 17–8)

ganglioside Any glycolipid having one or more sialic acid residues in its structure. Found in the plasma membrane of eukaryotic cells and especially abundant in nerve cells. (Chapter 10) (Figure 10–16)

gap gene In *Drosophila* development, a gene that is expressed in specific broad regions along the anteroposterior axis of the early embryo, and which helps designate the main divisions of the insect body. (Chapter 21) (Figure 21–18)

gap junction Communicating channel-forming cell–cell junction present in most animal tissues that allows ions and small molecules to pass from the cytoplasm of one cell to the cytoplasm of the next. (Chapter 19)

gastrulation Important stage in animal embryogenesis during which the embryo is transformed from a ball of cells to a structure with a gut (a gastrula). (Chapter 21)

gated transport Movement of proteins between the cytosol and the nucleus through the nuclear pore complexes in the nuclear envelope; these complexes function as selective gates. (Chapter 12)

geminin Protein that prevents the formation of the protein–DNA complexes required to initiate DNA replication forks during S phase and mitosis, thus ensuring that the chromosomes are replicated only once in each cell cycle. (Chapter 17)

gene Region of DNA that is transcribed as a single unit and carries information for a discrete hereditary characteristic, usually corresponding to (1) a single protein (or set of related proteins generated by variant post-transcriptional processing) or (2) a single RNA (or set of closely related RNAs). (Chapters 1, 7)

gene control region The set of linked DNA sequences regulating expression of a particular gene. Includes promoter and *cis*-regulatory sequences required to initiate transcription of the gene and control the rate of transcription. (Chapter 7) (Figure 7–20)

gene conversion Process by which DNA sequence information can be transferred from one DNA helix (which remains unchanged) to another DNA helix whose sequence is altered. It often accompanies general recombination events. (Chapter 5) (Figure 5–57)

gene family The set of genes in an organism related in DNA sequence because of their derivation from the same ancestor. (Chapter 1)

general transcription factor Any of the proteins whose assembly at all promoters of a given type is required for the binding and activation of RNA polymerase and the initiation of transcription. (Chapter 6) (Table 6–3, p. 333)

genetic code The set of rules specifying the correspondence between nucleotide triplets (codons) in DNA or RNA and amino acids in proteins. (Chapter 6) (Figure 6–52)

genetic instability Abnormally increased spontaneous mutation rate, such as occurs in cancer cells. (Chapter 20)

genetic screen Procedure for discovery of genes affecting a specific phenotype by surveying large numbers of mutagenized individuals. (Chapter 8)

genetics The study of the genes of an organism on the basis of heredity and variation. (Chapter 8)

genome The totality of genetic information belonging to a cell or an organism; in particular, the DNA that carries this information. (Chapters 1, 4)

genome annotation Process attempting to mark out all the genes (protein-coding and noncoding) in a genome and ascribing functions to each. (Chapter 8)

genomic imprinting Phenomenon in which a gene is either expressed or not expressed in the offspring depending on which parent it is inherited from. (Chapter 7) (Figure 7–52)

genomic library Collection of cloned DNA molecules representing an entire genome. (Chapter 8)

genotype Genetic constitution of an individual cell or organism. The particular combination of alleles found in a specific individual. (Chapter 8) (Panel 8–1, pp. 520–521)

germ cell A cell in the germ line of an organism, which includes the haploid gametes and their specified diploid precursor cells. Germ cells contribute to the formation of a new generation of organisms and are distinct from somatic cells, which form the body and leave no descendants. (Chapter 5)

germ layer One of the three primary tissue layers (endoderm, mesoderm, and ectoderm) of an animal embryo. (Chapter 21) (Figure 21–3)

glial cell Supporting non-neural cell of the nervous system. Includes oligodendrocytes and astrocytes in the vertebrate central nervous system and Schwann cells in the peripheral nervous system. (Chapter 11)

glycerophospholipid Phospholipid derived from glycerol, abundant in biomembranes. (Chapter 10) (Figures 10–2 and 10–3)

glycogen Polysaccharide composed exclusively of glucose units. Used to store energy in animal cells. Large granules of glycogen are especially abundant in liver and muscle cells. (Chapter 2) (Figure 2–52 and Panel 2–4, pp. 100–101)

glycolipid Lipid molecule with a sugar residue or oligosaccharide attached. (Chapter 10) (Panel 2–5, pp. 102–103)

glycolysis Ubiquitous metabolic pathway in the cytosol in which sugars are incompletely degraded with production of ATP. Literally, “sugar splitting.” (Chapter 2) (Figure 2–46 and Panel 2–8, pp. 108–109)

glycoprotein Any protein with one or more sugars or oligosaccharide chains covalently linked to amino acid side chains. Most secreted proteins and most proteins exposed on the outer surface of the plasma membrane are glycoproteins. (Chapter 12)

glycosaminoglycan (GAG) Long, linear, highly charged polysaccharide composed of a repeating pair of sugars, one of which is always an amino sugar. Mainly found covalently linked to a protein core in extracellular matrix proteoglycans. Examples include chondroitin sulfate, hyaluronan, and heparin. (Chapter 19) (Figure 19–32)

glycosylphosphatidylinositol anchor (GPI anchor) Lipid linkage by which some membrane proteins are bound to the membrane. The protein is joined, via an oligosaccharide linker, to a phosphatidylinositol anchor during its travel through the endoplasmic reticulum. (Chapters 10, 12) (Figure 12–30)

Golgi apparatus (Golgi complex) Complex organelle in eukaryotic cells, centered on a stack of flattened, membrane-enclosed spaces, in which proteins and lipids transferred from the endoplasmic reticulum are modified and sorted. It is also the site of synthesis of many cell-wall polysaccharides in plants and extracellular matrix glycosaminoglycans in animal cells. (Chapter 13) (Figure 13–27)

GPCR kinase (GRK) Member of a family of enzymes that phosphorylates multiple serines and threonines on a GPCR to produce receptor desensitization. (Chapter 15) (Figure 15–43)

G protein (heterotrimeric GTP-binding protein) A heterotrimeric GTP-binding protein with intrinsic GTPase activity that couples GPCRs to enzymes or ion channels in the plasma membrane. (Chapter 15) (Table 15–3, p. 907)

G-protein-coupled receptor (GPCR) A seven-pass cell-surface receptor that, when activated by its extracellular ligand, activates a G protein, which in turn activates either an enzyme or ion channel in the plasma membrane. (Chapter 15) (Figures 15–6 and 15–22)

Gram negative Description for bacteria that do not stain with Gram stain as a result of having a thin peptidoglycan cell wall outside their inner (plasma) membrane that is covered by a second lipid-containing outer membrane. (Chapter 23) (Figure 23–3)

Gram positive Description for bacteria that stain positive with Gram stain because of a thick layer of peptidoglycan cell wall outside their inner (plasma) membrane. (Chapter 23)

Gram staining A technique for classifying bacteria that is based on differences in the structure of the bacterial cell wall and outer surface. (Chapter 23)

granulocyte Category of white blood cell distinguished by conspicuous cytoplasmic granules. Includes neutrophils, basophils, and eosinophils. Arises from a granulocyte/macrophage (GM) progenitor cell. (Chapter 22) (Figure 22–11)

granulocyte/macrophage (GM) progenitor cell Committed progenitor cell in the bone marrow that gives rise to neutrophils and macrophages. (Figure 22–12)

green fluorescent protein (GFP) Fluorescent protein isolated from a jellyfish. Widely used as a marker in cell biology. (Chapters 8, 9) (Figure 9–16)

growth cone Migrating motile tip of a growing nerve cell axon or dendrite. (Chapter 16)

growth factor Extracellular signal protein that can stimulate a cell to grow. Growth factors often have other functions as well, including stimulating cells to survive or proliferate. Examples include epidermal growth factor (EGF) and platelet-derived growth factor (PDGF). (Chapter 17)

growth hormone (GH) Mammalian hormone secreted by the pituitary gland into the bloodstream that stimulates growth throughout the body. (Chapter 15)

GTP (guanosine 5'-triphosphate) Nucleoside triphosphate produced by the phosphorylation of GDP (guanosine diphosphate). Like ATP, it releases a large amount of free energy on hydrolysis of its terminal phosphate group. Has a special role in microtubule assembly, protein synthesis, and cell signaling. (Figure 2–59)

GTP-binding protein Also called GTPase; an enzyme that converts GTP to GDP. (Chapters 3, 15)

GTPase An enzyme that converts GTP to GDP. GTPases fall into two large families. Large *G proteins (heterotrimeric G proteins)* are composed of three different subunits and mainly couple GPCRs to enzymes or ion channels in the plasma membrane. Small monomeric *GTP-binding proteins* (also called *monomeric GTPases*) consist of a single subunit and help relay signals from many types of cell-surface receptors and have roles in intracellular signaling pathways, regulating intracellular vesicle trafficking, and signaling to the cytoskeleton. Both heterotrimeric G proteins and monomeric GTPases cycle between an active GTP-bound form and an inactive GDP-bound form and frequently act as molecular switches in intracellular signaling pathways. (Chapter 3) (Figure 3–63)

GTPase-activating protein (GAP) Protein that binds to a GTPase and inhibits it by stimulating its GTPase activity, causing the enzyme to hydrolyze its bound GTP to GDP. (Chapter 15) (Figure 15–8)

guanine nucleotide exchange factor (GEF) Protein that binds to a GTPase and activates it by stimulating it to release its tightly bound GDP, thereby allowing it to bind GTP in its place. (Chapter 15) (Figure 15–8)

haplotype block Combination of alleles and DNA markers that has been inherited in a large, linked block on one chromosome of a homologous pair—undisturbed by genetic recombination—across many generations. (Chapter 8)

Hedgehog protein Secreted extracellular signal molecule that has many different roles controlling cell differentiation and gene expression in animal embryos and adult tissues. Excessive Hedgehog signaling can lead to cancer. (Chapter 15)

helper T cell (T_H cell) Type of T cell that helps activate B cells to make antibodies, cytotoxic T cells to become effector cells, and macrophages to kill ingested pathogens. They can also help activate dendritic cells. (Chapter 24)

hematopoietic stem cells Stem cells located in the bone marrow that give rise to almost all blood-cell types. (Chapter 22)

hepatocyte The main liver cell type that carries out a variety of functions including blood protein synthesis, carbohydrate and lipid metabolism, and detoxification of harmful substances. (Chapter 22)

heterochromatin Chromatin that is highly condensed even in interphase; generally transcriptionally inactive. (*Compare with euchromatin.*) (Chapter 4)

heterochronic Describes genes involved in developmental timing; mutation results in cells of a specific fate behaving as cells at a different stage of development. (Chapter 21)

heterotrimeric GTP-binding protein See **G protein**. (Chapter 15)

high-mannose oligosaccharides Broad class of *N*-linked oligosaccharides, attached to mammalian glycoproteins in the endoplasmic reticulum, containing two *N*-acetylglucosamine residues and many mannose residues. (Chapter 13)

high-performance liquid chromatography (HPLC) Type of chromatography that uses columns packed with tiny beads of matrix; the solution to be separated is pushed through under high pressure. (Chapter 8)

histone One of a group of small abundant proteins, rich in arginine and lysine, that combine to form the nucleosome cores around which DNA is wrapped in eukaryotic chromosomes. (Chapter 4) (Figure 4–24)

histone chaperone (chromatin assembly factor) Protein that binds free histones, releasing them as they are incorporated into newly replicated chromatin. (Chapter 5) (Figure 4–27)

histone H1 “Linker” (as opposed to “core”) histone protein that binds to DNA where it exits from a nucleosome and helps to compact nucleosomes. (Chapter 4) (Figure 4–30)

Holliday junction (cross-strand exchange) X-shaped structure formed in DNA molecules undergoing recombination, in which the two DNA molecules are held together by exchanging one of their two strands; also called a cross-strand exchange. (Chapter 5) (Figure 5–54)

homeotic selector gene In *Drosophila* development, a gene that defines and preserves the differences between body segments. (Chapter 21)

homolog One of two or more genes that are similar in sequence as a result of derivation from the same ancestral gene. The term covers both orthologs and paralogs. (Chapter 1) (Figure 1–20) See **homologous chromosomes**.

homologous chromosomes (homologs) The maternal and paternal copies of a particular chromosome in a diploid cell. (Chapter 4)

homologous recombination (general recombination) Genetic exchange between a pair of identical or very similar DNA sequences, often those located on two copies of the same chromosome. Provides an error-free mechanism for repairing DNA double-strand breaks. (Chapter 5) (Figures 5–47, 5–49, and 5–53)

homophilic Binding between molecules of the same kind, especially those involved in cell–cell adhesion. (Chapter 19) (Figure 19–5)

horizontal gene transfer Gene transfer between bacteria or archaea via natural transformation by released naked DNA, transduction by bacteriophages, or sexual exchange by conjugation. (Chapter 23)

hormone Signal molecule secreted by an endocrine cell into the bloodstream, which can then carry the signal to distant target cells. (Chapters 15, 21)

Hox complex A gene complex consisting of a series of *Hox* genes. (Chapter 21)

Hox genes Genes coding for transcription regulators, each gene containing a homeodomain, and specifying body-region differences. *Hox* mutations typically cause homeotic transformations. (Chapter 21)

Hox proteins Transcription regulatory proteins encoded by *Hox* genes; possess a highly conserved, 60-amino-acid-long DNA-binding homeodomain. (Chapter 21)

HPV Human papillomavirus; infects the cervical epithelium and is important as a cause of carcinoma of the uterine cervix. (Chapter 20)

hyaluronan (hyaluronic acid) Type of nonsulfated glycosaminoglycan with a regular repeating sequence of up to 25,000 identical disaccharide units and not linked to a core protein. Found in the fluid lubricating joints and in many other tissues. (Chapter 19) (Figures 19–33 and 19–34)

hybridization In molecular biology, the process whereby two complementary nucleic acid strands form a base-paired duplex DNA–DNA, DNA–RNA, or RNA–RNA molecule. Forms the basis of a powerful technique for detecting specific nucleotide sequences. (Chapter 8) (Figure 8–31)

hybridoma Hybrid cell line generated by fusion of a tumor cell and another cell type. Monoclonal antibodies are produced by hybridoma lines obtained by fusing antibody-secreting B cells with cells of a B-lymphocyte tumor. (Chapter 8) (Figure 8–3)

hydrogen bond Noncovalent bond in which an electropositive hydrogen atom is partially shared by two electronegative atoms. (Chapter 2) (Panel 2–3, pp. 98–99)

hydronium ion (H₃O⁺) Water molecule associated with an additional proton. The form generally taken by protons in aqueous solution. (Chapter 2)

hydrophilic Dissolving readily in water. Literally, “water loving.” (Chapters 2, 10)

hydrophobic (lipophilic) Not dissolving readily in water. Literally, “water-fearing.” (Chapters 2, 10)

hydrophobic force Force exerted by the hydrogen-bonded network of water molecules that brings two nonpolar surfaces together by excluding water between them. (Panel 2–3, pp. 98–99)

hyperpolarization Deviation in the electrical potential across the plasma membrane towards a more negative value. (Chapter 11)

hypervariable region In immunology: any of the three small parts of the variable region of an immunoglobulin or T cell receptor chain that show the highest variability from molecule to molecule and contribute to the antigen-binding site. (Chapter 24) (Figure 24–26)

IκB Inhibitory proteins that bind tightly to NFκB dimers and hold them in an inactive state within the cytoplasm of unstimulated cells. (Chapter 15)

Ig gene segments In immunology: short DNA sequences that are joined together during B cell and T cell development to produce the coding sequences for immunoglobulins and T cell receptors, respectively. (Chapter 24) (Figure 24–28)

Ig superfamily Large and diverse family of proteins that contain immunoglobulin or immunoglobulin-like domains. Most are involved in cell–cell interactions or antigen recognition. (Chapter 24) (Figure 24–48)

IgA Immunoglobulin A; the principal class of antibody in secretions, including saliva, tears, milk, and respiratory and intestinal secretions. (Chapter 24)

IgD Immunoglobulin D; produced by immature naïve B cells after leaving the bone marrow. Transmembrane IgD and IgM proteins, with the same antigen-binding site, form the B cell receptors (BCRs) on these cells. (Chapter 24)

IgE Immunoglobulin E; binds with high affinity via its tail region to a class of Fc receptors on the surface of mast cells (tissues) or basophils (blood), where it acts as an antigen receptor; antigen binding stimulates the secretion of cytokines and biologically active amines, which help attract white blood cells, antibodies, and complement proteins to the site of activation. (Chapter 24)

IgG Immunoglobulin G; the major antibody class in the blood, produced in especially large quantities during secondary antibody responses. The tail region of some IgG subclasses can bind to specific Fc receptors on macrophages and neutrophils. Antigen–IgG complexes can activate complement. (Chapter 24)

IgM Immunoglobulin M; the first class of immunoglobulin that a developing B cell in the bone marrow makes, forming B-cell receptors on its surface. IgM antibodies are the major class of antibody secreted into the blood in the early stages of a primary antibody response on first exposure to an antigen, where their pentameric structure (with 10 antigen-binding sites) allows strong binding to pathogens. When bound to antigen, it is highly efficient at activation of complement. (Chapter 24)

image processing Computer-based techniques in microscopy that process digital images in order to extract latent information. Enables compensation for some optical faults in microscopes, enhanced contrast to improve detection of small differences in light intensity, and subtraction of background irregularities in the optical system. (Chapter 9)

imaginal disc Group of cells that are set aside, apparently undifferentiated, in the *Drosophila* embryo and which will develop into an adult structure (for example, eye, leg, wing). Overt differentiation occurs at metamorphosis. (Chapter 21) (Figure 21–65)

immunization Method of inducing adaptive immune responses to pathogens or foreign molecules, usually involving the co-injection of an adjuvant, a molecule (often of microbial origin) that helps activate innate immune responses required for the adaptive responses. (Chapter 24)

immunoblotting See **western blotting**. (Chapter 8)

immunoglobulin (Ig) See **antibody**. (Chapters 3, 24)

immunoglobulin (Ig) domain Characteristic protein domain of about 110 amino acids that is found in immunoglobulin light and heavy chains. Similar domains, known as immunoglobulin-like (Ig-like) domains, are present in many other proteins, which, together with Igs, constitute the Ig superfamily. (Chapter 24) (Figure 24–27)

immunoglobulin (Ig) superfamily Large and diverse family of proteins that contain immunoglobulin domains or immunoglobulin-like domains. Most are involved in cell–cell interactions or antigen recognition. (Chapter 19) (Figure 24–48)

immunogold electron microscopy Method to localize specific macromolecules using a primary antibody that binds to the molecule of interest and is then detected with a secondary antibody to which a colloidal gold particle has been attached. The gold particle is electron-dense and can be seen as a black dot in the electron microscope. (Figure 9–43)

immunological memory Long-lived property of the adaptive immune system that follows a primary immune response to many antigens, such that a subsequent encounter with the same antigen will provoke a more rapid and stronger secondary immune response. (Chapter 24) (Figure 24–16)

immunological self-tolerance The lack of response of the adaptive immune system to an antigen. Tolerance to self molecules is crucial to avoid autoimmune diseases. (Chapter 24) (Figure 24–21)

immunological synapse The highly organized interface that develops between a T cell and an antigen-presenting cell (APC) or target cell it is in contact with, formed by T-cell receptors binding to antigen–MHC complexes on the APC and cell adhesion proteins binding to their counterparts on the APCs. (Chapter 24)

induced fit A principle for increasing the specificity of substrate recognition by proteins and RNAs. In protein synthesis, the ribosome folds around a codon–anticodon interaction, and only when the match is correct does the subsequent reaction proceed efficiently. (Chapter 6)

induced pluripotent stem cells (iPS cells) Cells that are induced by artificial expression of specific transcription regulators to look and behave like the pluripotent embryonic stem cells that are derived from embryos. (Chapters 7, 22)

induced regulatory T cell (induced T_{reg} cell) A regulatory T cell (T_{reg} cell) that develops from naïve helper T cells when they are activated in the presence of TGFβ in the absence of IL6. (Chapter 24)

inflammasome Intracellular protein complex formed after activation of cytoplasmic NOD-like receptors with adaptor proteins. It contains a caspase enzyme that cleaves pro-inflammatory cytokines from their precursor proteins. (Chapter 24)

inflammatory response Local response of a tissue to injury or infection—characterized clinically by redness, swelling, heat, and pain. Caused by invasion of white blood cells, which are attracted by and secrete various cytokines. (Chapter 24)

inhibitors of apoptosis (IAPs) Intracellular protein inhibitors of apoptosis. (Chapter 18)

inhibitory G protein (G_i) Heterotrimeric G protein that can regulate ion channels and inhibit the enzyme adenylyl cyclase in the plasma membrane. See also **G protein**. (Chapter 15) (Table 15–3, p. 907)

inhibitory neurotransmitter Neurotransmitter that opens transmitter-gated Cl[−] or K⁺ channels in the postsynaptic membrane of a nerve or muscle cell and thus tends to inhibit the generation of an action potential. (Chapter 11)

initial segment Specialized membrane region at the base of a nerve axon (adjacent to the cell body) that is rich in voltage-gated Na⁺ channels plus other classes of ion channels that all contribute to the encoding of membrane depolarization into action potential frequency. (Chapter 11)

initiator caspases Apoptotic caspases that begin the apoptotic process, activating the executioner caspases. (Chapter 18)

initiator tRNA Special tRNA that initiates translation. It always carries the amino acid methionine, forming the complex Met–tRNA_i. (Chapter 6) (Figure 6–74)

innate immune response An early immune response in all organisms to a pathogen, which includes the production of antimicrobial molecules and the activation of phagocytic cells. Such a response is not specific for the pathogen, in contrast to an adaptive immune response. (Chapter 24)

innate immune system A variety of defense mechanisms that help prevent and fight infection. Unlike adaptive immune

mechanisms, innate immune mechanisms act from the start of an infection, do not adapt to a specific pathogen, and do not generate immunological memory. (Chapter 24)

inner mitochondrial membrane Mitochondrial membrane that encloses the matrix space and forms extensive invaginations called cristae. (Chapters 12, 14)

inner nuclear membrane One of two concentric membranes comprising the nuclear envelope; contains specific proteins as anchoring sites for chromatin and the nuclear lamina. (Chapter 12)

inositol 1,4,5-trisphosphate (IP₃) Small intracellular signaling molecule produced during activation of the inositol phospholipid signaling pathway. Acts to release Ca²⁺ from the endoplasmic reticulum. (Chapter 15) (Figures 15–29 and 15–30)

inositol phospholipid signaling pathway Intracellular signaling pathway that starts with the activation of phospholipase C and the generation of IP₃ and diacylglycerol (DAG) from inositol phospholipids in the plasma membrane. The DAG helps to activate protein kinase C. (Chapter 15) (Figures 15–29 and 15–30)

integrin Transmembrane adhesion protein that is involved in the attachment of cells to the extracellular matrix and to each other. (Chapter 19) (Figure 19–3 and Table 19–1, p. 1107)

interaction domain Compact protein module, found in many intracellular signaling proteins, that binds to a particular structural motif (for example, a short peptide sequence, a covalent modification, or another protein domain) in another protein or lipid. (Chapter 15)

interferon- α (IFN α) and interferon- β (IFN β) Cytokines (type I interferons) produced by mammalian cells as a general response to a viral infection. (Chapter 24)

intermembrane space The compartment in a mitochondrion between the outer and inner mitochondrial membranes. (Chapters 12, 14)

internal ribosome entry site (IRES) Specific site in a eukaryotic mRNA, other than at the 5' end, at which translation can be initiated. (Chapter 7) (Figure 7–72)

interphase Long period of the cell cycle between one mitosis and the next. Includes G₁ phase, S phase, and G₂ phase. (Chapter 17) (Figure 17–4)

intestinal epithelium The single cell layer of polarized epithelial cells that makes up the lining of the small and large intestine, where it acts as a barrier to the lumen of the gut and mediates absorption of nutrients. (Chapter 22)

intracellular pathogens Pathogens, including all viruses and many bacteria and protozoa, that enter and replicate inside host cells to cause disease. (Chapter 23)

intrinsic pathway (mitochondrial pathway) Pathway of apoptosis activated from inside the cell in response to stress or developmental signals; depends on the release into the cytosol of mitochondrial proteins normally resident in the mitochondrial intermembrane space. (Chapter 18)

intron Noncoding region of a eukaryotic gene that is transcribed into an RNA molecule but is then excised by RNA splicing during production of the mRNA or other functional RNA. (Chapters 4, 6) (Figure 4–15)

invadopodia Actin-rich protrusions extending in three dimensions that are important for cells to cross tissue barriers by degrading the extracellular matrix. (Chapter 16)

ion channel Transmembrane protein complex that forms a water-filled channel across the lipid bilayer through which specific inorganic ions can diffuse down their electrochemical gradients. (Chapter 11) (Figure 11–22)

ion-sensitive indicators Molecules whose light emission reflects the local concentration of a particular ion; some are luminescent (emitting light spontaneously) while others are fluorescent (emitting light on exposure to light). (Figure 9–21)

ionotropic receptor (transmitter-gated ion channel) Ion channel found at chemical synapses in the postsynaptic plasma membranes of nerve and muscle cells. Opens only in response to the binding of a specific extracellular neurotransmitter. The resulting inflow of ions leads to the generation of a local electrical signal in the postsynaptic cell. (Figures 11–38 and 15–6)

IP₃ receptor (IP₃-gated Ca²⁺-release channel) Gated Ca²⁺ channel in the ER membrane that opens on binding cytosolic IP₃, releasing stored Ca²⁺ into the cytosol. (Chapter 15) (Figure 15–30)

iron-sulfur cluster Electron-transporting group consisting of either two or four iron atoms bound to an equal number of sulfur atoms, found in a class of electron-transport proteins. (Figure 14–16)

iron-sulfur protein A protein that contains one or more iron-sulfur clusters, typically using those cofactors for electron transport. (Chapter 14)

J gene segment Short DNA sequence that encodes part of the variable region of light and heavy immunoglobulin chains and of α and β chains of T cell receptors. (Chapter 24) (Figures 24–28 and 24–29)

JAK-STAT signaling pathway Signaling pathway activated by cytokines and some hormones, providing a rapid route from the plasma membrane to the nucleus to alter gene transcription. Involves cytoplasmic Janus kinases (JAKs), and signal transducers and activators of transcription (STATs). (Chapter 15)

Janus kinases (JAKs) Cytoplasmic tyrosine kinases associated with cytokine receptors, which phosphorylate and activate transcription regulators called STATs. (Chapter 15)

junctional diversification The random loss and gain of nucleotides at joining sites during V(D)J recombination that occurs during B and T cell development when the cells are assembling the gene segments that encode their antigen receptors. It enormously increases the diversity of V-region coding sequences. (Chapter 24)

K⁺ leak channel K⁺-transporting ion channel in the plasma membrane of animal cells that remains open even in a “resting” cell. (Chapter 11)

K_m The Michaelis constant, equal to the concentration of an enzyme's substrate that allows that enzyme to produce product at one-half of its maximum rate. (Chapter 3)

karyotype Display of the full set of chromosomes of a cell, arranged with respect to size, shape, and number. (Chapter 4)

keratin Type of intermediate filament, commonly produced by epithelial cells. (Chapter 16)

kinase cascade Intracellular signaling pathway in which one protein kinase, activated by phosphorylation, phosphorylates the next protein kinase in the sequence, and so on, relaying the signal onward. (Chapter 15)

kinesin Member of one of the two main classes of motor proteins that use the energy of ATP hydrolysis to move along microtubules. (Chapter 16) (Figure 16–51)

kinesin-1 Motor protein associated with microtubules that transports cargo within the cell; also called “conventional kinesin.” (Chapter 16)

kinetic proofreading A principle for increasing the specificity of catalysis. In the synthesis of DNA, RNA, and proteins, it refers to a time delay that begins with an irreversible step (such as ATP or GTP hydrolysis) and during which incorrect base pairs are more likely to dissociate than correct pairs. (Chapter 6)

kinetochore Large protein complex that connects the centromere of a chromosome to microtubules of the mitotic spindle. (Chapter 17) (Figure 17–33)

kinetochore microtubule In the mitotic or meiotic spindle, a microtubule that connects the spindle pole to the kinetochore of a chromosome. (Chapter 17)

lagging strand One of the two newly synthesized strands of DNA found at a replication fork. The lagging strand is made in discontinuous lengths that are later joined covalently. (Chapter 5) (Figure 5–7)

lamellipodium (plural **lamellipodia**) Flattened, sheetlike protrusion supported by a meshwork of actin filaments, which is extended at the leading edge of a crawling animal cell. (Chapter 16) (Figures 16–18 and 16–19)

laminin Extracellular matrix fibrous protein found in basal laminae, where it forms a sheetlike network. (Chapter 19) (Figures 19–53 and 19–54)

lampbrush chromosome Huge chromosome paired in preparation for meiosis, found in immature amphibian eggs; consisting of large loops of chromatin extending out from a linear central axis. (Chapter 4) (Figure 4–45)

late endosome Compartment formed from a bulbous, vacuolar portion of early endosomes by a process called endosome maturation; late endosomes fuse with one another and with lysosomes to form endolysosomes that degrade their contents. (Chapter 13)

LDL-receptor-related protein (LRP) Co-receptor bound by Wnt proteins in the regulation of β -catenin proteolysis. (Chapter 15)

leading strand One of the two newly synthesized strands of DNA found at a replication fork. The leading strand is made by continuous synthesis in the 5′-to-3′ direction. (Chapter 5) (Figure 5–7)

lectin Protein that binds tightly to a specific sugar. Abundant lectins from plant seeds are used as affinity reagents to purify glycoproteins or to detect them on the surface of cells. (Chapter 10)

Legionnaires' disease Type of pneumonia resulting from infection with *Legionella pneumophila*, a bacterial parasite of freshwater amoebae that is spread to humans by air-conditioning systems that harbor infected amoebae. (Chapter 23)

lethal factor One of the subunits of anthrax toxin; a protease that cleaves several activated members of the MAP kinase family and causes a large fall in blood pressure and death on entry into the bloodstream of an animal. (Chapter 23)

leucine-rich repeat (LRR) receptor kinases Common type of receptor serine/threonine kinase in plants that contains

a tandem array of leucine-rich repeat sequences in its extracellular portion. (Chapter 15)

leukemia Cancer of white blood cells. (Chapter 20)

leukocyte General name for all the nucleated blood cells lacking hemoglobin. Also called white blood cells. Includes lymphocytes, granulocytes, and monocytes. (Chapter 22) (Figure 22–11)

ligand Any molecule that binds to a specific site on a protein or other molecule. From Latin *ligare*, “to bind.” (Chapter 3)

light microscope One of a class of microscopes that uses visible light to create the image. (Chapter 9)

lignin Network of cross-linked phenolic compounds that forms a supporting network throughout the cell walls of xylem and woody tissue in plants. (Chapter 19)

limit of resolution In microscopy, the smallest distance apart at which two point objects can be resolved as separate. Just under 0.2 μm for conventional light microscopy, a limit determined by the wavelength of light. (Chapter 9)

linkage In ligand binding, the conformational coupling between two separate ligand-binding sites on a protein, such that a conformational change in the protein induced by binding of one ligand affects the binding of a second ligand. (Chapter 3)

lipid bilayer (phospholipid bilayer) Thin double sheet of phospholipid molecules that forms the core structure of all cell membranes. The two layers of lipid molecules are packed with their hydrophobic tails pointing inward and their hydrophilic heads outward, exposed to water. (Chapter 10) (Figure 10–1 and Panel 2–5, pp. 102–103)

lipid droplets Storage form in cells for excess lipids; composed of a single monolayer of phospholipids and proteins that surrounds neutral lipids that can be retrieved from droplets as required by the cell. (Chapter 10)

lipid raft Small region of a membrane enriched in sphingolipids and cholesterol. (Chapter 10) (Figure 10–13)

liposome Artificial phospholipid bilayer vesicle formed from an aqueous suspension of phospholipid molecules. (Chapter 10) (Figure 10–9)

local mediator Extracellular signal molecule that acts on neighboring cells. (Chapter 15)

long noncoding RNA (lncRNA) One of a large group (~5000 in humans) of RNAs longer than 200 nucleotides and not coding for protein. The functions, if any, of most lncRNAs are unknown, but individual lncRNAs are known to play important roles in the cell; for example, in telomerase function and genomic imprinting. In a general sense, lncRNAs are believed to act as scaffolds, holding together proteins and nucleic acids to speed up a wide variety of reactions in the cell. (Chapter 7)

long-term depression (LTD) A long-lasting (hours or more) decrease in the sensitivity of certain synapses in the brain triggered by NMDA-receptor activation. As the opposing process to long-term potentiation, it is thought to be involved in learning and memory. (Chapter 11)

long-term potentiation (LTP) Long-lasting increase (days to weeks) in the sensitivity of certain synapses in the brain, induced by a short burst of repetitive firing in the presynaptic neurons. (Chapter 11) (Figure 11–46)

loss of heterozygosity The result of errant homologous recombination that uses the homolog from the other parent instead of the sister chromatid as the template, converting the

sequence of the repaired DNA to that of the other homolog. (Chapter 5)

low-density lipoprotein (LDL) Large complex composed of a single protein molecule and many esterified cholesterol molecules, together with other lipids. The form in which cholesterol is transported in the blood and taken up into cells. (Chapter 13) (Figure 13–53)

lumen The space inside a hollow structure. In cells: the cavity enclosed by an organelle membrane. In tissues: the cavity enclosed by a sheet of cells. (Chapters 10, 12)

lymphocyte White blood cell responsible for the specificity of adaptive immune responses. Two main types: B cells, which produce antibody, and T cells, which interact directly with other effector cells of the immune system and with infected cells. T cells develop in the thymus and are responsible for cell-mediated immunity. B cells develop in the bone marrow in mammals and are responsible for the production of circulating antibodies. (Chapter 24)

lymphoid cells Immune cells including B and T lymphocytes and natural killer cells. (Chapter 22)

lymphoid organ An organ containing large numbers of lymphocytes. Lymphocytes are produced in central (or primary) lymphoid organs and respond to antigen in peripheral (or secondary) lymphoid organs. (Chapter 24) (Figure 24–12)

lymphoma Cancer of lymphocytes, in which the cancer cells are mainly found in lymphoid organs (rather than in the blood, as in leukemias). (Chapter 20)

lysosomal storage diseases Genetic diseases resulting from defects in or a lack of one or more functional hydrolases in lysosomes of some cells, leading to accumulation of undigested substrates in lysosomes and consequent cell pathology. (Chapter 13)

lysosome Membrane-enclosed organelle in eukaryotic cells containing digestive enzymes, which are typically most active at the acid pH found in the lumen of lysosomes. (Chapter 13) (Figure 13–62)

lysozyme Enzyme that catalyzes the cutting of polysaccharide chains in the cell walls of bacteria. (Chapter 3)

M-Cdk (M-phase Cdk) Cyclin–Cdk complex formed in vertebrate cells by an M-cyclin and the corresponding cyclin-dependent kinase (Cdk). (Chapter 17) (Figure 17–10 and Table 17–1, p. 1034)

M-cyclin A cyclin found in all eukaryotic cells that promotes the events of mitosis. (Chapter 17) (Figure 17–10)

M6P receptor proteins Transmembrane receptor proteins present in the *trans* Golgi network that recognize the mannose 6-phosphate (M6P) groups added exclusively to lysosomal enzymes, marking the enzymes for packaging and delivery to early endosomes. (Chapter 13)

macromolecule Polymers constructed of long chains of covalently linked, small organic (carbon-containing) molecules. The principal building blocks from which a cell is constructed and the components that confer the most distinctive properties of living things. (Chapter 2)

macrophage Phagocytic cell derived from blood monocytes, resident in most tissues but able to roam. It has both scavenger and antigen-presenting functions in immune responses. (Chapter 13)

macropinocytosis Clathrin-independent, dedicated degradative endocytic pathway induced in most cell types

by cell-surface receptor activation by specific cargoes. (Chapter 13)

malaria Protozoal disease caused by any one of four species of *Plasmodium*, which are transmitted to humans by the bite of the female *Anopheles* mosquito. (Chapter 23)

malignant Of tumors and tumor cells: invasive and/or able to undergo metastasis. A malignant tumor is a cancer. (Chapter 20) (Figure 20–3)

MAP kinase module (mitogen-activated protein kinase module) An intracellular signaling module composed of three protein kinases, acting in sequence, with MAP kinase as the third. Typically activated by a Ras protein in response to extracellular signals. (Chapter 15) (Figure 15–50)

master transcription regulator A transcription regulator specifically required for formation of a particular cell type. Artificial expression of master transcription regulators (alone or in combination with others) can often convert one cell type into another. (Chapter 7)

maternal inheritance A form of inheritance observed in animals and plants, caused by the fact that mitochondrial DNA is inherited only through the female germ line. (Chapter 14)

maternal effect Describes a gene that acts in the mother to specify maternal mRNAs and proteins in the egg (that is, maternal-effect gene). Maternal-effect mutations affect the development of the embryo even if the embryo itself has not inherited the mutated gene. (Chapter 21)

maternal-zygotic transition (MZT) Event in animal development where the embryo's own genome largely takes over control of development from maternally deposited macromolecules. (Chapter 21)

matrix metalloprotease Ca^{2+} - or Zn^{2+} -dependent proteolytic enzyme present in the extracellular matrix that degrades matrix proteins. Includes the collagenases. (Chapter 19)

matrix space The large internal compartment of the mitochondrion. (Chapter 12)

Mcm helicase Six-subunit protein complex that serves as the replicative helicase in eukaryotic DNA replication, unwinding the DNA to enable DNA synthesis. (Chapters 5, 17)

mechanosensitive channels Transmembrane ion channels that open in response to a mechanical stress on the lipid bilayer in which they are embedded. (Chapter 11)

megakaryocyte Large myeloid cell with a multilobed nucleus that remains in the bone marrow when mature. Buds off platelets from long cytoplasmic processes. (Chapter 22)

meiosis I The first of two rounds of chromosome segregation after meiotic chromosome duplication; segregates the homologs, each composed of a tightly linked pair of sister chromatids. (Chapter 17)

meiosis II The second of two rounds of chromosome segregation after meiotic chromosome duplication; segregates the sister chromatids of each homolog. (Chapter 17)

membrane-associated protein Membrane protein not extending into the hydrophobic interior of the lipid bilayer but bound to either face of the membrane by noncovalent interactions with other membrane proteins. (Chapter 10) (Figure 10–17)

membrane-bending proteins Attach to specific membrane regions as needed, where they act to control local membrane

curvature and thus confer on these regions their three-dimensional shapes. (Chapter 10)

membrane-bound ribosome Ribosome attached to the cytosolic face of the endoplasmic reticulum. The site of synthesis of proteins that enter the endoplasmic reticulum. (Chapter 12) (Figure 12–21)

membrane potential Voltage difference across a membrane due to a slight excess of positive ions on one side and of negative ions on the other. A typical membrane potential for an animal cell plasma membrane is -60 mV (inside negative relative to the surrounding fluid). (Chapter 11) (Figure 11–23)

membrane protein Amphiphilic protein of diverse structure and function that associates with the lipid bilayer of cell membranes. (Chapter 10) (Figure 10–17)

membrane transport protein Membrane protein that mediates the passage of ions or molecules across a membrane. The two main classes are transporters (also called carriers or permeases) and channels. (Chapter 11) (Figure 11–3)

membraneless organelle An assembly of specific proteins held together by multivalent, low-affinity interactions. Also referred to as a *biomolecular condensate*. (Chapter 12)

memory cell In immunology: a T or B lymphocyte generated after antigen stimulation that is more easily and more quickly induced to become an effector cell or another memory cell by a later encounter with the same antigen. (Chapter 24) (Figure 24–17)

mesenchymal cell migration A mode of cell locomotion typical of fibroblasts that is characterized by protrusion of actin-rich lamellipodia at the leading edge, formation of integrin-based attachments to the underlying substratum, and actomyosin-based contraction at the rear of the cell. (Chapter 16)

mesoderm Embryonic tissue that is the precursor to muscle, connective tissue, skeleton, and many of the internal organs. (Chapter 21) (Figure 21–3)

messenger RNA (mRNA) RNA molecule that specifies the amino acid sequence of a protein. Produced in eukaryotes by processing of an RNA molecule made by RNA polymerase as a complementary copy of DNA. It is translated into protein in a process catalyzed by ribosomes. (Chapter 6) (Figure 6–21)

metabolism The sum total of the chemical processes that take place in living cells. All of catabolism plus anabolism. (Chapter 2) (Figure 2–14)

metabotropic receptors Neurotransmitter receptors that regulate ion channels indirectly through the activation of second-messenger molecules. (Chapter 11)

metaphase The stage of mitosis when all the chromosomes have been moved to the center of the cell, having been positioned there by the mitotic spindle. Metaphase follows prophase and prometaphase; it is succeeded by anaphase. (Chapter 17)

metaphase plate Imaginary plane at right angles to the mitotic spindle and midway between the spindle poles; the plane in which chromosomes are positioned at metaphase. (Chapter 17) (Panel 17–1, pp. 1048–1049)

metaphase-to-anaphase transition Transition in the eukaryotic cell cycle preceding sister-chromatid separation at anaphase. If the cell is not ready to proceed to anaphase, the cell cycle is halted at this point. (Chapter 17) (Figure 17–8 and Panel 17–1, pp. 1048–1049)

metastases Secondary tumors, at sites in the body additional to that of the primary tumor; they result from cancer cells

breaking loose, entering blood or lymphatic vessels, and colonizing separate environments. (Chapter 20)

metastasis The spread of cancer cells from their site of origin to other sites in the body. (Chapter 20) (Figures 20–1 and 20–20)

MHC genes Cluster of genes in one vertebrate chromosome (chromosome 6 in humans) that code for a set of highly polymorphic cell-surface glycoproteins (MHC proteins). (Figure 24–37)

MHC protein Cell-surface glycoprotein encoded within the major histocompatibility complex (MHC) of genes. The proteins are highly polymorphic and exist in two main classes—class I and class II MHC proteins, both of which function to present fragments of foreign proteins on the surface of antigen-presenting cells to T cells. (Chapter 24)

microbiome The combined genomes of the various species of a defined microbiota. (Chapter 23)

microbiota The collective of microorganisms that reside in or on an organism. (Chapter 23)

microelectrode A piece of fine glass tubing, pulled to an even finer tip, that is used to inject electric current into cells or to study the intracellular concentrations of common inorganic ions (such as H^+ , Na^+ , K^+ , Cl^- , and Ca^{2+}) in a single living cell by insertion of its tip directly into the cell interior through the plasma membrane. (Figure 11–36)

microRNAs (miRNAs) Short (~21 nucleotide) eukaryotic RNAs, produced by the processing of specialized RNA transcripts coded in the genome, that regulate gene expression through base-pairing with mRNA. (Chapters 7, 21) (Figure 7–78)

microsome Small vesicle derived from endoplasmic reticulum that is produced by fragmentation when cells are homogenized. (Chapter 12) (Figure 12–17)

microtubule flux Movement of individual tubulin molecules in the microtubules of the spindle toward the poles by loss of tubulin at their minus ends. Helps to generate the poleward movement of sister chromatids after they separate in anaphase. (Chapter 17) (Figure 17–37)

microtubule-associated protein (MAP) Any protein that binds to microtubules and modifies their properties. Many different kinds have been found, including structural proteins, such as MAP2, and motor proteins, such as dynein. [Not to be confused with the “MAP” (mitogen-activated protein) of “MAP kinase.”] (Chapter 16)

microtubule-organizing center (MTOC) Region in a cell, such as a centrosome or a basal body, from which microtubules grow. (Chapter 16)

midbody Structure formed at the end of cleavage that can persist for some time as a tether between the two daughter cells in animals. (Chapter 17) (Figure 17–44)

mitochondrial hsp70 Part of a multisubunit protein assembly bound to the matrix side of the TIM23 complex that acts as a motor to pull mitochondrial precursor proteins into the matrix space. (Chapter 12)

mitochondrial matrix Large internal compartment of the mitochondrion. The corresponding compartment in a chloroplast is known as the stroma. (Chapter 14)

mitochondrial outer membrane permeabilization (MOMP) The change in the outer mitochondrial membrane that releases cytochrome c and other soluble proteins from the

intermembrane space into the cytosol—a critical step in the intrinsic pathway of apoptosis. (Chapter 18)

mitochondrial precursor proteins Proteins that are first fully synthesized in the cytosol and then translocated into mitochondrial subcompartments, as directed by one or more signal sequences. (Chapter 12)

mitochondrion (plural **mitochondria**) Membrane-bounded organelle, about the size of a bacterium, that carries out oxidative phosphorylation and produces most of the ATP in eukaryotic cells. (Chapters 12, 14) (Figure 1–25)

mitogen Extracellular signal molecule that stimulates cells to proliferate. (Chapter 17)

mitotic chromosome Highly condensed duplicated chromosome as seen at mitosis, consisting of two sister chromatids held together at the centromere. (Chapter 4)

mitotic spindle Bipolar array of microtubules and associated molecules that forms in a eukaryotic cell during mitosis; it serves to position duplicated chromosomes and then segregate their sister chromatids to opposite spindle poles. (Chapter 17) (Figure 17–26 and Panel 17–1, pp. 1048–1049)

model organism A species that has been studied intensively over a long period and thus serves as a “model” for deriving fundamental biological principles. (Chapter 1)

molecular chaperone (**chaperone**) Protein that helps guide the proper folding of other proteins or helps them avoid misfolding. Includes heat-shock proteins. (Chapter 6)

monoallelic gene expression Expression of only one of the two copies of a gene in a diploid genome, occurring, for example, as a result of imprinting or X-inactivation. (Chapter 7)

monoclonal antibody Antibody secreted by a hybridoma cell line. Because the hybridoma is generated by the fusion of a single B cell with a single tumor cell, each hybridoma produces antibodies that are all identical. (Chapters 4, 8)

monocyte Type of white blood cell that leaves the bloodstream and matures into a macrophage in tissues. (Chapter 22) (Figure 22–11)

monomeric GTPase A single-subunit enzyme that converts GTP to GDP (also called monomeric GTP-binding protein). Cycles between an active GTP-bound form and an inactive GDP-bound form and frequently acts as a molecular switch in intracellular signaling pathways. (Chapter 15)

morphogen Diffusible signal molecule that can impose a pattern on a field of cells by causing cells in different places to adopt different fates. (Chapter 21) (Figure 21–8)

morphogenesis Developmental process in which cells undergo movements and deformations in order to assemble into tissues and organs with specific shapes and sizes. (Chapter 21)

motor protein Protein that uses energy derived from nucleoside triphosphate hydrolysis to propel itself along a linear track (protein filament or other polymeric molecule). (Chapter 16)

mRNA degradation control Cell regulation of gene expression by selectively preserving or destroying certain mRNA molecules in the cytoplasm. (Chapter 7)

mTOR The mammalian version of the large protein kinase called TOR, involved in cell signaling; mTOR exists in two functionally distinct multiprotein complexes. (Chapter 15)

multidrug resistance An observed phenomenon in which cells exposed to one anticancer drug evolve a resistance not

only to that drug but also to other drugs to which they have never been exposed. (Chapter 20)

multidrug resistance (MDR) protein Type of ABC transporter protein that can pump hydrophobic drugs (such as some anticancer drugs) out of the cytoplasm of eukaryotic cells. (Chapter 11)

multipass transmembrane protein Membrane protein in which the polypeptide chain crosses the lipid bilayer more than once. (Chapter 10) (Figure 10–17)

multivesicular bodies Intermediates in the endosome maturation process; early endosomes that are on their way to becoming late endosomes. (Chapter 13)

mutation Heritable change in the nucleotide sequence of a chromosome. (Chapter 1) (Panel 8–1, pp. 520–521)

mutation rate The rate at which changes (mutations) occur in DNA sequences. (Chapter 5)

mutualism Ecologic relationship between microbes and their host in which both the microbe and host benefit. (Chapter 23)

Myc Transcription regulatory protein that is activated when a cell is stimulated to grow and divide by extracellular signals. It activates the transcription of many genes, including those that stimulate cell growth. (Chapter 17) (Figure 17–59)

myelin sheath Insulating layer of specialized cell membrane wrapped around vertebrate axons. Produced by oligodendrocytes in the central nervous system and by Schwann cells in the peripheral nervous system. (Chapter 11) (Figure 11–35)

myeloid cell Any white blood cell other than a lymphocyte. (Chapter 22) (Figure 22–12)

myeloma Cancer of antibody-producing (B) cells found in bone marrow. (Chapter 20)

myoblast Mononucleated, undifferentiated muscle precursor cell. A skeletal muscle cell is formed by the fusion of multiple myoblasts. (Chapter 22)

myofibril Long, highly organized bundle of actin, myosin, and other proteins in the cytoplasm of muscle cells that contracts by a sliding filament mechanism. (Chapter 16)

myosin Type of motor protein that uses the energy of ATP hydrolysis to move along actin filaments. (Chapter 16)

Na⁺-K⁺ pump (Na⁺-K⁺ ATPase) Transmembrane carrier protein found in the plasma membrane of most animal cells that pumps Na⁺ out of and K⁺ into the cell, using energy derived from ATP hydrolysis. (Chapter 11) (Figure 11–15)

NAD⁺/NADH (nicotinamide adenine dinucleotide/reduced nicotinamide adenine dinucleotide) Electron carrier system that participates in oxidation–reduction reactions, such as the oxidation of food molecules. NAD⁺ accepts the equivalent of a hydride ion (H[−], a proton plus two electrons) to become the activated carrier NADH. The NADH formed donates its high-energy electrons to the ATP-generating process of oxidative phosphorylation. (Chapter 2) (Figure 2–36)

NADH dehydrogenase complex First of the three electron-driven proton pumps in the mitochondrial respiratory chain, also known as Complex I. It accepts electrons from NADH and passes them to a quinone. (Chapter 14) (Figure 14–18)

NADP⁺/NADPH (nicotinamide adenine dinucleotide phosphate/reduced nicotinamide adenine dinucleotide phosphate) Electron carrier system closely related to

NAD⁺/NADH, but used almost exclusively in reductive biosynthetic, rather than catabolic, pathways. (Chapter 2) (Figure 2–36)

naïve cell In immunology: a T or B lymphocyte that proliferates and differentiates into an effector cell or memory cell when it encounters its specific foreign antigen for the first time. (Chapter 24) (Figure 24–17)

natural killer cell (NK cell) Cytotoxic cell of the innate immune system that can kill virus-infected cells and some cancer cells. (Chapter 24)

natural regulatory T cell (natural T_{reg} cell) A regulatory T cell (T_{reg} cell) that develops in the thymus and helps maintain self-tolerance. (Chapter 24)

negative selection Process by which thymocytes expressing a T cell receptor with high affinity for a self peptide bound to a self MHC protein are eliminated by undergoing apoptosis. (Chapter 24)

negative staining A technique in electron microscopy enabling fine detail of isolated macromolecules to be seen. Samples are prepared such that a very thin film of heavy-metal salt covers everywhere except where excluded by the presence of macromolecules, which allow electrons to pass through, creating a reverse or negative image of the molecule. (Chapter 9)

Nernst equation Equation that relates the equilibrium electrical potential (membrane potential in volts) to differences in ion concentrations across a membrane. (Chapter 11)

neural crest Collection of cells located along the line where the neural tube pinches off from the surrounding epidermis in the vertebrate embryo. Neural crest cells migrate to give rise to a variety of tissues, including neurons and glia of the peripheral nervous system, pigment cells of the skin, and the bones of the face and jaws. (Chapter 21) (Figure 19–8)

neural tube Tube of ectoderm that will form the brain and spinal cord in a vertebrate embryo. (Chapter 21)

neurofilament Type of intermediate filament found in nerve cells. (Chapter 16) (Figure 16–64)

neuromuscular junction Specialized chemical synapse between an axon terminal of a motor neuron and a skeletal muscle cell. (Chapter 11) (Figure 11–39)

neuron (nerve cell) Impulse-conducting cell of the nervous system, with extensive processes specialized to receive, conduct, and transmit signals. (Chapter 11) (Figure 11–29)

neurotransmitter Small signal molecule secreted by the presynaptic nerve cell at a chemical synapse to relay the signal to the postsynaptic cell. Examples include acetylcholine, glutamate, GABA, glycine, and many neuropeptides. (Chapters 11, 15)

neutrophil White blood cell that is specialized for the uptake of particulate material by phagocytosis. Enters tissues that become infected or inflamed. (Chapter 13) (Figure 24–5)

NFκB protein Latent transcription regulator that is activated by various intracellular signaling pathways when cells are stimulated during immune, inflammatory, or stress responses. Also has important roles in animal development. (Chapter 15) (Figure 15–63)

nitric oxide (NO) Gaseous signal molecule that is widely used in cell–cell communication in both animals and plants. (Chapter 15) (Figure 15–41)

nitrogen fixation Biochemical process carried out by certain bacteria that reduces atmospheric nitrogen (N₂) to ammonia,

leading eventually to various nitrogen-containing metabolites. (Chapter 2)

NMDA receptor Subclass of glutamate-gated ion channel in the mammalian central nervous system critical for long-term potentiation and long-term depression. NMDA-receptor channels are doubly gated, opening only when glutamate is bound to the receptor and, simultaneously, the membrane is strongly depolarized. (Chapter 11)

NO synthase (NOS) Enzyme that synthesizes nitric oxide (NO) by the deamination of arginine. (Chapter 15) (Figure 15–41B)

NOD-like receptors (NLRs) Large family of pattern recognition receptors (PRRs) with leucine-rich repeat motifs; they are exclusively cytoplasmic and recognize a distinct set of microbial molecules. (Chapter 24)

nonclassical cadherins Large family of cadherins that are more distantly related in sequence than classical cadherins and include proteins involved in adhesion (including protocadherins, desmocollins, and desmogleins) and signaling. (Chapter 19)

noncoding RNA An RNA molecule that is the final product of a gene and does not code for protein. These RNAs serve as enzymatic, structural, and regulatory components for a wide variety of processes in the cell. (Chapter 6)

noncovalent bond Weak bonds that require multiple sets to hold two molecules together. Includes hydrogen bonds, electrostatic attractions, and van der Waals attractions. (Chapter 2)

nondisjunction Event occurring occasionally during meiosis in which a pair of homologous chromosomes fails to separate so that the resulting germ cell has either too many or too few chromosomes. (Chapter 17)

nonenveloped virus Virus consisting of a nucleic acid core and a protein capsid only. (Chapter 23) (Figure 23–19C and D)

nonhomologous end joining A DNA repair mechanism for rejoining the ends at double-strand breaks in which the two broken ends of DNA are brought together and rejoined by DNA ligation, generally with the loss of one or more nucleotides at the site of joining. (Chapter 5)

non-kinetochore microtubule In the mitotic or meiotic spindle, a microtubule that forms between the spindle poles and is not attached directly to kinetochores; it can cross-link at the spindle equator with an antiparallel microtubule from the other pole. Also called interpolar microtubule. (Chapter 17)

nonretroviral retrotransposon A type of transposable element that moves by being first transcribed into an RNA copy that is converted to DNA by reverse transcriptase and then inserted elsewhere in the genome. Its mechanism of insertion differs from that of the retroviral-like transposons. (Chapter 5) (Table 5–4, p. 308)

nonsense-mediated mRNA decay Mechanism for degrading aberrant mRNAs containing in-frame internal stop codons before they can be translated into protein. (Chapter 6) (Figure 6–80)

Notch Transmembrane receptor protein (and latent transcription regulator) involved in many cell-fate choices in animal development, for example in the specification of nerve cells from ectodermal epithelium. Its ligands are cell-surface proteins such as Delta and Serrate. (Chapter 15) (Figure 15–60)

NSF Hexameric ATPase that disassembles a complex of a v-SNARE and a t-SNARE. (Chapter 13) (Figure 13–21)

nuclear envelope The double membrane (two bilayers) surrounding the nucleus. Consists of an outer and inner membrane and is perforated by nuclear pores. The outer membrane is continuous with the endoplasmic reticulum. (Chapter 12) (Figures 4–10 and 12–54)

nuclear export receptors Proteins that bind to both the export signal and nuclear pore complex proteins, so as to guide their cargo through the nuclear pore complex to the cytosol. (Chapter 12)

nuclear export signal Sorting signal contained in the structure of those protein molecules and RNA–protein complexes that are to be transported from the nucleus to the cytosol through nuclear pore complexes; includes nuclear RNPs and new ribosomal subunits. (Chapter 12) (Figure 12–61)

nuclear import receptors Proteins that recognize nuclear localization signals to initiate the nuclear import of proteins containing the appropriate nuclear localization signal. (Chapter 12)

nuclear lamin Protein subunit of the intermediate filaments that form the nuclear lamina. (Chapter 12)

nuclear lamina Fibrous meshwork of proteins on the inner surface of the inner nuclear membrane. It is made up of a network of intermediate filaments formed from nuclear lamins. (Chapter 12)

nuclear localization signal (NLS) Signal sequence or signal patch found in proteins destined for the nucleus that enables their selective transport into the nucleus from the cytosol through the nuclear pore complexes. (Chapter 12) (Figures 12–56 and 12–61)

nuclear magnetic resonance (NMR) spectroscopy NMR is the resonant absorption of electromagnetic radiation at a specific frequency by atomic nuclei in a magnetic field, due to flipping of the orientation of their magnetic dipole moments. The NMR spectrum provides information about the chemical environment of the nuclei. NMR is used widely to determine the three-dimensional structure of small proteins and other small molecules. The principles of NMR are also used for medical diagnostic purposes in magnetic resonance imaging (MRI). (Chapter 8) (Figure 8–21)

nuclear pore complex (NPC) Large multiprotein structure forming an aqueous channel (the nuclear pore) through the nuclear envelope; it allows selected molecules to move between nucleus and cytoplasm. (Chapters 6, 12) (Figure 12–55)

nuclear receptor superfamily Intracellular receptors for hydrophobic signal molecules such as steroid and thyroid hormones and retinoic acid. The receptor–ligand complex acts as a transcription factor in the nucleus. (Chapter 15) (Figure 15–66)

nuclear transport The process by which cargo molecules are moved into and out of the nucleus through nuclear pore complexes. (Chapter 12)

nuclear transport receptor (karyopherin) Protein that escorts macromolecules either into or out of the nucleus: nuclear import receptor or nuclear export receptor. (Chapter 12) (Figure 12–61)

nucleolus A prominent structure in the nucleus where rRNA is transcribed and ribosomal subunits are assembled. (Figures 6–44 and 6–46)

nucleoporin Any of a number of different proteins that make up nuclear pore complexes. (Chapter 12)

nucleosome Beadlike structure in eukaryotic chromatin, composed of a short length of DNA wrapped around an octameric core of histone proteins. The fundamental structural unit of chromatin. (Chapter 4) (Figures 4–22 and 4–23)

nucleotide Nucleoside with a phosphate group joined in ester linkage to the sugar moiety. DNA and RNA are polymers of nucleotides. (Chapter 1) (Panel 2–6, pp. 104–105)

nucleotide excision repair Type of DNA repair that corrects irreversible damage of the DNA double helix, such as that caused by certain chemicals or UV light, by cutting out the damaged region on one strand and resynthesizing it using the undamaged strand as template. *Compare* **base excision repair**. (Chapter 5) (Figure 5–41)

O-linked glycosylation Addition of one or more sugars to a hydroxyl group on a protein. (Chapter 13)

obligate pathogens Bacteria that can only replicate inside their host. (Chapter 23)

olfactory receptors G-protein-coupled receptors on the modified cilia of olfactory receptor neurons that recognize odors. The receptors activate adenylyl cyclase via an olfactory-specific G protein (G_{olf}), and resultant increases in cAMP open cyclic-AMP-gated cation channels, allowing Na^+ influx and depolarization and initiation of a nerve impulse. (Chapter 15)

oligodendrocyte Glial cell in the vertebrate central nervous system that forms a myelin sheath around axons. *Compare* **Schwann cell**. (Chapter 11)

oligosaccharyl transferase Endoplasmic reticulum (ER) enzyme complex that transfers core oligosaccharides from dolichol lipid anchors to selected asparagine side chains in newly synthesized proteins as they enter the ER lumen. (Chapter 12)

oncogene An altered gene whose product can act in a dominant fashion to help make a cell cancerous. Typically, an oncogene is a mutant form of a normal gene (proto-oncogene) involved in the control of cell growth or division. (Chapter 20) (Figure 20–21)

open reading frame (ORF) A continuous nucleotide sequence free from stop codons in at least one of the three reading frames (and thus with the potential to code for protein). (Chapters 7, 8)

opportunistic pathogens Microbes of the normal flora that can cause disease only if the immune systems are weakened or if they gain access to a normally sterile part of the body. (Chapter 23)

optogenetics Use of genetically engineered channelrhodopsin and other light-responsive ion channels and transporters to modulate neuron function and hence analyze the neurons and circuits underlying complex functions, including behaviors in whole animals. (Chapter 11) (Figure 11–47)

organelle Subcellular compartment or large macromolecular complex, often but not always membrane-enclosed, that has a distinct structure, composition, and function. Examples of membrane-enclosed organelles are the nucleus, mitochondrion, ER, and Golgi apparatus; examples of organelles that form as biomolecular condensates and lack a membrane are the nucleolus and centrosomes. (Chapters 1, 12) (Figure 1–21)

organelle contact site Region of contact between two organelles stabilized by specific tethering proteins. (Chapter 12)

Organizer Specialized tissue at the dorsal lip of the blastopore in an amphibian embryo; a source of signals that

help to orchestrate formation of the embryonic body axis. (Chapter 21)

organoid A miniaturized and simplified organ produced in three-dimensional cell culture that possesses realistic microanatomy. (Chapter 22)

origin recognition complex (ORC) Large protein complex that is bound to the DNA at origins of replication in eukaryotic chromosomes throughout the cell cycle. (Chapters 5, 17) (Figure 5–31)

orthologs Genes or proteins from different species that are similar in sequence because they are descendants of the same gene in the last common ancestor of those species; orthologs often have the same or a very similar function in each organism. *Compare* **paralogs**. (Chapter 1) (Figure 1–20)

outer mitochondrial membrane The mitochondrial membrane that separates the organelle from the cytosol; surrounds the inner mitochondrial membrane. (Chapters 12, 14)

outer nuclear membrane One of two concentric membranes comprising the nuclear envelope; surrounds the inner nuclear membrane and is continuous with the inner nuclear membrane and the membrane of the endoplasmic reticulum. (Chapter 12)

OXA complex Protein translocator in the inner mitochondrial membrane that mediates insertion of inner membrane proteins. (Chapter 12)

oxidation (verb **oxidize**) Loss of electrons from an atom, as occurs during the addition of oxygen to a molecule or when a hydrogen is removed. Opposite of reduction. (Chapter 2) (Figure 2–20)

oxidative phosphorylation Process in bacteria and mitochondria in which ATP formation is driven by the transfer of electrons through an electron-transport chain to molecular oxygen. Involves the intermediate generation of an electrochemical proton gradient across a membrane and a chemiosmotic coupling of that gradient to drive ATP production by the ATP synthase. (Chapters 2, 14) (Figure 14–12)

P-type pumps A class of ATP-driven pumps comprising structurally and functionally related multipass transmembrane proteins that phosphorylate themselves during the pumping cycle. The class includes many of the ion pumps responsible for setting up and maintaining gradients of Na^+ , K^+ , H^+ , and Ca^{2+} across cell membranes. (Chapter 11) (Figure 11–12)

p53 A transcription regulatory protein that is activated by damage to DNA and is involved in blocking further progression through the cell cycle until the damage can be repaired. The *p53* gene is a tumor suppressor gene that is mutated in about half of human cancers. (Chapters 17, 20) (Figure 20–31)

pair-rule gene In *Drosophila* development, a gene expressed in a series of regular transverse stripes along the body of the embryo and which helps to determine its segments. (Chapter 21) (Figure 21–18)

pairing In meiosis, the lining up of the two homologous chromosomes along their length. (Chapter 17) (Figure 17–53)

pancreatic β cell Insulin-secreting cells of the pancreas. (Chapter 22)

papillomaviruses Class of viruses responsible for human warts and a prime example of DNA tumor viruses, being a cause of cancer of the uterine cervix. (Chapter 20)

paracrine signaling Short-range cell–cell communication via secreted signal molecules that act on neighboring cells. (Chapter 15) (Figure 15–2)

paralogs Genes or proteins that are similar in sequence because they are the result of a gene duplication event occurring in an ancestral organism. Those in two different organisms are less likely to have the same function than are orthologs. *Compare* **orthologs**. (Chapter 1) (Figure 1–20)

parasitism Ecologic relationship between microbes and their host in which the microbe benefits to the detriment of the host, as is often the case for pathogens. (Chapter 23)

passenger mutations Mutations that have occurred in the same cell as driver mutations, but which are irrelevant to the development of the cancer. (Chapter 20)

passive transport (facilitated diffusion) Transport of a solute across a membrane down its concentration gradient or its electrochemical gradient, using only the energy stored in the gradient. (Chapter 11) (Figure 11–4)

patch-clamp recording Electrophysiological technique in which a tiny electrode tip is sealed onto a patch of cell membrane, thereby making it possible to record the flow of current through individual ion channels in the patch. (Chapter 11) (Figure 11–36)

Patched Transmembrane protein predicted to cross the plasma membrane 12 times; located both in intracellular vesicles and on the cell surface where it binds the Hedgehog protein. (Chapter 15)

pathogen (adjective **pathogenic**) An organism, cell, virus, or prion that causes disease. (Chapters 23, 24)

pathogen-associated molecular patterns (PAMPs) Microbe-associated molecules, either not present or kept sequestered in the host organism, that often occur in a repeating pattern that is recognized by pattern recognition receptors (PRRs) of the innate immune system. PAMPs can be present in various microbial molecules, including nucleic acids, lipids, polysaccharides, and proteins. (Chapter 24)

pattern recognition receptor (PRR) Receptor present on or in cells of the innate immune system that recognizes and is activated by microbial pathogen-associated molecular patterns (PAMPs). (Chapter 24)

PDZ domain Protein-binding domain present in many scaffold proteins and often used as a docking site for the intracellular tails of transmembrane proteins. (Chapter 19) (Figure 19–22)

pectin Mixture of polysaccharides rich in galacturonic acid that forms a highly hydrated matrix in which cellulose is embedded in plant cell walls. (Chapter 19) (Figure 19–64)

peripheral (secondary) lymphoid organ Lymphoid organ in which T cells and B cells interact and respond to foreign antigens. Examples are spleen, lymph nodes, and mucosal-associated lymphoid organs. (Chapter 24) (Figure 24–12)

peroxins The proteins that form a protein translocator that participates in the import of proteins into peroxisomes. (Chapter 12)

peroxisome Small membrane-bounded organelle that uses molecular oxygen to oxidize organic molecules. Contains some enzymes that produce and others that degrade hydrogen peroxide (H_2O_2). (Chapter 12) (Figure 12–43)

pH scale Common measure of the acidity of a solution: “p” refers to power of 10, “H” to hydrogen. Defined as the negative logarithm of the hydrogen ion concentration in moles per liter (M). $\text{pH} = -\log[\text{H}^+]$. Thus a solution of pH 3 will contain 10^{-3} M hydrogen ions. pH less than 7 is acidic and pH greater than 7 is alkaline. (Chapter 2)

phagocytosis Process by which unwanted cells, debris, and other bulky particulate material is endocytosed (“eaten”) by a cell. Prominent in carnivorous cells, such as *Amoeba proteus*, and in vertebrate macrophages and neutrophils. From Greek *phagein*, “to eat.” (Chapter 13)

phagosome Large intracellular membrane-enclosed vesicle that is formed as a result of phagocytosis. Contains ingested extracellular material. (Chapter 13) (Figure 13–70)

phase-contrast microscope Type of light microscope that exploits the interference effects that occur when light passes through material of different refractive indices. Used to view living cells. (Chapter 9) (Figure 9–6)

phase variation The random switching of phenotype of an infectious agent that is caused by changes in expression of proteins at frequencies much higher than mutation rates. (Chapter 5)

phenotype The observable character (including both physical appearance and behavior) of a cell or organism. (Chapter 8) (Panel 8–1, pp. 520–521)

phosphatidylinositol 4,5-bisphosphate [PI(4,5)P₂, PIP₂] Membrane inositol phospholipid (a phosphoinositide) that is cleaved by phospholipase C into IP₃ and diacylglycerol at the beginning of the inositol phospholipid signaling pathway. It can also be phosphorylated by PI 3-kinase to produce PIP₃ docking sites for signaling proteins in the PI-3-kinase–Akt signaling pathway. (Chapter 15) (Figures 15–28 and 15–54)

phosphatidylserine Negatively charged phospholipid that is normally confined to the cytosolic leaflet of the lipid bilayer of the plasma membrane; in apoptotic cells, it accumulates in the outer leaflet, where it serves as an “eat me” signal to neighboring phagocytic cells. (Chapter 18)

phosphoinositide A lipid containing a phosphorylated inositol derivative. Minor component of the plasma membrane, but important in demarking different membranes and for intracellular signal transduction in eukaryotic cells. (Chapter 15) (Figure 15–53)

phosphoinositide 3-kinase (PI 3-kinase) Membrane-bound enzyme that is a component of the PI-3-kinase–Akt intracellular signaling pathway. It phosphorylates phosphatidylinositol 4,5-bisphosphate at the 3 position on the inositol ring to produce PIP₃ docking sites in the membrane for other intracellular signaling proteins. (Chapter 15) (Figure 15–54)

phosphoinositides (phosphatidylinositol phosphates, or PIPs) A lipid containing a phosphorylated inositol derivative. Minor component of the plasma membrane, but important in demarking different membranes and for intracellular signal transduction in eukaryotic cells. (Chapter 13) (Figure 13–10)

phospholipase C (PLC) Membrane-bound enzyme that cleaves inositol phospholipids to produce IP₃ and diacylglycerol in the inositol phospholipid signaling pathway. PLC β is activated by GPCRs via specific G proteins, while PLC γ is activated by RTKs. (Chapter 15) (Figure 15–56)

phospholipid The main category of lipids used to construct biomembranes. Generally composed of two fatty acids linked through glycerol (or sphingosine) phosphate to one of a variety of polar groups. (Chapter 10) (Figure 10–3 and Panel 2–5, pp. 102–103)

phosphorylation Reaction in which a phosphate group is covalently coupled to another molecule. (Chapters 2, 3, 15)

photoactivation Technique for studying intracellular processes in which an inactive form of a molecule of interest

is introduced into the cell and is then activated by a focused beam of light at a precise spot in the cell. (Chapter 9)

photochemical reaction center The part of a photosystem that converts light energy into chemical energy in photosynthesis. (Chapter 14) (Figure 14–46)

photorespiration A wasteful form of metabolism conducted by photosynthetic plants in low CO₂ environments that consumes O₂ and liberates CO₂, but does not result in the production of an energy carrier useful to the plant. (Chapter 14)

photosynthetic electron-transfer reactions Light-driven reactions in photosynthesis in which electrons move along an electron-transport chain in a membrane, generating ATP and NADPH. (Chapter 14)

photosystem Multiprotein complex involved in photosynthesis that captures the energy of sunlight and converts it to useful forms of energy: a reaction center plus an antenna (Chapter 14) (Figure 14–46)

phototropin Photoprotein associated with the plant plasma membrane that senses blue light and is partly responsible for phototropism. (Chapter 15)

phragmoplast Structure made of microtubules and actin filaments that forms in the prospective plane of division of a plant cell and guides formation of the cell plate. (Chapter 17) (Figure 17–48)

phytochrome Plant photoprotein that senses light via a covalently attached light-absorbing chromophore, which changes its shape in response to light and then induces a change in the protein's conformation. Plant phytochromes are cytoplasmic serine/threonine kinases, which respond differentially and reversibly to red and far-red light to alter cell behavior. (Chapter 15)

PI-3-kinase–Akt pathway Intracellular signaling pathway that stimulates animal cells to survive and grow. (Chapter 15) (Figure 15–54)

pinocytosis Literally, “cell drinking.” Type of endocytosis in which soluble materials are continually taken up from the environment in small vesicles and moved into endosomes along with the membrane-bound molecules. *Compare phagocytosis.* (Chapter 13) (Figure 13–51)

piRNAs (piwi-interacting RNAs) A class of small noncoding RNAs made in the germ line that, in complex with Piwi proteins, keep the movement of transposable elements in check by transcriptionally silencing transposon genes and destroying RNAs produced by them. (Chapter 7)

planar cell polarity Type of cellular asymmetry seen in some epithelia, such that each cell has a polarity vector oriented in the plane of the epithelium. (Chapter 21) (Figure 21–50)

plant growth regulator (plant hormone) Signal molecule that helps coordinate growth and development. Examples are ethylene, auxins, gibberellins, cytokinins, abscisic acid, and the brassinosteroids. (Chapter 15)

plasma membrane The membrane that surrounds a living cell. (Chapters 1, 10) (Figure 10–1)

plasmid vector Small, circular molecules of double-stranded DNA that have been derived from plasmids that occur naturally in bacterial cells; widely used for gene cloning. (Chapter 8)

plasmodesma (plural **plasmodesmata**) Plant equivalent of a gap junction. Communicating cell–cell junction in plants in which a channel of cytoplasm lined by plasma membrane

connects two adjacent cells through a small pore in their cell walls. (Chapter 19)

platelet Cell fragment, lacking a nucleus, that breaks off from a megakaryocyte in the bone marrow and is found in large numbers in the bloodstream. Helps initiate blood clotting when blood vessels are injured. (Figure 22–12)

pleckstrin homology domain (PH domain) Protein domain found in some intracellular signaling proteins. Some PH domains in intracellular signaling proteins bind to phosphatidylinositol 3,4,5-trisphosphate produced by PI 3-kinase, bringing the signaling protein to the plasma membrane when PI 3-kinase is activated. (Chapter 15)

pluripotent Describes a cell that has the potential to give rise to all or almost all of the cell types of the adult body. (Chapters 21, 22)

podosomes Actin-rich protrusive and adhesive structures on the cell surface important for crossing tissue barriers. (Chapter 16)

point spread function The distribution of light intensity within the three-dimensional, blurred image that is formed when a single point source of light is brought to a focus with a lens. (Chapter 9)

polarized In epithelia, the fact that the basal end of a cell, adherent to the basal lamina below, differs from the apical end, exposed to the medium above; thus, all epithelia and their individual cells are structurally polarized. (Chapter 19)

Polycomb group Set of proteins critical for cell memory for some genes. They form complexes as part of the chromatin of the *Hox* complex, where they maintain a repressed state in cells where *Hox* genes have not been activated. (Chapters 4, 21) (Figure 4–40)

polymerase chain reaction (PCR) Technique for amplifying specific regions of DNA by the use of sequence-specific primers and multiple cycles of DNA synthesis, each cycle being followed by a brief heat treatment to separate complementary strands. (Chapter 8) (Figure 8–34)

polymorphisms Describes genome sequences that coexist as two or more sequence variants at high frequency in a population. (Chapter 8)

polypeptide Linear polymer of amino acids. Proteins are large polypeptides, and the two terms can be used interchangeably. (Figure 3–1)

polypeptide backbone Repeating sequence of atoms along the core of the polypeptide chain. (Figure 3–1)

polyribosome mRNA engaged with multiple ribosomes in the act of translation. (Figure 6–77)

polytene chromosome Giant chromosome in which the DNA has undergone repeated replication and the many copies have stayed together in precise alignment. (Chapter 4) (Figures 4–47 and 4–48)

porins Channel-forming proteins of the outer membranes of bacteria, mitochondria, and chloroplasts. (Chapter 12)

position effect variegation Alteration in gene expression resulting from a change in the position of the gene on a chromosome, caused by the spread of neighboring heterochromatic domains. When an active gene is placed next to heterochromatin, the inactivating influence of the heterochromatin can spread to affect the gene to a variable degree, giving rise to a variegated phenotype. (Figure 4–32)

positional value A cell's internal record of its positional information in a multicellular organism; an intrinsic character that differs according to a cell's location. (Chapter 21)

positive feedback Control mechanism whereby the end product of a reaction or pathway stimulates its own production or activation. (Figure 7–42)

positive selection In immunology: a process of thymocyte maturation in which thymocytes expressing a T cell receptor with appropriate affinity for a self peptide bound to a self MHC protein is signaled to survive and continue development. (Chapter 24)

post-transcriptional controls Any control on gene expression that is exerted at a stage after transcription has begun. (Chapter 7) (Figure 7–57)

post-translational Occurring after completion of translation, thus after the completion of protein synthesis. (Chapter 12)

preprophase band Circumferential band of microtubules and actin filaments that forms around a plant cell under the plasma membrane prior to mitosis and cell division. (Chapter 17) (Figure 17–48)

primary cell wall The first cell wall produced by a developing plant cell; it is thin and flexible, allowing room for cell growth. (Chapter 19) (Figure 19–64)

primary cilium Short, single, nonmotile cilium lacking dynein that arises from a centriole and projects from the surface of many animal cell types. Involved in cell signaling, with some signaling proteins being concentrated in these (Chapter 15) (Figure 15–39)

primary Ig repertoire The billions of IgM and IgD immunoglobulin molecules made by the B cells of an adaptive immune system in the absence of antigen stimulation. (Chapter 24)

primary immune response Adaptive immune response to an antigen that is made on first encounter with that antigen. (Chapter 24) (Figure 24–16)

primary pathogens Pathogens that can cause overt disease in most healthy people. Some cause acute, life-threatening epidemic infections and spread rapidly between hosts; other potential primary pathogens may persistently infect a single individual for years without causing overt disease, the host often being unaware of being infected. (Chapter 23)

primary structure Linear sequence of monomer units in a polymer, such as the amino acid sequence of a protein. (Chapter 3)

primary tumor Tumor at the original site at which a cancer first arose. Secondary tumors develop elsewhere by metastasis. (Chapter 20)

prion disease Transmissible spongiform encephalopathy—such as kuru and Creutzfeldt–Jakob disease (CJD) in humans, scrapie in sheep, and bovine spongiform encephalopathy (BSE, or “mad cow disease”) in cows—that is caused and transmitted by an infectious, abnormally folded protein (prion). (Figure 3–33)

pro-apoptotic Bcl2 family effectors Pro-apoptotic proteins of the intrinsic pathway of apoptosis that in response to an apoptotic stimulus become activated and aggregate to form oligomers in the mitochondrial outer membrane, inducing the release of cytochrome *c* and other intermembrane proteins. Bax and Bak are the main effector Bcl2 family proteins in mammalian cells. (Chapter 18)

pro-inflammatory cytokine Any cytokine that stimulates an inflammatory response. (Chapter 24)

prokaryote Single-celled microorganism whose cells lack a well-defined, membrane-enclosed nucleus. Either a bacterium or an archaeon. (Chapter 1) (Figure 1–9)

promoter Nucleotide sequence in DNA to which RNA polymerase binds to begin transcription. (Chapters 6, 7) (Figure 6–11)

prophase The initial stage of mitosis, during which the chromosomes condense and the mitotic spindle begins to assemble outside of the nucleus. It is followed by prometaphase—where the nuclear envelope disassembles, and the microtubules from each spindle pole engage with the chromosomes. (Panel 17–1, pp. 1048–1049)

proteasome Large protein complex in the cytosol with proteolytic activity that is responsible for degrading proteins that have been marked for destruction by ubiquitylation or by some other means. (Chapter 6) (Figures 6–86 and 6–87)

protein The major macromolecular constituent of cells. A linear polymer of amino acids linked together by peptide bonds in a specific sequence. (Chapters 1, 3) (Figure 3–1)

protein activity control The selective activation, inactivation, degradation, or compartmentalization of specific proteins after they have been made. One of the means by which a cell controls which proteins are active at a given time or location in the cell. (Chapter 7)

protein degradation control The means by which the concentration of a protein in the cell can be reduced by selectively degrading it in response to external signals or to stages of the cell division cycle. (Chapter 7)

protein domain See **domain**. (Chapter 3)

protein glycosylation The process of transferring either a single saccharide or a preformed precursor oligosaccharide to proteins. (Chapter 12)

protein kinase Enzyme that transfers the terminal phosphate group of ATP to one or more specific amino acids (serine, threonine, or tyrosine) of a target protein. (Chapters 3, 15)

protein kinase C (PKC) Ca^{2+} -dependent protein kinase that, when activated by diacylglycerol and an increase in the concentration of cytosolic Ca^{2+} , phosphorylates target proteins on specific serine and threonine residues. (Chapter 15) (Figure 15–30)

protein phosphatase Enzyme that catalyzes phosphate removal from amino acids of a target protein. (Chapters 3, 15)

protein phosphatase 2A (PP2A) Protein phosphatase, composed of three subunits, that dephosphorylates Cdk substrates and thereby helps to govern cell-cycle progression. (Chapter 17)

protein phosphorylation The covalent addition of a phosphate group to a serine, threonine, or tyrosine side chain of a protein. (Chapter 3)

protein subunit An individual protein chain in a protein composed of more than one chain. (Chapter 3)

protein translocation The process of moving a protein across a membrane. (Chapter 12)

protein translocator Any membrane-bound protein that mediates the transport of another protein across a membrane. (Chapter 12) (Figure 12–48)

protein tyrosine phosphatase Enzyme that removes phosphate groups from phosphorylated tyrosine residues on proteins. (Chapter 15)

proteoglycan Molecule consisting of one or more glycosaminoglycan chains attached to a core protein. (Chapters 13, 19) (Figure 19–36)

proto-oncogene Normal gene, usually concerned with the regulation of cell proliferation, that can be converted into a cancer-promoting oncogene by mutation. (Chapter 20)

protofilament Linear string of subunits joined end to end; multiple protofilaments associate with one another laterally to construct and provide strength and adaptability to microtubules. (Chapter 16)

proton (H^+) Positively charged subatomic particle that forms part of an atomic nucleus. Hydrogen has a nucleus composed of a single proton (H^+). (Chapter 2)

proton-motive force The force exerted by the electrochemical proton gradient that moves protons across a membrane. (Chapter 14)

protozoan parasite Parasitic, nonphotosynthetic, single-celled, motile eukaryotic organism; for example, *Plasmodium*. (Chapter 23)

pseudogene Nucleotide sequence of DNA that has accumulated multiple mutations that have rendered an ancestral gene inactive and nonfunctional. (Chapter 4)

pseudopodia Three-dimensional actin-based protrusions at the leading edge of rapidly migrating cells. (Chapter 16)

purified cell-free system Fractionated cell homogenate that retains a particular biological function of the intact cell, and in which biochemical reactions and cell processes can be most easily studied. (Chapter 8)

purifying selection Natural selection operating in a population to slow genome changes and reduce divergence by eliminating individuals carrying deleterious mutations. (Chapter 4)

Q cycle The cyclic series of reactions that allow the quinone (Q) electron carrier to transfer an electron to the cytochrome *c* reductase complex and then recapture it, in order to pump two rather than one proton per electron across the membrane in a respiratory system; a similar Q cycle occurs as part of the electron transfers critical for photosynthesis. (Figure 14–23)

quantitative RT-PCR (reverse transcription–polymerase chain reaction) Technique in which a population of mRNAs is converted into cDNAs via reverse transcription, and the cDNAs are then amplified by PCR. The quantitative part relies on a direct relationship between the rate at which the PCR product is generated and the original concentration of the mRNA species of interest. Widely used for detecting viral infection. (Chapter 8)

quaternary structure Three-dimensional relationship of the different polypeptide chains in a multisubunit protein or protein complex. (Chapter 3)

quinone (Q) Small, lipid-soluble, mobile electron carrier molecule found in the respiratory and photosynthetic electron-transport chains. (Chapter 14) (Figure 14–17)

Rab cascade An ordered recruitment of sequentially acting Rab proteins into Rab domains on membranes, which changes the identity of an organelle and determines membrane dynamics. (Chapter 13)

Rab effectors Molecules that bind activated, membrane-bound Rab proteins and act as downstream mediators of vesicle transport, membrane tethering, and fusion. (Chapter 13)

Rab proteins Monomeric GTPases in the Ras superfamily present in plasma and organelle membranes in their

GTP-bound state and as soluble cytosolic proteins in their GDP-bound state. Involved in conferring specificity on vesicle docking. (Chapter 13) (Table 15–5, p. 915)

Rac (Rac protein) Member of the Rho family of monomeric GTPases that regulate the actin and microtubule cytoskeletons, cell-cycle progression, gene transcription, and membrane transport. (Chapter 15)

Rad51 protein Eukaryotic protein that catalyzes the pairing of homologous DNA strands during recombination and repair processes. Analogous to the RecA protein in *E. coli* and other bacteria. (Chapter 5)

Ran (Ran protein) Monomeric GTPase of the Ras superfamily present in both cytosol and nucleus. It is required for the active transport of macromolecules into and out of the nucleus through nuclear pore complexes. (Chapter 12) (Table 15–5, p. 915)

rapidly inactivating K⁺ channel Neuronal voltage-gated K⁺ channel, open when the membrane is depolarized, with a specific voltage sensitivity and kinetics of inactivation that induce a reduced rate of action potential firing at levels of stimulation only just above the threshold required, thereby resulting in a firing rate proportional to the strength of the depolarizing stimulus. (Chapter 11)

Ras A small family of proto-oncogenes that are frequently mutated in cancers, each of which produces a Ras monomeric GTPase. (Chapter 20)

Ras (Ras protein) Monomeric GTPase of the Ras superfamily that helps to relay signals from cell-surface receptor tyrosine kinase receptors to the nucleus, frequently in response to signals that stimulate cell division. Named for the *Ras* gene, first identified in viruses that cause rat sarcomas. (Chapters 3, 15, 17) (Figure 3–64)

Ras GAPs Ras GTPase-activating proteins; increase the rate of hydrolysis of bound GTP by Ras, thereby inactivating Ras. (Chapter 15)

Ras GEFs Ras guanine nucleotide exchange factors; stimulate the dissociation of GDP and the subsequent uptake of GTP from the cytosol, thereby activating Ras. (Chapter 15)

Ras–MAP-kinase signaling pathway Intracellular signaling pathway that relays signals from activated receptor tyrosine kinases to effector proteins in the cell, including transcription regulators in the nucleus. (Chapter 15)

Ras superfamily Large superfamily of monomeric GTPases (also called small GTP-binding proteins) of which Ras is the prototypical member. (Chapter 15) (Table 15–5, p. 915)

Rb A cell-cycle control protein that inhibits E2F transcription regulatory proteins and entry into S phase of the cell cycle. The *Rb* gene is a tumor suppressor that is defective in both copies in individuals with retinoblastoma. (Chapter 20) (Figures 17–59 and 20–24)

reading frame The phase in which nucleotides are read in sets of three to encode a protein. An mRNA molecule can be read in any one of three reading frames, only one of which will give the required protein. (Chapter 6) (Figure 6–53)

RecA protein Prototype for a ubiquitous class of DNA-binding proteins that catalyze synapsis of DNA strands during genetic recombination in bacteria; analogous to Rad51 protein in eukaryotes. (Chapter 5) (Figure 5–48)

receptor Any protein that binds a specific signal molecule (ligand) and initiates a response in the cell. Some are on the

cell surface, while others are inside the cell. (Chapter 15) (Figure 15–3)

receptor editing Process by which a developing B cell that recognizes a self molecule changes its antigen receptors so that the cell no longer does so. (Chapter 24)

receptor-mediated endocytosis Internalization of receptor–ligand complexes from the plasma membrane by endocytosis, a type of receptor down-regulation. (Chapter 13) (Figure 13–54)

receptor serine/threonine kinase Cell-surface receptor with an extracellular ligand-binding domain and an intracellular kinase domain that phosphorylates signaling proteins on serine or threonine residues in response to ligand binding. The TGF receptor is an example. (Chapter 15) (Figure 15–58)

receptor tyrosine kinase (RTK) Cell-surface receptor with an extracellular ligand-binding domain and an intracellular kinase domain that phosphorylates signaling proteins on tyrosine residues in response to ligand binding. (Chapter 15) (Figure 15–44 and Table 15–4, p. 911)

recombinant DNA technology Collection of techniques by which DNA segments from different sources are combined to make a new DNA, often called a recombinant DNA. Recombinant DNAs are widely used in the cloning of genes, in the genetic modification of organisms, and in the production of large amounts of rare proteins. (Chapter 8)

recycling endosome Organelle that provides an intermediate stage on the passage of recycled receptors back to the cell membrane. Regulates plasma membrane insertion of some proteins. (Chapter 13) (Figure 13–56)

red blood cell Small hemoglobin-containing blood cell of vertebrates that transports oxygen to, and carbon dioxide from, tissues. Also called an erythrocyte. (Figures 22–11 and 22–12)

redox pair Pair of molecules in which one acts as an electron donor and one as an electron acceptor in an oxidation–reduction reaction; for example, NADH (electron donor) and NAD⁺ (electron acceptor). (Panel 14–1, p. 825)

redox potential The affinity of a redox pair for electrons, generally measured as the voltage difference between an equimolar mixture of the pair and a standard reference. NADH/NAD⁺ has a low redox potential, and O₂/H₂ has a high redox potential (high affinity for electrons). (Panel 14–1, p. 825)

redox reaction Reaction in which one component becomes oxidized and the other reduced; an oxidation–reduction reaction. (Chapter 14) (Panel 14–1, p. 825)

reduction (verb **reduce**) Addition of electrons to an atom, as occurs during the addition of hydrogen to a biological molecule or the removal of oxygen from it. Opposite of oxidation. (Chapter 2) (Figure 2–20)

regulated secretory pathway A secretory pathway found mainly in cells specialized for secreting products rapidly on demand—such as hormones, neurotransmitters, or digestive enzymes—in which soluble proteins and other substances are initially stored in secretory vesicles for later release. (Chapter 13) (Figure 13–38)

regulator of G protein signaling (RGS) A type of GAP protein that binds to a trimeric G protein and enhances its GTPase activity, thus helping to limit G-protein-mediated signaling. (Chapter 15) (Figure 15–8)

regulatory site Region of an enzyme surface to which a regulatory molecule binds and thereby influences the catalytic events at the separate active site. (Chapter 3)

regulatory T cell (T_{reg}) A type of T cell that suppresses the development, activation, or function of other immune cells via secreted cytokines or cell-surface inhibitory proteins. (Chapter 24)

replication fork Y-shaped region of a replicating DNA molecule; the point at which the two strands of the parent DNA helix are being separated and the daughter strands are being formed. (Chapter 5) (Figures 5–7 and 5–18)

replication origin A location on a DNA molecule at which duplication of the DNA begins by the formation of replication forks. (Chapters 4, 5) (Figures 4–19 and 5–24)

replicative cell senescence Phenomenon observed in primary cell cultures in which cell proliferation slows down and finally irreversibly halts. (Chapters 17, 20)

respiratory chain (electron-transport chain) Electron-transport chain present in the inner mitochondrial membrane that generates an electrochemical proton gradient across the membrane: this gradient is used to drive ATP synthesis by ATP synthase. (Chapter 14) (Figures 14–4 and 14–10)

resting membrane potential Electrical potential across the plasma membrane of a cell at rest; that is, a cell that has not been stimulated to open additional ion channels beyond those that are normally open. (Chapter 11)

restriction nuclease One of a large number of nucleases that can cleave a DNA molecule at any site where a specific short sequence of nucleotides occurs. Extensively used in recombinant DNA technology. (Chapter 8) (Figure 8–23)

restriction point Important transition at the end of G₁ in the eukaryotic cell cycle; passage past this point commits the cell to enter S phase. The term was originally used for this transition in the mammalian cell cycle; in this book we use the term “Start.” (Chapter 17) (Figure 17–8)

retinoblastoma A rare type of human cancer arising from cells in the retina of the eye that are converted to a cancerous state by an unusually small number of mutations. Studies of retinoblastoma led to the discovery of the first tumor suppressor gene. (Chapter 20)

retinoblastoma protein (Rb protein) Tumor suppressor protein involved in the regulation of cell division. Mutated in the cancer retinoblastoma, as well as in many other tumors. Its normal activity is to regulate the eukaryotic cell cycle by binding to and inhibiting the E2F proteins, thus blocking progression to DNA replication and cell division. (Chapter 17) (Figure 17–59)

retroviral-like retrotransposons A large family of transposons that move themselves in and out of chromosomes by a mechanism similar to that used by retroviruses, being first transcribed into an RNA copy that is converted to DNA by reverse transcriptase then inserted elsewhere in the genome. (Chapter 5) (Table 5–4, p. 308)

retrovirus RNA-containing virus that replicates in a cell by first making an RNA–DNA intermediate and then a double-strand DNA molecule that becomes integrated into the cell's DNA. (Chapters 5, 20) (Figure 5–61)

reverse genetics Approach to discovering gene function that starts from the DNA (gene) and its protein product and then creates mutants to analyze the gene's function. (Chapter 8)

reverse transcriptase Enzyme first discovered in retroviruses that makes a double-strand DNA copy from a single-strand RNA template molecule. (Chapter 5)

Reynolds number A dimensionless number that represents the ratio of inertial forces to viscous forces acting on an object moving through a fluid. The inertial force or momentum relates

to the size of an object and its speed, while viscous forces depend on the properties of the fluid. (Chapter 16)

RGD sequence Tripeptide sequence of arginine-glycine-aspartic acid that forms a binding site for integrins; present in fibronectin and some other extracellular proteins. (Chapter 19) (Figure 19–48C)

Rho family Family of monomeric GTPases within the Ras superfamily involved in signaling the rearrangement of the cytoskeleton. Includes Rho, Rac, and Cdc42. (Chapters 15, 16) (Table 15–5, p. 915)

Rho protein Member of the Rho family of monomeric GTPases that regulate the actin and microtubule cytoskeletons, cell-cycle progression, gene transcription, and membrane transport. (Chapter 15)

rhodopsin Seven-span membrane protein of the GPCR family that acts as a light sensor in rod photoreceptor cells in the vertebrate retina. Contains the light-sensitive prosthetic group retinol. (Chapter 15) (Figure 15–40)

ribonucleic acid See **RNA**. (Chapter 1)

ribosomal RNA (rRNA) Any one of a number of specific RNA molecules that form part of the structure of a ribosome and participate in the synthesis of proteins. Often distinguished by their sedimentation coefficient (for example, 28S rRNA, 5S rRNA). (Chapter 6)

ribosome Particle composed of rRNAs and ribosomal proteins that catalyzes the synthesis of protein using information provided by mRNA. (Chapter 6) (Figure 6–68)

ribozyme An RNA molecule with catalytic activity. (Chapter 6)

RNA (ribonucleic acid) Polymer formed from covalently linked ribonucleotide monomers. See also **messenger RNA**, **ribosomal RNA**, **transfer RNA**. (Chapter 1) (Figures 6–4 and 6–7)

RNA editing Type of RNA processing that alters the nucleotide sequence of an RNA transcript after it is synthesized by inserting, deleting, or altering individual nucleotides. (Chapter 7)

RNA exosome Large protein complex with an interior rich in 3′-to-5′ RNA exonucleases; degrades RNA molecules to produce ribonucleotides. (Chapter 6)

RNA interference (RNAi) As originally described, mechanism by which an experimentally introduced double-stranded RNA induces sequence-specific destruction of complementary mRNAs. The term RNAi is often used to include the inhibition of gene expression by microRNAs (miRNAs) and piwi-interacting RNAs (piRNAs), which are encoded in the cell's own genome. (Chapter 7)

RNA polymerase Enzyme that catalyzes the synthesis of an RNA molecule on a DNA template from ribonucleoside triphosphate precursors. (Chapter 6) (Figure 6–9)

RNA primer Short stretch of RNA synthesized on a DNA template. It is required by DNA polymerases to start their DNA synthesis. (Chapter 5)

RNA-processing control Regulation by a cell of gene expression by controlling the processing of RNA transcripts, which includes their splicing. (Chapter 7)

RNA splicing Process in which intron sequences are excised from RNA transcripts. A major process in the nucleus of eukaryotic cells leading to formation of messenger RNAs (mRNAs). (Chapter 6)

RNA transport and localization control Regulation by a cell of gene expression by selecting which completed mRNAs are exported from the nucleus to the cytosol and determining where in the cytosol they are localized. (Chapter 7)

RNA world Hypothesis that early life on Earth was based primarily on RNA molecules that both stored genetic information and catalyzed biochemical reactions. (Chapter 6)

RNA-seq Sequencing the entire repertoire of RNA from a cell or tissue; also known as deep RNA sequencing. (Chapter 8)

robustness The ability of biological regulatory systems to function normally in the face of perturbations such as exposure to frequent and/or extreme variations in external conditions or the concentrations or activities of key components. (Chapter 8)

rod photoreceptor (rod) Photoreceptor cell in the vertebrate retina that is responsible for noncolor vision in dim light. (Chapter 15)

rough ER (rough endoplasmic reticulum) Endoplasmic reticulum with ribosomes on its cytosolic surface. Involved in the synthesis of secreted and membrane-bound proteins. (Chapter 12) (Figure 12–15)

rRNA gene Gene that specifies a ribosomal RNA (rRNA). (Chapter 6)

ryanodine receptor A regulated Ca^{2+} channel in the ER membrane that opens in response to rising Ca^{2+} levels and thus amplifies a Ca^{2+} signal. (Chapter 15)

SAM complex Protein translocator that helps β -barrel proteins to fold properly in the outer mitochondrial membrane. (Chapter 12)

Sanger sequencing See **dideoxy sequencing**. (Chapter 8)

Sar1 protein (Sar1 GEF) Monomeric GTPase responsible for regulating COPII coat assembly at the endoplasmic reticulum membrane. (Chapter 13)

sarcoma Cancer of connective tissue. (Chapter 20)

scaffold protein Protein that binds groups of intracellular proteins into a complex, often anchoring the complex at a specific location in the cell. (Chapters 3, 15) (Figures 3–76 and 15–10)

scanning electron microscope (SEM) Type of electron microscope that produces an image of the surface of an object. (Chapter 9) (Figure 9–45)

S-Cdk Cyclin–Cdk complex formed in vertebrate cells by an S-cyclin and the corresponding cyclin-dependent kinase (Cdk). (Chapter 17) (Figure 17–10 and Table 17–1, p. 1034)

SCF Family of ubiquitin ligases formed as a complex of several different proteins. One is involved in regulating the eukaryotic cell cycle, directing the destruction of inhibitors of S-Cdks in late G_1 and thus promoting the activation of S-Cdks and DNA replication. (Chapters 3, 17) (Figure 3–67)

Schwann cell Glial cell responsible for forming myelin sheaths in the peripheral nervous system. *Compare oligodendrocyte*. (Chapter 11) (Figure 11–35)

S-cyclin Member of a class of cyclins that accumulate during late G_1 phase and bind Cdks soon after progression through Start; they help stimulate DNA replication and chromosome duplication. Levels remain high until late mitosis, after which these cyclins are destroyed. (Chapter 17) (Figure 17–10)

Sec61 complex Three-subunit core of the protein translocator that transfers polypeptide chains across the endoplasmic reticulum membrane. (Chapter 12)

second messenger (small intracellular mediator) Small intracellular signaling molecule that is formed or released for action in response to an extracellular signal and helps to relay the signal within the cell. Examples include cyclic AMP, cyclic GMP, IP_3 , Ca^{2+} , and diacylglycerol. (Chapter 15)

secondary cell wall Permanent rigid cell wall that is laid down underneath the thin primary cell wall in certain plant cells that have completed their growth. (Chapter 19)

secondary Ig repertoire Immunoglobulins produced by B cells after somatic hypermutation and class switching induced by antigen and helper T cells. Compared to the primary Ig repertoire, these Igs have a greatly increased diversity of both Ig classes and antigen-binding sites and have increased affinity for antigen. (Chapter 24)

secondary immune response The adaptive immune response that occurs in response to a second or subsequent exposure to an antigen. The response is more rapid in onset and stronger than the primary immune response. (Chapter 24) (Figure 24–16)

secondary structure Regular local folding pattern of a polymeric molecule; in proteins, α helices and β sheets. (Chapter 3)

secretory vesicle Membrane-enclosed organelle in which molecules destined for secretion are stored prior to release. Sometimes called secretory granule because darkly staining contents make the organelle visible as a small solid object. (Chapter 13) (Figure 13–43)

securin Protein that binds to the protease separase and thereby prevents its cleavage of the protein linkages that hold sister chromatids together in early mitosis. Securin is destroyed at the metaphase-to-anaphase transition. (Chapter 17) (Figure 17–39)

segment Divisions of an insect body along its anteroposterior axis, each forming highly specialized structures, but all built according to a similar fundamental plan. (Chapter 21)

segment-polarity gene In *Drosophila* development, a gene involved in specifying the anteroposterior organization of each body segment. (Chapter 21) (Figure 21–18)

segmentation clock The gene expression oscillator controlling regular segmentation during vertebrate embryonic development. (Chapter 21)

segmentation genes Genes expressed by subsets of cells in the embryo that refine the pattern of gene expression so as to define the boundaries and ground plan of the individual body segments. (Chapter 21)

selectin Member of a family of cell-surface carbohydrate-binding proteins that mediate transient, Ca^{2+} -dependent cell–cell adhesion in the bloodstream; for example, between white blood cells and the endothelium of the blood vessel wall. (Chapter 19) (Figure 19–28)

selectivity filter The part of an ion channel structure that determines which ions it can transport. (Chapter 11) (Figures 11–24 and 11–25)

sensory bristles Miniature sense organs present on most exposed surfaces of *Drosophila*, consisting of a sensory neuron and supporting cells and responding to chemical or mechanical stimuli. (Chapter 21)

separase Protease that cleaves the cohesin protein linkages that hold sister chromatids together. Acts at anaphase, enabling chromatid separation and segregation. (Chapter 17) (Figure 17–39)

septum Structure formed during bacterial cell division by the inward growth of the cell wall and plasma membrane; it divides the cell into two. (Chapter 16)

sequential induction Development process that generates a progressively more complicated pattern. A series of local inductions whereby one of two cell types present in a developing tissue can produce a signal to induce neighboring cells to specialize in a third way; the third cell type can then signal back to the other two cell types nearby to generate a fourth and a fifth cell type, and so on. (Figure 21–12)

serine protease A type of protease that has a reactive serine in the active site. (Chapters 3, 19) (Figures 3–13 and 3–37)

serine/threonine kinase Enzyme that uses ATP to phosphorylate specific proteins on serine or threonines. (Chapter 15)

SH2 domain Src homology region 2, a protein domain present in many signaling proteins. Binds a short amino acid sequence containing a phosphotyrosine. (Chapter 15) (Panel 3–2, pp. 150–151)

side chain The part of an amino acid that differs between amino acid types. The side chains give each type of amino acid its unique physical and chemical properties. (Chapter 3) (Panel 3–1, pp. 118–119)

signal patch Protein-sorting signal that consists of a specific three-dimensional arrangement of amino acids on the folded protein's surface. (Chapter 12) (Figure 13–41)

signal peptidase Enzyme that removes a terminal signal sequence from a protein once the sorting process is complete. (Figures 12–27 and 12–49)

signal sequence Short continuous sequence of amino acids that determines the eventual location of a protein in the cell. An example is the N-terminal sequence of 20 or so amino acids that directs nascent secretory and transmembrane proteins to the endoplasmic reticulum. (Chapter 12) (Figure 12–13)

signal-recognition particle (SRP) Ribonucleoprotein particle that binds an ER signal sequence on a partially synthesized polypeptide chain and directs the polypeptide and its attached ribosome to the endoplasmic reticulum. (Chapter 12) (Figure 12–19)

signaling center Cluster of specialized cells in developing tissues that serves as a source of developmental signals; for example, for the generation of a morphogen gradient. (Chapter 21)

single-molecule localization microscopy (SMLM) Any one of a number of superresolution microscopy techniques for the visualization of biological systems with an optical resolution measured in the tens of nanometers. These methods, which include PALM and STORM, allow the accurate localization of single fluorescently labeled molecules by imaging sufficient photons, and then calculating the center of the resulting point spread functions. (Chapter 9)

single-nucleotide polymorphism (SNP) A variation between individuals in a population due to a relatively common difference in a specific nucleotide at a defined point in the DNA sequence. (Chapter 8)

single-particle reconstruction Computational procedure in electron microscopy in which images of many identical molecules are obtained and digitally combined to produce an averaged three-dimensional image, thereby revealing structural details that are hidden by noise in the original images; critical for attaining atomic-level resolution by cryo-electron microscopy (Chapter 9) (Panel 9–1, p. 594)

single-pass transmembrane protein Membrane protein in which the polypeptide chain crosses the lipid bilayer only once. (Chapter 10) (Figure 10–24)

single-strand DNA-binding (SSB) protein Protein that binds to the single strands of the opened-up DNA double helix, preventing helical structures from re-forming while the DNA is being replicated. (Chapter 5) (Figure 5–15)

sister chromatids Tightly linked pair of chromosomes that arise from chromosome duplication during S phase. They separate during M phase and segregate into different daughter cells. (Chapter 17) (Figures 4–18 and 17–24)

skeletal muscle Body muscles made up of myofibril-containing striated muscle fibers that are under the voluntary control of the somatic nervous system. (Chapter 22)

sliding clamp Ring-shaped protein complex that holds the DNA polymerase on DNA during DNA replication. (Chapter 5) (Figure 5–17)

Smad family Latent transcription regulators that are phosphorylated and activated by receptor serine/threonine kinases to carry a signal from the cell surface to the nucleus. (Chapter 15) (Figure 15–58)

small interfering RNAs (siRNAs) Short (21–26 nucleotide) double-stranded RNAs that inhibit gene expression by directing destruction of complementary mRNAs. Production of siRNAs is usually triggered by exogenously introduced double-stranded RNA. (Chapter 7) (Figure 7–80)

small nuclear RNA (snRNA) Small RNA molecules that are complexed with proteins to form the ribonucleoprotein particles (small nuclear ribonucleoproteins; snRNPs) involved in RNA splicing. (Chapter 6) (Figures 6–29 and 6–30)

small nucleolar RNA (snoRNA) Small RNAs found in the nucleolus, with various functions, including guiding the modifications of precursor rRNA. (Chapter 6) (Table 6–1, p. 327, and Figure 6–43)

SMC protein complex A large ring-shaped protein complex that uses ATP hydrolysis energy to produce the looped domains that organize chromosomes; includes the eukaryotic cohesion and condensin proteins (Figures 4–56, 4–57, and 17–25)

smooth ER (smooth endoplasmic reticulum) Region of the endoplasmic reticulum not associated with ribosomes. Involved in detoxification reactions, Ca^{2+} storage, and lipid synthesis. (Chapter 12) (Figure 12–15)

Smoothed Seven-pass transmembrane protein with a structure very similar to a GPCR; it is controlled by the Patched and iHog proteins. (Chapter 15)

SNARE proteins (SNAREs) Members of a large family of transmembrane proteins present in organelle membranes and the vesicles derived from them. SNAREs catalyze the many membrane fusion events in cells. They exist in pairs—a v-SNARE in the vesicle membrane that binds specifically to a complementary t-SNARE in the target membrane. (Chapter 13)

sodium dodecyl sulfate polyacrylamide-gel electrophoresis (SDS-PAGE) Type of electrophoresis used to separate proteins by size. The protein mixture to be separated is first treated with a powerful negatively charged detergent (SDS) and with a reducing agent (β -mercaptoethanol), before being run through a polyacrylamide gel. The detergent and reducing agent unfold the proteins, free them from association with other molecules, and separate any polypeptide subunits. (Chapter 8)

somatic cell Any cell of a plant or animal other than cells of the germ line. From Greek *soma*, “body.” (Chapter 5)

somatic hypermutation In immunology: the targeted accumulation of point mutations in the assembled variable-region coding sequences of immunoglobulin genes that occurs when B cells are activated to form memory cells. Results in the production of antibodies with altered antigen-binding sites, some of which bind antigen with increased affinity; it is important for affinity maturation in antibody responses. (Chapter 24)

somatic mutations In cancer, one or more detectable abnormalities in the DNA sequence of tumor cells that distinguish them from the normal somatic cells from which the tumor was derived. (Chapter 20)

somite One of a series of paired blocks of mesoderm that form during early development and lie on either side of the notochord in a vertebrate embryo. They give rise to the segments of the body axis, including the vertebrae, muscles, and associated connective tissue. (Chapter 21) (Figure 21–38)

sorting receptors Proteins involved in binding signal sequences or cargo proteins and bringing them to specific sites in the cell where cargo proteins are released. Sorting receptors pick up and deliver cargo proteins repeatedly. (Chapter 12)

sorting signal Signal sequence or signal patch that directs the delivery of a protein to a specific location, such as to a particular intracellular compartment. (Chapter 12)

spectrin Abundant protein associated with the cytosolic side of the plasma membrane in red blood cells, forming a network that supports the membrane. Also present in other cells. (Chapters 10, 16) (Figure 10–38)

S phase The period of a eukaryotic cell cycle in which DNA is synthesized. (Chapter 5) (Figure 17–4)

spindle assembly checkpoint Regulatory system that operates during mitosis to ensure that all chromosomes are properly attached to the spindle before sister-chromatid separation starts. (Chapter 17) (Figure 17–20 and Panel 17–1, pp. 1048–1049)

spliceosome Large assembly of RNA and protein molecules that performs pre-mRNA splicing in eukaryotic cells. (Chapter 6)

Src (Src protein family) Family of cytoplasmic tyrosine kinases (pronounced “sark”) that associate with the cytoplasmic domains of some enzyme-linked cell-surface receptors that lack intrinsic tyrosine kinase activity. They transmit a signal onward by phosphorylating the receptor itself and specific intracellular signaling proteins on tyrosines. (Figures 3–61 and 24–46)

SRP (signal-recognition particle) receptor Component in the endoplasmic reticulum (ER) membrane that guides the signal-recognition particle to the ER membrane. (Chapter 12)

standard free-energy change (ΔG°) The change in free energy under a standard condition, defined as that where the concentrations of all the reactants are set to the same fixed value of 1 mole/liter. (Chapter 2)

starch Polysaccharide composed exclusively of glucose units, used as an energy-storage material in plant cells. (Figure 2–53)

Start (restriction point) Important transition at the end of G_1 in the eukaryotic cell cycle. Passage through Start commits the cell to enter S phase. The term was originally used for this point in the yeast cell cycle only; the equivalent point in the mammalian cell cycle was called the restriction point. In this book we use Start for both. (Chapter 17) (Figure 17–8)

STAT (signal transducer and activator of transcription)

Latent transcription regulator that is activated by phosphorylation by Janus kinases (JAKs) and enters the nucleus in response to signaling from receptors of the cytokine receptor family. (Figure 15–57)

stem cell Undifferentiated cell that can continue dividing indefinitely, throwing off daughter cells that can either commit to differentiation or remain a stem cell (in the process of self-renewal). (Chapter 22) (Figure 22–2)

stem-cell niche The specialized microenvironment in a tissue in which self-renewing stem cells can be maintained. (Chapter 22)

steroid hormones Hormones, including cortisol, estrogen, and testosterone, that are hydrophobic lipid molecules derived from cholesterol that activate intracellular nuclear receptors. (Chapter 15)

stimulatory G protein (G_s) G protein that, when activated, activates the enzyme adenylyl cyclase and thus stimulates the production of cyclic AMP. See also **G protein**. (Chapter 15) (Table 15–3, p. 907)

stochastic Random. Involving chance, probability, or random variables. (Chapter 8)

strand exchange Reaction in which a single-strand 3' end from one duplex DNA molecule penetrates another duplex and finds a homologous sequence through base-pairing. Also called strand invasion. (Chapter 5)

strand-directed mismatch repair A proofreading system that removes DNA replication errors missed by the DNA polymerase proofreading exonuclease. It detects the DNA helix distortion from noncomplementary base pairs and then excises the mismatch in the newly synthesized strand specifically; the excised DNA segment is then replaced using the old strand as a template. (Chapter 5)

stress fibers Cortical fibers of contractile actin–myosin II bundles that connect the cell to the extracellular matrix or adjacent cells through focal adhesions or a circumferential belt and adherens junctions. (Chapter 16)

stroma (1) “Bedding”: the connective tissue in which a glandular or other epithelium is embedded. Stromal cells provide the environment necessary for the development of other cells within the tissue. (2) The large interior space of a chloroplast, containing enzymes that incorporate CO_2 into sugars. (Chapters 12, 14, 20) (Figures 14–38 and 20–19)

substrate Molecule on which an enzyme acts. (Chapters 2, 3)

superresolution Describes several approaches in light microscopy that bypass the limit imposed by the diffraction of light and successfully allow objects as small as 10 nm to be imaged and clearly resolved. (Chapter 9)

survival factor Extracellular signal that promotes cell survival by inhibiting apoptosis. (Chapter 18) (Figure 18–12)

symporter Carrier protein that transports two types of solute across the membrane in the same direction. (Chapter 11) (Figure 11–8)

synapse Communicating cell–cell junction that allows signals to pass from a nerve cell to another cell. In a chemical synapse, the signal is carried by a diffusible neurotransmitter. In an electrical synapse, a direct connection is made between the cytoplasm of the two cells via gap junctions. (Figures 11–38 and 19–23)

synaptic plasticity Changes in the strength with which a chemical synapse transmits a signal. It is thought to be

important in memory formation, where concentrations of postsynaptic AMPA receptor are modulated in response to a synapse's activity. (Chapter 11)

synaptic signaling Intercellular signaling performed by neurons that transmit signals electrically along their axons and release neurotransmitters at synapses, which are often located far away from the neuronal cell body. (Chapter 15)

synaptic vesicle Small neurotransmitter-filled secretory vesicle found at the axon terminals of nerve cells. Its contents are released into the synaptic cleft by exocytosis when an action potential reaches the axon terminal. (Chapter 13)

synaptonemal complex Structure that holds paired homologous chromosomes tightly together in pachytene of prophase I in meiosis and promotes the final steps of crossing-over. (Chapter 17) (Figures 17–54 and 17–55)

syncytium Mass of cytoplasm containing many nuclei enclosed by a single plasma membrane. Typically the result either of cell fusion or of a series of incomplete division cycles in which the nuclei divide but the cell does not. (Chapters 17, 21)

T cell receptor (TCR) Transmembrane receptor for antigen on the surface of T lymphocytes, consisting of an immunoglobulin-like heterodimer. (Chapter 24) (Figure 24–32)

T cell-mediated immune response Any adaptive immune response mediated by antigen-specific T cells. (Chapter 24)

tail-anchored protein Protein containing a transmembrane segment very close to its C-terminus. (Chapter 12)

TATA box Sequence in the promoter region of many eukaryotic genes that binds a general transcription factor (TFIID) and hence specifies the position at which transcription is initiated. (Chapter 6) (Figure 6–15)

telomerase Enzyme that elongates the telomere sequences in DNA, which occur at the ends of eukaryotic chromosomes. (Chapters 5, 17) (Figure 5–33)

telomere End of a chromosome, associated with a characteristic DNA sequence that is replicated in a special way. Counteracts the tendency of the chromosome otherwise to shorten with each round of replication. From Greek *telos*, “end,” and *meros*, “portion.” (Chapters 4, 17)

telophase Final stage of mitosis in which the two sets of separated chromosomes decondense and become enclosed by nuclear envelopes. (Chapter 17) (Panel 17–1, pp. 1048–1049)

template Single strand of DNA or RNA whose nucleotide sequence acts as a guide for the synthesis of a complementary strand. (Chapter 4) (Figure 1–3)

terminal differentiation The limit of cell determination when a cell forms one of the highly specialized cell types of the adult body. (Chapter 21)

terminally differentiated A cell at the limit of cell determination, being one of the highly specialized cell types of the adult body. (Chapter 21)

terminator Signal in bacterial DNA that halts transcription; in eukaryotes, transcription terminates after cleavage and polyadenylation of the newly synthesized RNA. (Chapter 6)

tertiary structure Complex three-dimensional form of a folded polymer chain, especially a protein or RNA molecule. (Chapter 3)

T_{FH} cell See **follicular helper T cell**. (Chapter 24)

T_H1 cell A type of effector helper T cell that secretes IFN γ to help activate macrophages and induces B cells to switch the class of antibody they make. (Chapter 24) (Figure 24–44)

T_H17 cell A type of effector helper T cell that secretes IL17, which recruits neutrophils and stimulates an inflammatory response. (Chapter 24) (Figure 24–44)

T_H2 cell A type of effector helper T cell that helps activate B cells to produce antibodies, to undergo somatic hypermutation, and switch the class of immunoglobulin produced. (Chapter 24) (Figure 24–44)

thylakoid Flattened sac of membrane inside a chloroplast that contains chlorophyll and other pigments and carries out the light-trapping reactions of photosynthesis. Stacks of thylakoids form the grana of chloroplasts. (Chapters 12, 14) (Figures 14–37 and 14–38)

thylakoid membrane Chloroplast membrane system that contains the large membrane protein complexes for photosynthesis and photophosphorylation. (Chapter 14)

thymocytes Developing T cells in the thymus. (Chapter 24)

tight junction Cell–cell junction that seals adjacent epithelial cells together, preventing the passage of most dissolved molecules from one side of the epithelial sheet to the other. (Chapter 19) (Figures 19–2 and 19–21)

TIM complexes Protein translocators in the mitochondrial inner membrane. The TIM23 complex mediates the transport of proteins into the matrix and the insertion of some proteins into the inner membrane; the TIM22 complex mediates the insertion of a subgroup of proteins into the inner membrane. (Chapter 12) (Figure 12–48)

tissue-specific stem cells (or adult stem cells) Stem cells that replenish the differentiated cell types of a particular tissue. They are found in small numbers and produce only a subset of cell types compared to embryonic stem cells. (Chapter 22)

Toll A transmembrane receptor protein. On the ventral side of the *Drosophila* egg membrane, its activation controls the distribution of Dorsal, a transcription regulator of the NF κ B family. (Chapter 21)

Toll-like receptors (TLRs) Family of pattern recognition receptors (PRRs) on or in cells of the innate immune system. They recognize pathogen-associated immunostimulants (that is, PAMPs) associated with microbes. (Chapter 24) (Figure 24–4)

TOM complex Multisubunit protein complex that transports proteins across the mitochondrial outer membrane. (Chapter 12) (Figure 12–48)

TOR Large serine/threonine protein kinase that is activated by the PI-3-kinase–Akt signaling pathway and promotes cell growth. (Chapter 15)

totipotent Describes a cell that is able to give rise to all the different cell types in an organism. (Chapters 21, 22)

trans face Face on the other (far) side. (Chapter 13)

trans Golgi network (TGN) Network of interconnected tubular and cisternal structures closely associated with the *trans* face of the Golgi apparatus and the compartment from which proteins and lipids exit the Golgi, bound for the cell surface or another compartment. (Chapter 13)

transcellular transport Transport of solutes, such as nutrients, across an epithelium, by means of membrane transport proteins in the apical and basal faces of the epithelial cells. (Chapter 11) (Figure 11–11)

transcription (DNA transcription) Copying of one strand of DNA into a complementary RNA sequence by the enzyme RNA polymerase. (Chapters 1, 6) (Figures 6–1 and 6–8)

transcription regulators General name for any protein that binds to a specific DNA sequence (known as a *cis*-regulatory sequence) to influence the transcription of a gene. (Chapter 7)

transcriptional control Regulation by a cell of gene expression by controlling when and how often a given gene is transcribed. (Chapter 7)

transcytosis Uptake of material at one face of a cell by endocytosis, its transfer across a cell in vesicles, and discharge from another face by exocytosis. (Chapter 13) (Figure 13–56)

transfer RNA (tRNA) Set of small RNA molecules used in protein synthesis as an interface (adaptor) between mRNA and amino acids. Each type of tRNA molecule is covalently linked to a particular amino acid. (Chapter 6) (Figure 6–54)

transferrin receptor Cell-surface receptor for transferrin (a soluble protein that carries iron); delivers iron to the cell interior via receptor-mediated endocytosis and recycling of the receptor–transferrin complex. (Chapter 13)

transformed A cell with an altered phenotype that behaves in many ways like a cancer cell (that is, unregulated proliferation, anchorage-independent growth in culture). (Chapter 20)

transforming growth factor- β superfamily (TGF β superfamily) Large family of structurally related secreted proteins that act as hormones and local mediators to control a wide range of functions in animals, including during development. It includes the TGF β /activin and bone morphogenetic protein (BMP) subfamilies. (Chapter 15) (Figure 15–58)

transgene The foreign or modified gene that has been added to create a transgenic organism. (Chapter 8)

transgenic organism Plant or animal that has stably incorporated one or more genes from another cell or organism (through insertion, deletion, and/or replacement) and can pass them on to successive generations. (Chapter 8) (Figures 8–55 and 8–71)

transit amplifying cell Cell derived from a stem cell that divides a limited number of times before terminally differentiating. (Figure 22–3)

transition state Structure that forms transiently in the course of a chemical reaction and has the highest free energy of any reaction intermediate. Its formation is a rate-limiting step in the reaction. (Chapter 3) (Figure 3–45)

translation (RNA translation) Process by which the sequence of nucleotides in an mRNA molecule directs the incorporation of amino acids into protein. Occurs on a ribosome. (Chapters 1, 6) (Figures 6–1 and 6–68)

translational control Regulation by a cell of gene expression by selecting which mRNAs in the cytoplasm are translated by ribosomes. (Chapter 7)

transmembrane adhesion proteins Transmembrane proteins with one end linking to the cytoskeleton inside the cell and the other end linking to other structures outside the cell. (Chapter 19)

transmembrane protein Membrane protein that extends through the lipid bilayer, with part of its mass on either side of the membrane. (Chapter 10) (Figure 10–17)

transmembrane segment The region of a transmembrane protein that spans the lipid bilayer, often as an α helix. (Chapter 12)

transmitter-gated ion channel (ionotropic receptor) Ion channel found at chemical synapses in the postsynaptic plasma

membranes of nerve and muscle cells. Opens only in response to the binding of a specific extracellular neurotransmitter. The resulting inflow of ions leads to the generation of a local electrical signal in the postsynaptic cell. (Chapter 11) (Figures 11–38 and 15–6)

transport vesicle Membrane-enclosed transport containers that bud from specialized coated regions of a donor membrane and pass from one cell compartment to another as part of the cell's membrane transport processes; vesicles can be spherical, tubular, or irregularly shaped. (Chapter 13)

transporter (carrier protein, permease) Membrane transport protein that binds to a solute and transports it across the membrane by undergoing a series of conformational changes. Transporters can either transport ions or molecules passively down an electrochemical gradient or they can link the conformational changes to a source of metabolic energy such as ATP hydrolysis to drive active transport. *Compare channel.* See also **membrane transport protein.** (Chapter 11) (Figure 11–3)

transposable element (transposon) Segment of DNA that can move from one chromosomal position to another by transposition. (Chapter 5) (Table 5–4, p. 308)

transposition (transpositional recombination) Movement of a DNA sequence from one chromosomal site to another. (Chapter 5) (Table 5–4, p. 308)

transposon See **transposable element.** (Chapter 5)

treadmilling Process by which a polymeric protein filament is maintained at constant length by addition of protein subunits at one end and loss of subunits at the other. (Panel 16–2, pp. 960–961)

Trithorax group Set of proteins critical for cell memory that maintains the transcription of *Hox* genes in cells where transcription has already been switched on. (Chapter 21)

t-SNAREs Transmembrane SNARE protein, usually composed of three proteins and found on target membranes where it interacts with v-SNAREs on vesicle membranes. (Chapter 13)

tubulin The protein subunit of microtubules. (Chapter 16) (Panel 16–1, p. 950, and Figure 16–36)

γ -tubulin ring complex (γ -TuRC) Protein complex containing γ -tubulin and other proteins that is an efficient nucleator of microtubule formation; it caps microtubule minus ends. (Chapter 16)

tumor progression Process by which an initial mildly disordered cell behavior gradually evolves into a full-blown cancer by the selection for cell proliferation. (Chapter 20) (Figures 20–8 and 20–9)

tumor suppressor gene Gene that appears to help prevent formation of a cancer. Loss-of-function mutations in such genes favor the development of cancer. (Chapter 20) (Figure 20–21)

tumor virus Virus that can help make the cell it infects cancerous. (Chapter 20)

turgor pressure Large hydrostatic pressure developed inside a plant cell as the result of the intake of water by osmosis; it is the force driving cell expansion in plant growth and it maintains the rigidity of plant stems and leaves. (Chapter 19)

two-dimensional gel electrophoresis Technique combining two different separation procedures—separation by charge (isoelectric focusing) in the first dimension, then separation by size in a direction at a right angle to that of the first step—to

resolve up to 2000 proteins in the form of a two-dimensional protein map. (Chapter 8)

type III fibronectin repeat The major repeat domain in fibronectin, it is about 90 amino acids long and occurs at least 15 times in each subunit. The repeat is among the most common of all protein domains in vertebrates. (Chapter 19)

type III secretion system One of several secretion systems in Gram-negative bacteria; delivers effector proteins into host cells in a contact-dependent manner. (Chapter 23) (Figure 23–7)

type IV collagen An essential component of mature basal laminae consisting of three long protein chains twisted into a rope-like superhelix with multiple bends. Separate molecules assemble into a flexible, felt-like network that gives the basal lamina tensile strength. (Chapter 19)

tyrosine kinase Enzyme that phosphorylates specific proteins on tyrosines. *See also* **cytoplasmic tyrosine kinase, receptor tyrosine kinase (RTK)**. (Chapter 15)

tyrosine-kinase-associated receptor Cell-surface receptor that functions similarly to receptor tyrosine kinases (RTKs), except that the kinase domain is encoded by a separate gene and is noncovalently associated with the receptor polypeptide chain. (Chapter 15)

ubiquitin Small, highly conserved protein present in all eukaryotic cells that can be covalently attached to lysines of other proteins. Attachment of a short chain of ubiquitins to such a lysine can tag a protein for intracellular proteolytic destruction by a proteasome. (Chapter 3) (Figure 3–65)

ubiquitin ligase Any one of a large number of enzymes that attach ubiquitin to a protein, often marking it for destruction in a proteasome. The process catalyzed by a ubiquitin ligase is called ubiquitylation. (Chapter 3) (Figure 3–67)

unfolded protein response The cellular response triggered by an accumulation of misfolded proteins in the endoplasmic reticulum. It causes an expansion of the ER and increased transcription of genes that code for endoplasmic reticulum chaperones and degradative enzymes, as well as other changes. (Chapter 12) (Figure 12–36)

uniporter Carrier protein that transports a single solute from one side of the membrane to the other. (Chapter 11) (Figure 11–8)

V gene segment A DNA sequence encoding most of the variable region of an immunoglobulin or T cell receptor polypeptide chain. There are many different V gene segments, one of which becomes joined to a D or J gene segment by somatic recombination when an individual lymphoid progenitor cell begins to differentiate into a B or T lymphocyte. (Chapter 24) (Figure 24–28)

vaccination The process of injecting a substance (called an antigen) into an animal in order to elicit an adaptive immune response that protects the animal against pathogens containing the antigen; critical for combating the COVID-19 pandemic. (Chapter 24)

vacuole Large fluid-filled compartment found in most plant and fungal cells, often occupying more than a third of the cell volume. (Chapter 13) (Figure 13–66)

van der Waals attraction Type of (individually weak) noncovalent bond that is formed at close range between nonpolar atoms. (Table 2–1, p. 51, and Panel 2–3, pp. 98–99)

variable region Region of an immunoglobulin or T cell receptor polypeptide chain that is the most variable and

contributes to the antigen-binding site. (Chapter 24) (Figures 24–25 and 24–32)

vascular endothelial growth factor (VEGF) Secreted protein that stimulates the growth of blood vessels. (Table 15–4, p. 911)

V(D)J recombination Somatic recombination process by which gene segments are brought together to form a functional gene for a polypeptide chain of an immunoglobulin or T cell receptor. (Chapter 24) (Figure 24–28)

vesicular transport The transport of proteins from one cell compartment to another by means of membrane-bounded intermediaries, such as vesicles or organelle fragments. (Chapter 12)

virulence factor Protein, encoded by a virulence gene, that contributes to an organism's ability to cause disease. (Chapter 23)

virulence gene Gene that contributes to an organism's ability to cause disease. (Chapter 23)

virus Particle consisting of nucleic acid (RNA or DNA) enclosed in a protein coat and capable of replicating within a host cell and spreading from cell to cell. (Chapters 1, 23) (Figure 23–11)

virus receptor Molecule on the host-cell surface to which virus surface proteins bind to enable the binding of virus to the cell and its subsequent entry into the cytoplasm. (Chapter 23)

voltage-gated cation channel Type of ion channel found in the membranes of electrically excitable cells (such as nerve, endocrine, egg, and muscle cells). Opens in response to a shift in membrane potential past a threshold value. (Chapter 11)

voltage-gated K⁺ channel Ion channel in the membrane of nerve cells that opens in response to membrane depolarization, enabling K⁺ efflux and rapid restoration of the negative membrane potential. (Chapter 11)

voltage-gated Na⁺ channel Ion channel in the membrane of nerve and skeletal muscle cells that opens in response to a stimulus causing sufficient depolarization, allowing Na⁺ to enter the cell down its electrochemical gradient. (Chapter 11)

v-SNAREs Transmembrane SNARE protein, comprising a single polypeptide chain, usually found in vesicle membranes where it interacts with t-SNAREs in target membranes. (Chapter 13)

V-type pumps Turbine-like protein machines constructed from multiple different subunits that use the energy of ATP hydrolysis to drive transport across a membrane. A V-type proton pump transfers H⁺ into organelles such as lysosomes to acidify their interior. (Chapter 11) (Figure 11–12)

water channel *See* **aquaporin**. (Chapter 11)

Wee1 Protein kinase that inhibits Cdk activity by phosphorylating amino acids in the Cdk active site. Important for regulating entry into M phase of the cell cycle. (Chapter 17)

western blotting Technique by which proteins are separated by electrophoresis and immobilized on a paper sheet and then analyzed, usually by means of a labeled antibody. Also called immunoblotting. (Chapter 8)

white blood cell General name for all the nucleated blood cells lacking hemoglobin. Also called leukocytes. Includes lymphocytes, granulocytes, and monocytes. (Table 22–1, p. 1286)

Wnt protein Member of a family of secreted signal proteins that have many different roles in controlling cell differentiation,

proliferation, and gene expression in animal embryos and adult tissues. (Chapter 15)

Wnt/ β -catenin pathway Signaling pathway activated by binding of a Wnt protein to its cell-surface receptors. The pathway has several branches. In the major (canonical) branch, activation causes increased amounts of β -catenin to enter the nucleus, where it regulates the transcription of genes controlling cell differentiation and proliferation. Overactivation of the Wnt/ β -catenin pathway can lead to cancer. (Chapter 15) (Figure 15–61)

XIAP An inhibitor of apoptosis protein (IAP) that is encoded on the X chromosome. (Chapter 18)

X-inactivation Inactivation of one copy of the X chromosome in the somatic cells of female mammals. (Chapter 7)

X-inactivation center (XIC) Site in an X chromosome at which inactivation is initiated and spreads outward. (Figure 7–55)

Xist A long (20,000 nucleotide) noncoding RNA responsible for inactivating one of the two X chromosomes in female mammals. (Chapter 7)

x-ray crystallography Technique for determining the three-dimensional arrangement of atoms in a molecule on the basis of the diffraction pattern of x-rays passing through a crystal of the molecule. (Chapter 8) (Figure 8–20)

zygote Diploid cell produced by fusion of a male and female gamete. A fertilized egg. (Chapter 21)

